

Continuity and Differentiability

Chapter 5 · All Questions and Complete Solutions



Part 1 · Examples

Q

1

Example:
 $f(x) = 2x + 3$



Check the continuity of the function f given by $f(x) = 2x + 3$ at $x = 1$.

Conclusion

First note that the function is defined at the given point $x = 1$ and its value is 5.

Then find the limit of the function at $x = 1$.

Clearly

Thus

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (2x + 3) = 2(1) + 3 = 5$$

$$\lim_{x \rightarrow 1} f(x) = 5 = f(1)$$

Hence, f is continuous at $x = 1$.

Q

2

Example:
 $f(x) = x^2$



Examine whether the function f given by $f(x) = x^2$ is continuous at $x = 0$.

Conclusion

First note that the function is defined at the given point $x = 0$ and its value is 0.

Then find the limit of the function at $x = 0$.

Clearly

Thus

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} x^2 = 0^2 = 0$$

$$\lim_{x \rightarrow 0} f(x) = 0 = f(0)$$

Hence, f is continuous at $x = 0$.

Q

3

Example:
 $f(x) = |x|$



Discuss the continuity of the function f given by $f(x) = |x|$ at $x = 0$.

Conclusion

By definition

$$f(x) = \begin{cases} -x, & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases}$$

Clearly the function is defined at 0 and $f(0) = 0$.

Left hand limit of f at 0 is

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (-x) = 0$$

Similarly, the right hand limit of f at 0 is

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x = 0$$

Thus, the left hand limit, right hand limit and the value of the function coincide at $x = 0$.

Hence, f is continuous at $x = 0$.

Q

4

Example:
piecewise
function



Show that the function f given by

$$f(x) = \begin{cases} x^3 + 3, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$$

is not continuous at $x = 0$.

Conclusion

The function is defined at $x = 0$ and its value at $x = 0$ is 1.

When $x \neq 0$, the function is given by a polynomial.

Hence,

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} (x^3 + 3) = 0^3 + 3 = 3$$

Since the limit of f at $x = 0$ does not coincide with $f(0)$, the function is not continuous at $x = 0$.

It may be noted that $x = 0$ is the only point of discontinuity for this function.

Q

5

Example:
 $f(x)=k$



Check the points where the constant function $f(x) = k$ is continuous.

Conclusion

The function is defined at all real numbers and by definition, its value at any real number equals k .

Let c be any real number.

Then

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} k = k$$

Since $f(c) = k = \lim_{x \rightarrow c} f(x)$ for any real number c , the function f is continuous at every real number.

Q

6

Example:
 $f(x)=x$



Prove that the identity function on real numbers given by $f(x) = x$ is continuous at every real number.

Given

The function is clearly defined at every point, and $f(c) = c$ for every real number c .

Also,

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} x = c$$

Thus, $\lim_{x \rightarrow c} f(x) = c = f(c)$, so the function is continuous at every real number.

We have now defined continuity at a single point.

Next, we extend this to continuity of a function as a whole.

Definition 2 A real function f is said to be continuous if it is continuous at every point in the domain of f .

Conclusion

This needs a little more explanation.

Suppose f is a function defined on a closed interval $[a, b]$.

For f to be continuous, it must be continuous at every point in $[a, b]$, including the end points a and b .

Continuity of f at a means

$$\lim_{x \rightarrow a^+} f(x) = f(a)$$

and continuity of f at b means

$$\lim_{x \rightarrow b^-} f(x) = f(b)$$

Note that $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow b^+} f(x)$ do not make sense here.

It follows that if f is defined at only one point, it is continuous there — i.

e.

, if the domain of f is a single point, f is a continuous function.

Q

7

Example:
 $f(x)=|x|$



Is the function defined by $f(x) = |x|$, a continuous function?

Conclusion

We may rewrite f as

$$f(x) = \begin{cases} -x, & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases}$$

By Example 3, we know that f is continuous at $x = 0$.

Let c be a real number such that $c < 0$.

Then $f(c) = -c$.

Also

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (-x) = -c$$

Since $\lim_{x \rightarrow c} f(x) = f(c)$, f is continuous at all negative real numbers.

Now, let c be a real number such that $c > 0$.

Then $f(c) = c$.

Also

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} x = c$$

Since $\lim_{x \rightarrow c} f(x) = f(c)$, f is continuous at all positive real numbers.

Hence, f is continuous at all points.

Q

8

Example:
 $f(x) = x^3 + x^2 - 1$



Discuss the continuity of the function f given by $f(x) = x^3 + x^2 - 1$.

Conclusion

Clearly f is defined at every real number c and its value at c is $c^3 + c^2 - 1$.

We also know that

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (x^3 + x^2 - 1) = c^3 + c^2 - 1$$

Thus $\lim_{x \rightarrow c} f(x) = f(c)$, and hence f is continuous at every real number.

This means f is a continuous function.

Q

9

Example:
 $f(x) = 1/x, x \neq 0$



Discuss the continuity of the function f defined by $f(x) = \frac{1}{x}, x \neq 0$.

Given

Fix any non-zero real number c .

We have

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} \frac{1}{x} = \frac{1}{c}$$

Also, since for $c \neq 0$, $f(c) = \frac{1}{c}$, we have $\lim_{x \rightarrow c} f(x) = f(c)$.

So f is continuous at every point in the domain of f .

Thus f is a continuous function.

Here is a good place to explain the idea of infinity.

We do this by looking at the function $f(x) = \frac{1}{x}$ near $x = 0$.

As usual, we find the value of the function at real numbers close to 0. We are really trying to find the right hand limit of f at 0. We tabulate this below (Table 5.1).

Table 5.1							
x	1	0.3	0.2	$0.1 = 10^{-1}$	$0.01 = 10^{-2}$	$0.001 = 10^{-3}$	10^{-n}
$f(x)$	1	3.333...	5	10	$100 = 10^2$	$1000 = 10^3$	10^n

Substitute

As x gets closer to 0 from the right, the value of $f(x)$ keeps rising.

In other words, the value of $f(x)$ can be made larger than any given number by choosing a positive real number close enough to 0. In symbols, we write

$$\lim_{x \rightarrow 0^+} f(x) = +\infty$$

(read as: the right hand limit of $f(x)$ at 0 is plus infinity).

Note that $+\infty$ is NOT a real number, so the right hand limit of f at 0 does not exist as a real number.

In the same way, we can find the left hand limit of f at 0. The next table is self-explanatory.

Table 5.2							
x	-1	-0.3	-0.2	-10^{-1}	-10^{-2}	-10^{-3}	-10^{-n}
$f(x)$	-1	-3.333...	-5	-10	-10^2	-10^3	-10^n

Conclusion

From Table 5.2, the value of $f(x)$ can be made smaller than any given number by choosing a negative real number close enough to 0. In symbols, we write

$$\lim_{x \rightarrow 0^-} f(x) = -\infty$$

(read as: the left hand limit of $f(x)$ at 0 is minus infinity).

Again, $-\infty$ is NOT a real number, so the left hand limit of f at 0 does not exist as a real number.

The graph of the reciprocal function in Fig 5.3 shows this geometrically.

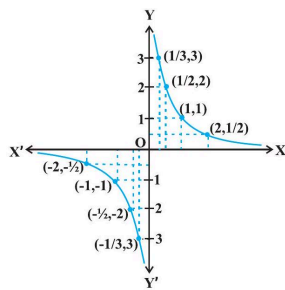


Fig 5.3

Q

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Example:
piecewise
function



Discuss the continuity of the function f defined by

$$f(x) = \begin{cases} x+2, & \text{if } x \leq 1 \\ x-2, & \text{if } x > 1 \end{cases}$$

Given

The function f is defined at all points of the real line.

Substitute

Case 1 If $c < 1$, then $f(c) = c+2$.

Therefore, $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (x+2) = c+2$ Thus, f is continuous at all real numbers less than 1.

Case 2 If $c > 1$, then $f(c) = c-2$.

Therefore,

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (x-2) = c-2 = f(c)$$

Thus, f is continuous at all points $x > 1$.

Conclusion

Case 3 If $c = 1$, then the left hand limit of f at $x = 1$ is

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x+2) = 1+2 = 3$$

The right hand limit of f at $x = 1$ is

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (x-2) = 1-2 = -1$$

Since the left and right hand limits of f at $x = 1$

do not coincide, f is not continuous at $x = 1$.

Hence $x = 1$ is the only point of discontinuity of f .

The graph of the function is given in Fig 5.4.

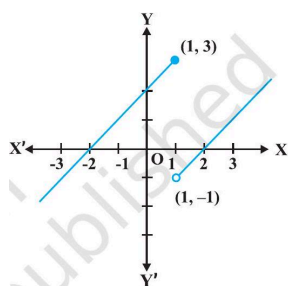


Fig 5.4

Q

11

Example:
piecewise
function

((

Find all the points of discontinuity of the function f defined by

$$f(x) = \begin{cases} x+2, & \text{if } x < 1 \\ 0, & \text{if } x = 1 \\ x-2, & \text{if } x > 1 \end{cases}$$

Conclusion

As in the previous example we find that f is continuous at all real numbers $x \neq 1$.

The left hand limit of f at $x = 1$ is

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x+2) = 1+2 = 3$$

The right hand limit of f at $x = 1$ is

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (x-2) = 1-2 = -1$$

Since, the left and right hand limits of f at $x = 1$ do not coincide, f is not continuous at $x = 1$.

Hence $x = 1$ is the only point of discontinuity of f .

The graph of the function is given in the Fig 5.5.

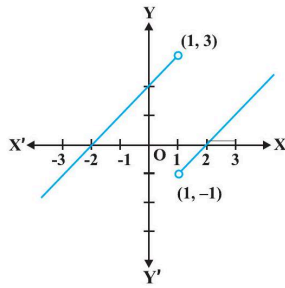


Fig 5.5

Q

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Example:
piecewise
function



Discuss the continuity of the function defined by

$$f(x) = \begin{cases} x+2, & \text{if } x < 0 \\ -x+2, & \text{if } x > 0 \end{cases}$$

Given

Observe that the function is defined at all real numbers except at 0. Domain of definition of this function is

$$D_1 \cup D_2 \text{ where } D_1 = \{x \in \mathbb{R} : x < 0\} \text{ and } D_2 = \{x \in \mathbb{R} : x > 0\}$$

Substitute

Case 1 If $c \in D_1$, then $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (x+2) = c+2 = f(c)$ and hence f is continuous in D_1 .

Case 2 If $c \in D_2$, then $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (-x+2) = -c+2 = f(c)$ and hence f is continuous in D_2 .

Since f is continuous at all points in the domain of f , we deduce that f is continuous.

Graph of this function is given in the Fig 5.6.

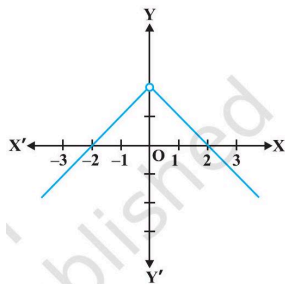


Fig 5.6

Conclusion

Note that to graph this function we need to lift the pen from the plane of the paper, but we need to do that only for those points where the function is not defined.

Q

13

Example:
piecewise
function



Discuss the continuity of the function f given by

$$f(x) = \begin{cases} x, & \text{if } x \geq 0 \\ x^2, & \text{if } x < 0 \end{cases}$$

Given

Clearly the function is defined at every real number.

The graph of the function is given in Fig 5.7.

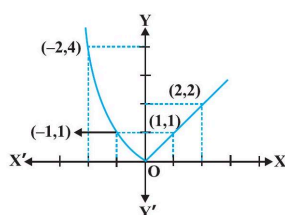


Fig 5.7

Substitute

Looking at the graph, it makes sense to split the domain of f into three disjoint parts of the real line.

Let $D_1 = \{x \in \mathbf{R}: x < 0\}$, $D_2 = \{0\}$ and $D_3 = \{x \in \mathbf{R}: x > 0\}$

Case 1 At any point in D_1 , we have $f(x) = x^2$, and it is easy to see that the function is continuous there (see Example 2).

Case 2 At any point in D_3 , we have $f(x) = x$, and it is easy to see that the function is continuous there (see Example 6).

Conclusion

Case 3 Now we check the function at $x = 0$.

The value of the function at 0 is $f(0) = 0$.

The left hand limit of f at 0 is

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} x^2 = 0^2 = 0$$

The right hand limit of f at 0 is

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x = 0$$

Thus $\lim_{x \rightarrow 0} f(x) = 0 = f(0)$, so f is continuous at 0. This means f is continuous at every point in its domain, so f is a continuous function.

Q

14

Example:
Show that
every
polynomial...



Show that every polynomial function is continuous.

Conclusion

Recall that a function p is a polynomial function if it is defined by $p(x) = a_0 + a_1 x + \dots + a_n x^n$ for some natural number n , $a_n \neq 0$ and $a_i \in \mathbf{R}$.

Clearly this function is defined for every real number.

For a fixed real number c , we have

$$\lim_{x \rightarrow c} p(x) = p(c)$$

By definition, p is continuous at c .

Since c is any real number, p is continuous at every real number and hence p is a continuous function.

Q

15

Example:
 $f(x) = [x]$

(())

Find all the points of discontinuity of the greatest integer function defined by $f(x) = [x]$, where $[x]$ denotes the greatest integer less than or equal to x .

Given

First, f is defined for all real numbers.

The graph of the function is given in Fig 5.8.

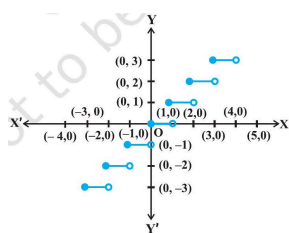


Fig 5.8

Substitute

From the graph, it looks like f is discontinuous at every integral point.

Below, we check if this is true.

Case 1 Let c be a real number that is not an integer.

The graph shows that for all real numbers close to c , the value of the function equals $[c]$; i.

e.

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} [x] = [c].$$

Also $f(c) = [c]$, so the function is continuous at every real number that is not an integer.

Conclusion

Case 2 Let c be an integer.

Then we can find a small enough real number $r > 0$ such that $[c-r] = c-1$, whereas $[c+r] = c$.

In terms of limits, this means

$$\lim_{x \rightarrow c^-} f(x) = c-1, \lim_{x \rightarrow c^+} f(x) = c$$

Since these two limits cannot be equal for any c , the function is discontinuous at every integral point.

Q

16

Example:
Prove that
every
rational...



Prove that every rational function is continuous.

Conclusion

Recall that every rational function f is given by

$$f(x) = \frac{p(x)}{q(x)}, q(x) \neq 0$$

where p and q are polynomial functions.

The domain of f is all real numbers except points at which q is zero.

Since polynomial functions are continuous (Example 14), f is continuous by (4) of Theorem 1.

Q

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Example:
Discuss the
continuity
of...



Discuss the continuity of sine function.

Conclusion

To see this we use the following facts

$$\lim_{x \rightarrow 0} \sin x = 0$$

We have not proved it, but is intuitively clear from the graph of $\sin x$ near 0.

Now, observe that $f(x) = \sin x$ is defined for every real number.

Let c be a real number.

Put $x = c+h$.

If $x \rightarrow c$ we know that $h \rightarrow 0$.

Therefore

$$\begin{aligned} \lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} \sin x \\ &= \lim_{h \rightarrow 0} \sin (c+h) \\ &= \lim_{h \rightarrow 0} [\sin c \cos h + \cos c \sin h] \\ &= \lim_{h \rightarrow 0} [\sin c \cos h] + \lim_{h \rightarrow 0} [\cos c \sin h] \\ &= \sin c + 0 = \sin c = f(c) \end{aligned}$$

Thus $\lim_{x \rightarrow c} f(x) = f(c)$ and hence f is a continuous function.

Remark

Remark A similar proof may be given for the continuity of cosine function.

Q

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Example:
 $f(x) = \tan x$



Prove that the function defined by $f(x) = \tan x$ is a continuous function.

Conclusion

The function is $f(x) = \tan x = \frac{\sin x}{\cos x}$.

This is defined for all real numbers such that $\cos x \neq 0$, i.

e.

$$x \neq (2n+1) \frac{\pi}{2}.$$

We have already shown that both sine and cosine are continuous.

So $\tan x$, being a quotient of two continuous functions, is continuous wherever it is defined.

Continuous functions behave in an interesting way under composition.

Recall that if f and g are two real functions, then

$$(f \circ g)(x) = f(g(x))$$

is defined whenever the range of g is a subset of the domain of f .

The next theorem (stated without proof) gives the continuity of composite functions.

Theorem 2 Suppose f and g are real valued functions such that $(f \circ g)$ is defined at c .

If g is continuous at c and f is continuous at $g(c)$, then $(f \circ g)$ is continuous at c .

The examples below illustrate this theorem.

Q

19

Example:
 $f(x) = \sin(x^2)$

()

Show that the function defined by $f(x) = \sin(x^2)$ is a continuous function.

Conclusion

Observe that the function is defined for every real number.

The function f may be thought of as a composition $g \circ h$ of the two functions g and h , where $g(x) = \sin x$ and $h(x) = x^2$.

Since both g and h are continuous functions, by Theorem 2, it can be deduced that f is a continuous function.

Q

20

Example:
 $f(x) = |1 - x + |x||$



Show that the function f defined by

$$f(x) = |1 - x + |x||,$$

where x is any real number, is a continuous function.

Conclusion

Define g by $g(x) = 1 - x + |x|$ and h by $h(x) = |x|$ for all real x .

Then

$$\begin{aligned}(h \circ g)(x) &= h(g(x)) \\ &= h(1 - x + |x|) \\ &= |1 - x + |x|| = f(x)\end{aligned}$$

In Example 7, we have seen that h is a continuous function.

Hence g being a sum of a polynomial function and the modulus function is continuous.

But then f being a composite of two continuous functions is continuous.

Q

21

Example:
 $f(x) = \sin(x^2)$



Find the derivative of the function given by $f(x) = \sin(x^2)$.

Conclusion

The given function is a composite of two functions.

Indeed, if $t = u(x) = x^2$ and $v(t) = \sin t$, then

$$f(x) = (v \circ u)(x) = v(u(x)) = v(x^2) = \sin x^2$$

$$\text{Put } t = u(x) = x^2.$$

$$\text{Note that } \frac{dv}{dt} = \cos t \text{ and } \frac{dt}{dx} = 2x \text{ both exist.}$$

So, by the chain rule,

$$\frac{df}{dx} = \frac{dv}{dt} \cdot \frac{dt}{dx} = \cos t \cdot 2x$$

We usually write the final result only in terms of x .

Thus

$$\frac{df}{dx} = \cos t \cdot 2x = 2x \cos x^2$$

Q

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Example: $\frac{dy}{dx}$



Find $\frac{dy}{dx}$ if $x - y = \pi$.

Conclusion

One way is to solve for y and rewrite the above as

$$y = x - \pi$$

But then

$$\frac{dy}{dx} = 1$$

Alternatively, directly differentiating the relationship w.

r.

t.

, x , we have

$$\frac{d}{dx}(x - y) = \frac{d\pi}{dx}$$

Recall that $\frac{d\pi}{dx}$ means to differentiate the constant function taking value π everywhere w.

r.

t.

, x.

Thus

$$\frac{d}{dx}(x) - \frac{d}{dx}(y) = 0$$

which implies that

$$\frac{dy}{dx} = \frac{dx}{dx} = 1$$

Q

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Example: $\frac{dy}{dx}$

(())

Find $\frac{dy}{dx}$, if $y + \sin y = \cos x$.

Conclusion

We differentiate the relationship directly with respect to x, i.

e.

,

$$\frac{dy}{dx} + \frac{d}{dx}(\sin y) = \frac{d}{dx}(\cos x)$$

which implies using chain rule

$$\frac{dy}{dx} + \cos y \cdot \frac{dy}{dx} = -\sin x$$

This gives

$$\frac{dy}{dx} = -\frac{\sin x}{1 + \cos y}$$

where

$$y \neq (2n+1)\pi$$

Q

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Example:
 $f(x) = \sin^{-1} x$

✧

Find the derivative of f given by $f(x) = \sin^{-1} x$ assuming it exists.

Conclusion

Let $y = \sin^{-1} x$.

Then, $x = \sin y$.

Differentiating both sides w.

r.

t.

x, we get

$$1 = \cos y \frac{dy}{dx}$$

which implies that

$$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\cos(\sin^{-1} x)}$$

Observe that this is defined only for $\cos y \neq 0$, i.

e.

$$\sin^{-1} x \neq -\frac{\pi}{2}, \frac{\pi}{2}, \text{ i.}$$

e.

$$x \neq -1, 1, \text{ i.}$$

e.

$$x \in (-1, 1).$$

To make this result a bit more attractive, we carry out the following manipulation.

Recall that for $x \in (-1, 1)$, $\sin(\sin^{-1} x) = x$ and hence

$$\cos^2 y = 1 - (\sin y)^2 = 1 - (\sin(\sin^{-1} x))^2 = 1 - x^2$$

Also, since $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$, $\cos y$ is positive and hence $\cos y = \sqrt{1-x^2}$. Thus, for

$x \in (-1, 1)$,

$$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-x^2}}$$

f(x)	$\sin^{-1} x$	$\cos^{-1} x$	$\tan^{-1} x$
$f^{-1}(x)$	$\frac{1}{\sqrt{(1-x^2)}}$	$\frac{-1}{\sqrt{(1-x^2)}}$	$\frac{1}{1+x^2}$
Domain off	$(-1, 1)$	$(-1, 1)$	$\mathbb{R}$

Q

25

Example:
 $x = e^{\log x}$



Is it true that $x = e^{\log x}$ for all real x ?

Conclusion

First, observe that the domain of log function is set of all positive real numbers.

So the above equation is not true for non-positive real numbers.

Now, let $y = e^{\log x}$.

If $y > 0$, we may take logarithm which gives us $\log y = \log(e^{\log x}) = \log x \cdot \log e = \log x$.

Thus $y = x$.

Hence $x = e^{\log x}$ is true only for positive values of x .

One of the striking properties of the natural exponential function in differential calculus is that it doesn't change during the process of differentiation.

This is captured in the following theorem whose proof we skip.

Theorem 5*

(1) The derivative of e^x w.

r.

t.

, x is e^x ; i.

e.

, $\frac{d}{dx}(e^x) = e^x$.

(2) The derivative of $\log x$ w.

r.

t.

, x is $\frac{1}{x}$; i.

e.

, $\frac{d}{dx}(\log x) = \frac{1}{x}$.

Q

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Example: e^{-x}



Differentiate the following w.r.t. x : (i) e^{-x} (ii) $\sin(\log x)$, $x > 0$ (iii) $\cos^{-1}(e^x)$ (iv) $e^{\cos x}$

Given

(i) Let $y = e^{-x}$.

Using chain rule, we have

$$\frac{dy}{dx} = e^{-x} \cdot \frac{d}{dx}(-x) = -e^{-x}$$

Substitute

(ii) Let $y = \sin(\log x)$.

Using chain rule, we have

$$\frac{dy}{dx} = \cos(\log x) \cdot \frac{d}{dx}(\log x) = \frac{\cos(\log x)}{x}$$

[^0]

(iii) Let $y = \cos^{-1}(e^x)$.

Using chain rule, we have

$$\frac{dy}{dx} = \frac{-1}{\sqrt{(1-e^{2x})}} \cdot \frac{d}{dx}(e^x) = \frac{-e^x}{\sqrt{(1-e^{2x})}}$$

Conclusion

(iv) Let $y = e^{\cos x}$.

Using chain rule, we have

$$\frac{dy}{dx} = e^{\cos x} \cdot (-\sin x) = -(\sin x) e^{\cos x}$$

Q

27

Example:
 $\sqrt{(x-3)}$
 $(x^2+4)^3$
 x^2+4x+5

()

Differentiate $\sqrt{\frac{(x-3)(x^2+4)}{3x^2+4x+5}}$ w.r.t. x .

Conclusion

$$\text{Let } y = \sqrt{\frac{(x-3)(x^2+4)}{3x^2+4x+5}}$$

Taking logarithm on both sides, we have

$$\log y = \frac{1}{2} [\log(x-3) + \log(x^2+4) - \log(3x^2+4x+5)]$$

Now, differentiating both sides w.

r.

t.

x, we get

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{1}{2} \left[\frac{1}{(x-3)} + \frac{2x}{x^2+4} - \frac{6x+4}{3x^2+4x+5} \right]$$

or

$$\begin{aligned} \frac{dy}{dx} &= \frac{y}{2} \left[\frac{1}{(x-3)} + \frac{2x}{x^2+4} - \frac{6x+4}{3x^2+4x+5} \right] \\ &= \frac{1}{2} \sqrt{\frac{(x-3)(x^2+4)}{3x^2+4x+5}} \left[\frac{1}{(x-3)} + \frac{2x}{x^2+4} - \frac{6x+4}{3x^2+4x+5} \right] \end{aligned}$$

Q

28

Example: a^x

✦

Differentiate a^x w.r.t. x , where a is a positive constant.

Conclusion

$$\text{Let } y = a^x.$$

Then

$$\log y = x \log a$$

Differentiating both sides w.

r.

t.

x, we have

$$\frac{1}{y} \cdot \frac{dy}{dx} = \log a$$

or

$$\frac{dy}{dx} = y \log a$$

Thus

$$\frac{d}{dx} (a^x) = a^x \log a$$

Alternatively

$$\begin{aligned} \frac{d}{dx} (a^x) &= \frac{d}{dx} (e^{x \log a}) = e^{x \log a} \cdot \frac{d}{dx} (x \log a) \\ &= e^{x \log a} \cdot \log a = a^x \log a \end{aligned}$$

Q

29

Example:
 $x^{\sin x}, x > 0$

✦

Differentiate $x^{\sin x}, x > 0$ w.r.t. x .

Conclusion

$$\text{Let } y = x^{\sin x}.$$

Taking logarithm on both sides, we have

$$\log y = \sin x \log x$$

Therefore

$$\frac{1}{y} \cdot \frac{dy}{dx} = \sin x \cdot \frac{d}{dx} (\log x) + \log x \cdot \frac{d}{dx} (\sin x)$$

or

$$\frac{1}{y} \cdot \frac{dy}{dx} = (\sin x) \cdot \frac{1}{x} + \log x \cos x$$

or

$$\begin{aligned}\frac{dy}{dx} &= y \left[\frac{\sin x}{x} + \cos x \log x \right] \\ &= x^{\sin x} \left[\frac{\sin x}{x} + \cos x \log x \right] \\ &= x^{\sin x - 1} \cdot \sin x + x^{\sin x} \cdot \cos x \log x\end{aligned}$$

Q

30

Example: $\frac{d}{dx} y^x$



Find $\frac{dy}{dx}$, if $y^x + x^y + x^x = a^b$.

Conclusion

Given that $y^x + x^y + x^x = a^b$.

Putting $u = y^x$, $v = x^y$ and $w = x^x$, we get $u + v + w = a^b$

Therefore

$$\frac{du}{dx} + \frac{dv}{dx} + \frac{dw}{dx} = 0$$

Now, $u = y^x$.

Taking logarithm on both sides, we have

$$\log u = x \log y$$

Differentiating both sides w.

r.

t.

x, we have

$$\begin{aligned}\frac{1}{u} \cdot \frac{du}{dx} &= x \frac{d}{dx} (\log y) + \log y \cdot \frac{d}{dx} (x) \\ &= x \cdot \frac{1}{y} \cdot \frac{dy}{dx} + \log y \cdot 1\end{aligned}$$

So

$$\frac{du}{dx} = u \left(\frac{x}{y} \cdot \frac{dy}{dx} + \log y \right) = y^x \left[\frac{x}{y} \cdot \frac{dy}{dx} + \log y \right]$$

Also $v = x^y$

Taking logarithm on both sides, we have

$$\log v = y \log x$$

Differentiating both sides w.

r.

t.

x, we have

$$\begin{aligned}\frac{1}{v} \cdot \frac{dv}{dx} &= y \frac{d}{dx} (\log x) + \log x \cdot \frac{dy}{dx} \\ &= y \cdot \frac{1}{x} + \log x \cdot \frac{dy}{dx}\end{aligned}$$

So

$$\begin{aligned}\frac{dv}{dx} &= v \left[\frac{y}{x} + \log x \cdot \frac{dy}{dx} \right] \\ &= x^y \left[\frac{y}{x} + \log x \cdot \frac{dy}{dx} \right]\end{aligned}$$

Again

$w = x^x$

Taking logarithm on both sides, we have

$$\log w = x \log x.$$

Differentiating both sides w.

r.

t x, we have

$$\begin{aligned}\frac{1}{w} \cdot \frac{dw}{dx} &= x \frac{d}{dx} (\log x) + \log x \cdot \frac{d}{dx} (x) \\ &= x \cdot \frac{1}{x} + \log x \cdot 1\end{aligned}$$

i.

e.

$$\begin{aligned}\frac{dw}{dx} &= w(1+\log x) \\ &= x^x(1+\log x)\end{aligned}$$

From (1), (2), (3), (4), we have

$$y^x \left(\frac{x}{y} - \frac{dy}{dx} + \log y \right) + x^y \left(\frac{y}{x} + \log x - \frac{dy}{dx} \right) + x^x(1+\log x) = 0$$

or

$$(x \cdot y^{x-1} + x^y \cdot \log x) \frac{dy}{dx} = -x^x(1+\log x) - y \cdot x^{y-1} - y^x \log y$$

Therefore

$$\frac{dy}{dx} = \frac{-[y^x \log y + y \cdot x^{y-1} + x^x(1+\log x)]}{x \cdot y^{x-1} + x^y \log x}$$

Q

31

Example: $\frac{dy}{dx}$

(())

Find $\frac{dy}{dx}$, if $x = a \cos \theta$, $y = a \sin \theta$.

Conclusion

Given that

$$x = a \cos \theta, y = a \sin \theta$$

Therefore

$$\frac{dx}{d\theta} = -a \sin \theta, \frac{dy}{d\theta} = a \cos \theta$$

Hence

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{a \cos \theta}{-a \sin \theta} = -\cot \theta$$

Q

32

Example: $\frac{dy}{dx}$

✧

Find $\frac{dy}{dx}$, if $x = a t^2$, $y = 2 a t$.

Conclusion

Given that $x = a t^2$, $y = 2 a t$

So

$$\frac{dx}{dt} = 2 a t \text{ and } \frac{dy}{dt} = 2 a$$

Therefore

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2 a}{2 a t} = \frac{1}{t}$$

Q

33

Example: $\frac{dy}{dx}$

✧

Find $\frac{dy}{dx}$, if $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$.

Conclusion

We have $\frac{dx}{d\theta} = a(1 + \cos \theta)$, $\frac{dy}{d\theta} = a(\sin \theta)$

Therefore

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{a \sin \theta}{a(1 + \cos \theta)} = \tan \frac{\theta}{2}$$

Note

- Note It may be noted here that $\frac{dy}{dx}$ is expressed in terms of parameter only without directly involving the main variables x and y .

Q

34

Example: $\frac{d}{dx} y/dx$ 

Find $\frac{dy}{dx}$, if $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$.

Conclusion

Let $x = a \cos^3 \theta$, $y = a \sin^3 \theta$.

Then

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = (a \cos^3 \theta)^{\frac{2}{3}} + (a \sin^3 \theta)^{\frac{2}{3}} \\ = a^{\frac{2}{3}} (\cos^2 \theta + \sin^2 \theta) = a^{\frac{2}{3}}$$

Hence, $x = a \cos^3 \theta$, $y = a \sin^3 \theta$ is parametric equation of $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$

Now

$$\frac{dx}{d\theta} = -3a \cos^2 \theta \sin \theta \text{ and } \frac{dy}{d\theta} = 3a \sin^2 \theta \cos \theta$$

Therefore

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{3a \sin^2 \theta \cos \theta}{-3a \cos^2 \theta \sin \theta} = -\tan \theta = -\left(\frac{y}{x}\right)$$

Q

35

Example: $\frac{d^2}{dx^2} yd^2 x^2$

()

Find $\frac{d^2 y}{dx^2}$, if $y = x^3 + \tan x$.

Conclusion

Given that $y = x^3 + \tan x$.

Then

$$\frac{dy}{dx} = 3x^2 + \sec^2 x$$

Therefore

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} (3x^2 + \sec^2 x) \\ = 6x + 2 \sec x \cdot \sec x \tan x = 6x + 2 \sec^2 x \tan x$$

Q

36

Example: $y = A \sin x + B \cos x$ 

If $y = A \sin x + B \cos x$, then prove that $\frac{d^2 y}{dx^2} + y = 0$.

Conclusion

We have

$$\frac{dy}{dx} = A \cos x - B \sin x$$

and

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} (A \cos x - B \sin x) \\ = -A \sin x - B \cos x = -y$$

Hence

$$\frac{d^2 y}{dx^2} + y = 0$$

Q

37

Example: $y = 3e^{2x} + 2e^{3x}$ 

If $y = 3e^{2x} + 2e^{3x}$, prove that $\frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 6y = 0$.

Conclusion

Given that $y = 3e^{2x} + 2e^{3x}$.

Then

$$\frac{dy}{dx} = 6e^{2x} + 6e^{3x} = 6(e^{2x} + e^{3x})$$

Therefore

$$\frac{d^2 y}{dx^2} = 12e^{2x} + 18e^{3x} = 6(2e^{2x} + 3e^{3x})$$

Hence

$$\frac{d^2 y}{d x^2} - 5 \frac{d y}{d x} + 6 y = 6(2 e^{2 x} + 3 e^{3 x})$$

$$-30(e^{2 x} + e^{3 x}) + 6(3 e^{2 x} + 2 e^{3 x}) = 0$$

Q

38

Example:
y = sin⁻¹ x



If $y = \sin^{-1} x$, show that $(1-x^2) \frac{d^2 y}{d x^2} - x \frac{d y}{d x} = 0$.

Conclusion

We have $y = \sin^{-1} x$.

Then

$$\frac{d y}{d x} = \frac{1}{\sqrt{(1-x^2)}}$$

or

$$\sqrt{(1-x^2)} \frac{d y}{d x} = 1$$

So

$$\frac{d}{d x} (\sqrt{(1-x^2)}) \cdot \frac{d y}{d x} = 0$$

or

$$\sqrt{(1-x^2)} \cdot \frac{d^2 y}{d x^2} + \frac{d y}{d x} \cdot \frac{d}{d x} (\sqrt{(1-x^2)}) = 0$$

or

$$\sqrt{(1-x^2)} \cdot \frac{d^2 y}{d x^2} - \frac{d y}{d x} \cdot \frac{2 x}{2 \sqrt{(1-x^2)}} = 0$$

Hence

$$(1-x^2) \frac{d^2 y}{d x^2} - x \frac{d y}{d x} = 0$$

Alternatively, Given that $y = \sin^{-1} x$, we have

$$y_1 = \frac{1}{\sqrt{(1-x^2)}}, \text{ i.e., } (1-x^2) y_1^2 = 1$$

So

$$(1-x^2) \cdot 2 y_1 y_2 + y_1^2 (0-2 x) = 0$$

Hence

$$(1-x^2) y_2 - x y_1 = 0$$

Q

39

Example:
sqrt 3
x+2+1sqrt 2
x^2+4

((

Differentiate w.r.t. x, the following function: (i) $\sqrt{(3 x+2)} + \frac{1}{\sqrt{(2 x^2+4)}}$ (ii) $\log_7(\log x)$

Given

$$(i) \text{ Let } y = \sqrt{(3 x+2)} + \frac{1}{\sqrt{(2 x^2+4)}} = (3 x+2)^{\frac{1}{2}} + (2 x^2+4)^{-\frac{1}{2}}$$

Note that this function is defined at all real numbers $x > -\frac{2}{3}$.

Therefore

$$\begin{aligned} \frac{d y}{d x} &= \frac{1}{2} (3 x+2)^{-\frac{1}{2}} \cdot \frac{d}{d x} (3 x+2) + (-\frac{1}{2}) (2 x^2+4)^{-\frac{3}{2}} \cdot \frac{d}{d x} (2 x^2+4) \\ &= \frac{1}{2} (3 x+2)^{-\frac{1}{2}} \cdot (3) - \frac{1}{2} (2 x^2+4)^{-\frac{3}{2}} \cdot 4 x \\ &= \frac{3}{2 \sqrt{(3 x+2)}} - \frac{2 x}{(2 x^2+4)^{\frac{3}{2}}} \end{aligned}$$

This is defined for all real numbers $x > -\frac{2}{3}$.

Conclusion

(ii) Let $y = \log_7(\log x) = \frac{\log(\log x)}{\log 7}$ (by change of base formula).

The function is defined for all real numbers $x > 1$.

Therefore

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{\log 7} \cdot \frac{d}{dx} (\log (\log x)) \\ &= \frac{1}{\log 7} \cdot \frac{1}{\log x} \cdot \frac{d}{dx} (\log x) \\ &= \frac{1}{x \log 7 \log x}\end{aligned}$$

Q

40

Example: $\cos^{-1}(\sin x)$



Differentiate the following w.r.t. x . (i) $\cos^{-1}(\sin x)$ (ii) $\tan^{-1}\left(\frac{\sin x}{1+\cos x}\right)$ (iii) $\sin^{-1}\left(\frac{2^{x+1}}{1+4^x}\right)$

Given

(i) Let $f(x) = \cos^{-1}(\sin x)$.

This function is defined for all real numbers.

We can rewrite it as

$$\begin{aligned}f(x) &= \cos^{-1}(\sin x) \\ &= \cos^{-1} \cos \frac{\pi}{2} - x \\ &= \frac{\pi}{2} - x\end{aligned}$$

Thus $f'(x) = -1$.

Substitute

(ii) Let $f(x) = \tan^{-1}\left(\frac{\sin x}{1+\cos x}\right)$.

This function is defined for all real numbers where $\cos x \neq -1$, i.e.,

, at all odd multiples of π .

We can rewrite it as

$$\begin{aligned}f(x) &= \tan^{-1}\left(\frac{\sin x}{1+\cos x}\right) \\ &= \tan^{-1}\left[\frac{2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)}{2 \cos^2 \frac{x}{2}}\right] \\ &= \tan^{-1}\left[\tan\left(\frac{x}{2}\right)\right] = \frac{x}{2}\end{aligned}$$

We can cancel $\cos\left(\frac{x}{2}\right)$ from both the numerator and denominator, since it is not zero.

Thus $f'(x) = \frac{1}{2}$.

Conclusion

(iii) Let $f(x) = \sin^{-1}\left(\frac{2^{x+1}}{1+4^x}\right)$.

To find the domain, we need all x such that $-1 \leq \frac{2^{x+1}}{1+4^x} \leq 1$.

Since the middle quantity is always positive, this means finding all x such that

$$\frac{2^{x+1}}{1+4^x} \leq 1, \text{ i.e.,}$$

e.

, all x such that $2^{x+1} \leq 1+4^x$.

This can be rewritten as $2 \leq \frac{1}{2^x} + 2^x$, which holds for all x .

So the function is defined at every real number.

Putting $2^x = \tan \theta$, we can rewrite this function as

$$\begin{aligned}
 f(x) &= \sin^{-1} \left[\frac{2^{x+1}}{1+4^x} \right] \\
 &= \sin^{-1} \frac{2^x \cdot 2}{1+(2^x)^2} \\
 &= \sin^{-1} \left[\frac{2 \tan \theta}{1+\tan^2 \theta} \right] \\
 &= \sin^{-1} [\sin 2\theta] \\
 &= 2\theta = 2 \tan^{-1}(2^x)
 \end{aligned}$$

Thus

$$\begin{aligned}
 f'(x) &= 2 \cdot \frac{1}{1+(2^x)^2} \cdot \frac{d}{dx} (2^x) \\
 &= \frac{2}{1+4^x} \cdot (2^x) \log 2 \\
 &= \frac{2^{x+1} \log 2}{1+4^x}
 \end{aligned}$$

Q

41

Example:
 $f'(x)$



Find $f'(x)$ if $f(x) = (\sin x)^{\sin x}$ for all $0 < x < \pi$.

Conclusion

The function $y = (\sin x)^{\sin x}$ is defined for all positive real numbers.

Taking logarithms, we have

$$\log y = \log (\sin x)^{\sin x} = \sin x \log (\sin x)$$

Then

$$\begin{aligned}
 \frac{1}{y} \cdot \frac{dy}{dx} &= \frac{d}{dx} (\sin x \log (\sin x)) \\
 &= \cos x \log (\sin x) + \sin x \cdot \frac{1}{\sin x} \cdot \frac{d}{dx} (\sin x) \\
 &= \cos x \log (\sin x) + \cos x \\
 &= (1 + \log (\sin x)) \cos x
 \end{aligned}$$

Thus

$$\frac{dy}{dx} = y(1 + \log (\sin x)) \cos x = (1 + \log (\sin x)) (\sin x)^{\sin x} \cos x$$

Q

42

Example: $\frac{dy}{dx}$



For a positive constant a find $\frac{dy}{dx}$, where

$$y = a^{t^+} \frac{1}{t}, \text{ and } x = \left(t + \frac{1}{t}\right)^a$$

Conclusion

Both y and x are defined for all real $t \neq 0$.

Clearly

$$\begin{aligned}
 \frac{dy}{dt} &= \frac{d}{dt} \left(a^{t^+} \frac{1}{t} \right) = a^{t^+} \frac{1}{t} \cdot \frac{d}{dt} \left(t + \frac{1}{t} \right) \cdot \log a \\
 &= a^{t^+} \frac{1}{t} \left(1 - \frac{1}{t^2} \right) \log a
 \end{aligned}$$

Similarly

$$\begin{aligned}
 \frac{dx}{dt} &= a \left[t + \frac{1}{t} \right]^{a-1} \cdot \frac{d}{dt} \left(t + \frac{1}{t} \right) \\
 &= a \left[t + \frac{1}{t} \right]^{a-1} \cdot \left(1 - \frac{1}{t^2} \right)
 \end{aligned}$$

$$\frac{dx}{dt} \neq 0 \text{ only if } t \neq \pm 1.$$

Thus for $t \neq \pm 1$,

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{a^{t^+} \frac{1}{t} \left(1 - \frac{1}{t^2} \right) \log a}{a \left[t + \frac{1}{t} \right]^{a-1} \left(1 - \frac{1}{t^2} \right)} \cdot 1 - \frac{1}{t^2} \\
 &= \frac{a^{t^+} \frac{1}{t} \log a}{a \left(t + \frac{1}{t} \right)^{a-1}}
 \end{aligned}$$

Q**43****Example:** $\sin^2 x$ **(())****Differentiate $\sin^2 x$ w.r.t. $e^{\cos x}$.****Conclusion**Let $u(x) = \sin^2 x$ and $v(x) = e^{\cos x}$.

$$\text{We want to find } \frac{\frac{d u}{d v}}{\frac{d v}{d x}} = \frac{\frac{d u}{d x}}{\frac{d v}{d x}} .$$

Clearly

$$\frac{d u}{d x} = 2 \sin x \cos x \text{ and } \frac{d v}{d x} = e^{\cos x}(-\sin x) = -(\sin x) e^{\cos x}$$

Thus

$$\frac{d u}{d v} = \frac{2 \sin x \cos x}{-\sin x e^{\cos x}} = -\frac{2 \cos x}{e^{\cos x}}$$

Q

1

Continuity:
 $f(x)=5x-3$



Prove that the function $f(x) = 5x-3$ is continuous at $x = 0$, at $x = -3$ and at $x = 5$.

Conclusion

The given function is $f(x) = 5x-3$

At $x = 0$, $f(0) = 5(0)-3 = -3$

$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} (5x-3) = 5(0)-3 = -3$

$\therefore \lim_{x \rightarrow 0} f(x) = f(0)$

Therefore, f is continuous at $x = 0$.

At $x = -3$, $f(-3) = 5(-3)-3 = -18$

$\lim_{x \rightarrow -3} f(x) = \lim_{x \rightarrow -3} (5x-3) = 5(-3)-3 = -18$

$\therefore \lim_{x \rightarrow -3} f(x) = f(-3)$

Therefore, f is continuous at $x = -3$.

At $x = 5$, $f(5) = 5(5)-3 = 22$

$\lim_{x \rightarrow 5} f(x) = \lim_{x \rightarrow 5} (5x-3) = 5(5)-3 = 22$

$\therefore \lim_{x \rightarrow 5} f(x) = f(5)$

Therefore, f is continuous at $x = 5$.

Q

2

Continuity:
 $f(x)=2x^2-1$



Examine the continuity of the function $f(x) = 2x^2-1$ at $x = 3$.

Conclusion

The given function is $f(x) = 2x^2-1$

At $x = 3$, $f(3) = 2(3)^2-1 = 17$

$\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} (2x^2-1) = 2(3^2)-1 = 17$

$\therefore \lim_{x \rightarrow 3} f(x) = f(3)$

Therefore, f is continuous at $x = 3$.

Q

3

Continuity:
 $f(x)=x-5$



Examine the following functions for continuity.

Part (a)

$f(x) = x-5$

Conclusion

The given function is $f(x) = x-5$

Clearly, f is defined at every real number k and its value at k is $k-5$.

It is also observed that

$\lim_{x \rightarrow k} f(x) = \lim_{x \rightarrow k} (x-5) = k-5 = f(k)$

$\therefore \lim_{x \rightarrow k} f(x) = f(k)$

Hence, f is continuous at every real number and therefore, it is a continuous function.

Part (b)

$f(x) = \frac{1}{x-5}, x \neq 5$

Conclusion

The given function is $f(x) = \frac{1}{x-5}, x \neq 5$

For any real number $k \neq 5$, we get

$\lim_{x \rightarrow k} f(x) = \lim_{x \rightarrow k} \frac{1}{x-5} = \frac{1}{k-5}$

Also,

$f(k) = \frac{1}{k-5}$ (As $k \neq 5$)

$\therefore \lim_{x \rightarrow k} f(x) = f(k)$

Hence, f is continuous at every point in the domain of f and therefore, it is a continuous function.

Part (c)

$f(x) = \frac{x^2-25}{x+5}, x \neq -5$

Conclusion

The given function is $f(x) = \frac{x^2-25}{x+5}, x \neq -5$

For any real number $c \neq -5$, we get

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} \frac{x^2 - 25}{x + 5} = \lim_{x \rightarrow c} \frac{(x+5)(x-5)}{x+5} = \lim_{x \rightarrow c} (x-5) = (c-5)$$

Also,

$$f(c) = \frac{(c+5)(c-5)}{c+5} = (c-5)$$

$$\therefore \lim_{x \rightarrow c} f(x) = f(c)$$

Hence, f is continuous at every point in the domain of f and therefore, it is a continuous function.

Part (d)

$$f(x) = |x-5|$$

Given

The given function is $f(x) = |x-5| = \begin{cases} 5-x, & \text{if } x < 5 \\ x-5, & \text{if } x \geq 5 \end{cases}$

This function f is defined at all points of the real line.

Let c be a point on a real line.

Then, $c < 5$, $c = 5$ or $c > 5$

Substitute

Case I: $c < 5$

Then, $f(c) = 5 - c$

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (5 - x) = 5 - c$$

$$\therefore \lim_{x \rightarrow c} f(x) = f(c)$$

Therefore, f is continuous at all real numbers less than 5.

Case II: $c = 5$

Then, $f(c) = f(5) = (5 - 5) = 0$

$$\lim_{x \rightarrow 5^-} f(x) = \lim_{x \rightarrow 5^-} (5 - x) = (5 - 5) = 0$$

$$\lim_{x \rightarrow 5^+} f(x) = \lim_{x \rightarrow 5^+} (x - 5) = 0$$

$$\therefore \lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = f(c)$$

Therefore, f is continuous at $x = 5$

Conclusion

Case III: $c > 5$

Then, $f(c) = f(5) = c - 5$

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (x - 5) = c - 5$$

$$\therefore \lim_{x \rightarrow c} f(x) = f(c)$$

Therefore, f is continuous at all real numbers greater than 5.

Hence, f is continuous at every real number and therefore, it is a continuous function.

Q

4

Continuity:
 $f(x) = x^n$

(())

Prove that the function $f(x) = x^n$ is continuous at $x = n$, where n is a positive integer.

Conclusion

The given function is $f(x) = x^n$

Notice that f is defined at all positive integers, n , and its value at n is n^n .

Then,

$$\lim_{x \rightarrow n} f(x) = \lim_{x \rightarrow n} (x^n) = n^n$$

$$\therefore \lim_{x \rightarrow n} f(x) = f(n)$$

Therefore, f is continuous at n , where n is a positive integer.

Q

5

Continuity:
piecewise function

⚙

Is the function f defined by

$$f(x) = \begin{cases} x, & \text{if } x \leq 1 \\ 5, & \text{if } x > 1 \end{cases}$$

continuous at $x = 0$? At $x = 1$? At $x = 2$? Find all points of discontinuity of f , where f is defined by

Conclusion

The given function is $f(x) = \begin{cases} x, & \text{if } x \leq 1 \\ 5, & \text{if } x > 1 \end{cases}$

At $x = 0$,

Clearly, f is defined at 0 and its value at 0 is 0.

Then,

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} (x) = 0$$

$$\therefore \lim_{x \rightarrow 0} f(x) = f(0)$$

Therefore, f is continuous at $x = 0$.

At $x = 1$,

Clearly, f is defined at 1 and its value at 1 is 1.

The left hand limit of f at $x = 1$ is,

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x) = 1$$

The right hand limit of f at $x = 1$ is,

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (5) = 5$$

$$\therefore \lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$$

Therefore, f is not continuous at $x = 1$.

At $x = 2$,

Clearly, f is defined at 2 and its value at 2 is 5.

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} (5) = 5$$

$$\therefore \lim_{x \rightarrow 2} f(x) = f(2)$$

Therefore, f is continuous at $x = 2$.

Q

6

Continuity: piecewise function

✧

$$f(x) = \begin{cases} 2x+3, & \text{if } x \leq 2 \\ 2x-3, & \text{if } x > 2 \end{cases}$$

Given

The given function is $f(x) = \begin{cases} 2x+3, & \text{if } x \leq 2 \\ 2x-3, & \text{if } x > 2 \end{cases}$

Clearly, the given function f is defined at all the points of the real line.

Let c be a point on the real line.

Then, three cases arise.

$$c < 2$$

$$c > 2$$

$$c = 2$$

Substitute

Case I: $c < 2$

$$f(c) = 2c+3$$

Then,

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (2x+3) = 2c+3$$

$$\therefore \lim_{x \rightarrow c} f(x) = f(c)$$

Therefore, f is continuous at all points x , such that $x < 2$.

Case II: $c > 2$

Then,

$$f(c) = 2c-3$$

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (2x-3) = 2c-3$$

$$\therefore \lim_{x \rightarrow c} f(x) = f(c)$$

Therefore, f is continuous at all points x , such that $x > 2$

Conclusion

Case III: $c = 2$

Then, the left hand limit of f at $x = 2$ is,

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (2x+3) = 2(2)+3 = 7$$

The right hand limit of f at $x = 2$ is,

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (2x-3) = 2(2)-3 = 1$$

The left and right hand limits of f at $x = 2$ do not coincide.

Therefore, f is not continuous at $x = 2$.

Hence, $x = 2$ is the only point of discontinuity of f .

Q

7

Continuity:
piecewise
function

$$f(x) = \begin{cases} |x| + 3, & \text{if } x \leq -3 \\ -2x, & \text{if } -3 < x < 3 \\ 6x + 2, & \text{if } x \geq 3 \end{cases}$$

Given

The given function is

$$f(x) = \begin{cases} |x| + 3, & \text{if } x \leq -3 \\ -2x, & \text{if } -3 < x < 3 \\ 6x + 2, & \text{if } x \geq 3 \end{cases}$$

The given function f is defined at all the points of the real line.Let c be a point on the real line.

Substitute

Case I:

If $c < -3$, then $f(c) = -c + 3$

$$\begin{aligned} \lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} (-x + 3) = -c + 3 \\ \therefore \lim_{x \rightarrow c} f(x) &= f(c) \end{aligned}$$

Therefore, f is continuous at all points x , such that $x < -3$.

Case II:

If $c = -3$, then $f(-3) = -(-3) + 3 = 6$

$$\begin{aligned} \lim_{x \rightarrow -3^-} f(x) &= \lim_{x \rightarrow -3^-} (-x + 3) = -(-3) + 3 = 6 \\ \lim_{x \rightarrow -3^+} f(x) &= \lim_{x \rightarrow -3^+} (-2x) = -2(-3) = 6 \\ \therefore \lim_{x \rightarrow -3} f(x) &= f(-3) \end{aligned}$$

Therefore, f is continuous at $x = -3$.

Case III:

If $-3 < c < 3$, then $f(c) = -2c$

$$\begin{aligned} \lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} (-2x) = -2c \\ \therefore \lim_{x \rightarrow c} f(x) &= f(c) \end{aligned}$$

Therefore, f is continuous in $(-3, 3)$.

Case IV:

If $c = 3$, then the left hand limit of f at $x = 3$ is,

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (-2x) = -2(3) = -6$$

The right hand limit of f at $x = 3$ is,

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} (6x + 2) = 6(3) + 2 = 20$$

The left and right hand limits of f at $x = 3$ do not coincide.Therefore, f is not continuous at $x = 3$.

Conclusion

Case V:

If $c > 3$, then $f(c) = 6c + 2$

$$\begin{aligned} \lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} (6x + 2) = 6c + 2 \\ \therefore \lim_{x \rightarrow c} f(x) &= f(c) \end{aligned}$$

Therefore, f is continuous at all points x , such that $x > 3$.Hence, $x = 3$ is the only point of discontinuity of f .

Q

8

Continuity:
piecewise
function

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$$f(x) = \begin{cases} \frac{|x|}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

Given

The given function is

$$f(x) = \begin{cases} \frac{|x|}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

It is known that, $x < 0 \Rightarrow |x| = -x$ and $x > 0 \Rightarrow |x| = x$

Therefore, the given function can be rewritten as

$$\begin{aligned} \frac{|x|}{x} &= \frac{-x}{x} = -1, & \text{if } x < 0 \\ f(x) &= \begin{cases} 0, & \text{if } x = 0 \\ \frac{|x|}{x} &= \frac{x}{x} = 1, & \text{if } x > 0 \end{cases} \end{aligned}$$

The given function f is defined at all the points of the real line.

Let c be a point on the real line.

Substitute

Case I:

If $c < 0$, then $f(c) = -1$

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (-1) = -1$$

$$\therefore \lim_{x \rightarrow c} f(x) = f(c)$$

Therefore, f is continuous at all points $x < 0$.

Case II:

If $c = 0$, then the left hand limit of f at $x = 0$ is,

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (-1) = -1$$

The right hand limit of f at $x = 0$ is,

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (1) = 1$$

The left and right hand limits of f at $x = 0$ do not coincide.

Therefore, f is not continuous at $x = 0$.

Conclusion

Case III:

If $c > 0$, then $f(c) = 1$

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (1) = 1$$

$$\therefore \lim_{x \rightarrow c} f(x) = f(c)$$

Therefore, f is continuous at all points x , such that $x > 0$.

Hence, $x = 0$ is the only point of discontinuity of f .

Q

9

Continuity:
piecewise
function



$$f(x) = \begin{cases} \frac{x}{|x|}, & \text{if } x < 0 \\ -1, & \text{if } x \geq 0 \end{cases}$$

Conclusion

The given function is

$$f(x) = \begin{cases} \frac{x}{|x|}, & \text{if } x < 0 \\ -1, & \text{if } x \geq 0 \end{cases}$$

It is known that $x < 0 \Rightarrow |x| = -x$

Therefore, the given function can be rewritten as

$$f(x) = \begin{cases} \frac{x}{|x|} = \frac{x}{-x} = -1, & \text{if } x < 0 \\ -1, & \text{if } x \geq 0 \end{cases}$$

$$\Rightarrow f(x) = -1 \text{ for all } x \in \mathbb{R}$$

Let c be any real number.

Then, $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (-1) = -1$

Also, $f(c) = -1 = \lim_{x \rightarrow c} f(x)$

Therefore, the given function is a continuous function.

Hence, the given function has no point of discontinuity.

Q

10

Continuity:
piecewise
function



$$f(x) = \begin{cases} x+1, & \text{if } x \geq 1 \\ x^2+1, & \text{if } x < 1 \end{cases}$$

Given

The given function is $f(x) = \begin{cases} x+1, & \text{if } x \geq 1 \\ x^2+1, & \text{if } x < 1 \end{cases}$

The given function f is defined at all the points of the real line.

Let c be a point on the real line.

Substitute

Case I:

If $c < 1$, then $f(c) = c^2+1$

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (x^2+1) = c^2+1$$

$$\therefore \lim_{x \rightarrow c} f(x) = f(c)$$

Therefore, f is continuous at all points x , such that $x < 1$.

Case II:

If $c = 1$, then $f(c) = f(1) = 1+1 = 2$

The left hand limit of f at $x = 1$ is,

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x^2 + 1) = 1^2 + 1 = 2$$

The right hand limit of f at $x = 1$ is,

$$\begin{aligned}\lim_{x \rightarrow 1^+} f(x) &= \lim_{x \rightarrow 1^+} (x + 1) = 1 + 1 = 2 \\ \therefore \lim_{x \rightarrow 1} f(x) &= f(1)\end{aligned}$$

Therefore, f is continuous at $x = 1$.

Conclusion

Case III:

If $c > 1$, then $f(c) = c + 1$

$$\begin{aligned}\lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} (x + 1) = c + 1 \\ \therefore \lim_{x \rightarrow c} f(x) &= f(c)\end{aligned}$$

Therefore, f is continuous at all points x , such that $x > 1$.

Hence, the given function f has no point of discontinuity.

Q

11

Continuity:
piecewise
function



$$f(x) = \begin{cases} x^3 - 3, & \text{if } x \leq 2 \\ x^2 + 1, & \text{if } x > 2 \end{cases}$$

Given

The given function is $f(x) = \begin{cases} x^3 - 3, & \text{if } x \leq 2 \\ x^2 + 1, & \text{if } x > 2 \end{cases}$

The given function f is defined at all the points of the real line.

Let c be a point on the real line.

Substitute

Case I:

If $c < 2$, then $f(c) = c^3 - 3$

$$\begin{aligned}\lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} (x^3 - 3) = c^3 - 3 \\ \therefore \lim_{x \rightarrow c} f(x) &= f(c)\end{aligned}$$

Therefore, f is continuous at all points x , such that $x < 2$.

Case II:

If $c = 2$, then $f(c) = f(2) = 2^3 - 3 = 5$

$$\begin{aligned}\lim_{x \rightarrow 2^-} f(x) &= \lim_{x \rightarrow 2^-} (x^3 - 3) = 2^3 - 3 = 5 \\ \lim_{x \rightarrow 2^+} f(x) &= \lim_{x \rightarrow 2^+} (x^2 + 1) = 2^2 + 1 = 5 \\ \therefore \lim_{x \rightarrow 2} f(x) &= f(2)\end{aligned}$$

Therefore, f is continuous at $x = 2$.

Conclusion

Case III:

If $c > 2$, then $f(c) = c^2 + 1$

$$\begin{aligned}\lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} (x^2 + 1) = c^2 + 1 \\ \therefore \lim_{x \rightarrow c} f(x) &= f(c)\end{aligned}$$

Therefore, f is continuous at all points x , such that $x > 2$.

Thus, the given function f is continuous at every point on the real line.

Hence, f has no point of discontinuity.

Q

12

Continuity:
piecewise
function

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$$f(x) = \begin{cases} x^{10} - 1, & \text{if } x \leq 1 \\ x^2, & \text{if } x > 1 \end{cases}$$

Given

The given function is $f(x) = \begin{cases} x^{10} - 1, & \text{if } x \leq 1 \\ x^2, & \text{if } x > 1 \end{cases}$

The given function f is defined at all the points of the real line.

Let c be a point on the real line.

Substitute

Case I:

If $c < 1$, then $f(c) = c^{10} - 1$

$$\begin{aligned}\lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} (x^{10} - 1) = c^{10} - 1 \\ \therefore \lim_{x \rightarrow c} f(x) &= f(c)\end{aligned}$$

Therefore, f is continuous at all points x , such that $x < 1$.

Case II:

If $c = 1$, then the left hand limit of f at $x = 1$ is,

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x^{10} - 1) = 1^{10} - 1 = 1 - 1 = 0$$

The right hand limit of f at $x = 1$ is,

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (x^2) = 1^2 = 1$$

The left and right hand limits of f at $x = 1$ do not coincide.

Therefore, f is not continuous at $x = 1$.

Conclusion

Case III:

If $c > 1$, then $f(c) = c^2$

$$\begin{aligned}\lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} (x^2) = c^2 \\ \therefore \lim_{x \rightarrow c} f(x) &= f(c)\end{aligned}$$

Therefore, f is continuous at all points x , such that $x > 1$.

So $x = 1$ is the only point of discontinuity of f .

Q

13

**Continuity:
piecewise
function**



Is the function defined by

$$f(x) = \begin{cases} x+5, & \text{if } x \leq 1 \\ x-5, & \text{if } x > 1 \end{cases}$$

a continuous function? Discuss the continuity of the function f , where f is defined by

Given

The given function is $f(x) = \begin{cases} x+5, & \text{if } x \leq 1 \\ x-5, & \text{if } x > 1 \end{cases}$

The given function f is defined at all the points of the real line.

Let c be a point on the real line.

Substitute

Case I:

If $c < 1$, then $f(c) = c+5$

$$\begin{aligned}\lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} (x+5) = c+5 \\ \therefore \lim_{x \rightarrow c} f(x) &= f(c)\end{aligned}$$

Therefore, f is continuous at all points x , such that $x < 1$.

Case II:

If $c = 1$, then $f(1) = 1+5 = 6$

The left hand limit of f at $x = 1$ is,

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x+5) = 1+5 = 6$$

The right hand limit of f at $x = 1$ is,

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (x-5) = 1-5 = -4$$

The left and right hand limits of f at $x = 1$ do not coincide.

Therefore, f is not continuous at $x = 1$.

Conclusion

Case III:

If $c > 1$, then $f(c) = c-5$

$$\begin{aligned}\lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} (x-5) = c-5 \\ \therefore \lim_{x \rightarrow c} f(x) &= f(c)\end{aligned}$$

Therefore, f is continuous at all points x , such that $x > 1$.

So, $x = 1$ is the only point of discontinuity of f .

Q

14

**Continuity:
piecewise
function**



$$f(x) = \begin{cases} 3, & \text{if } 0 \leq x \leq 1 \\ 4, & \text{if } 1 < x < 3 \\ 5, & \text{if } 3 \leq x \leq 10 \end{cases}$$

Given

The given function is

$$f(x) = \begin{cases} 3, & \text{if } 0 \leq x \leq 1 \\ 4, & \text{if } 1 < x < 3 \\ 5, & \text{if } 3 \leq x \leq 10 \end{cases}$$

The given function f is defined at all the points of the interval $[0, 10]$.

Let c be a point in the interval $[0, 10]$.

Substitute

Case I:

If $0 \leq c < 1$, then $f(c) = 3$

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} 3 = 3$$

$$\therefore \lim_{x \rightarrow c} f(x) = f(c)$$

Therefore, f is continuous in the interval $[0, 1)$.

Case II:

If $c = 1$, then $f(3) = 3$ The left hand limit of f at $x = 1$ is,

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} 3 = 3$$

The right hand limit of f at $x = 1$ is,

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} 4 = 4$$

The left and right hand limits of f at $x = 1$ do not coincide.Therefore, f is not continuous at $x = 1$.

Case III:

If $1 < c < 3$, then $f(c) = 4$

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} 4 = 4$$

$$\therefore \lim_{x \rightarrow c} f(x) = f(c)$$

Therefore, f is continuous at in the interval $(1, 3)$.

Case IV:

If $c = 3$, then $f(c) = 5$ The left hand limit of f at $x = 3$ is,

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} 4 = 4$$

The right hand limit of f at $x = 3$ is,

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} 5 = 5$$

The left and right hand limits of f at $x = 3$ do not coincide.Therefore, f is discontinuous at $x = 3$.**Conclusion**

Case V:

If $3 < c \leq 10$, then $f(c) = 5$

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} 5 = 5$$

$$\therefore \lim_{x \rightarrow c} f(x) = f(c)$$

Therefore, f is continuous at all points of the interval $(3, 10]$.Hence, f is discontinuous at $x = 1$ and $x = 3$.**Q****15****Continuity:
piecewise
function**

$$f(x) = \begin{cases} 2x, & \text{if } x < 0 \\ 0, & \text{if } 0 \leq x \leq 1 \\ 4x, & \text{if } x > 1 \end{cases}$$

Given

The given function is

$$f(x) = \begin{cases} 2x, & \text{if } x < 0 \\ 0, & \text{if } 0 \leq x \leq 1 \\ 4x, & \text{if } x > 1 \end{cases}$$

The given function f is defined at all the points of the real line.Let c be a point on the real line.**Substitute**

Case I:

If $c < 0$, then $f(c) = 2c$

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (2x) = 2c$$

$$\therefore \lim_{x \rightarrow c} f(x) = f(c)$$

Therefore, f is continuous at all points x , such that $x < 0$.

Case II:

If $c = 0$, then $f(c) = f(0) = 0$ The left hand limit of f at $x = 0$ is,

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (2x) = 2(0) = 0$$

The right hand limit of f at $x = 0$ is,

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (0) = 0$$

$$\therefore \lim_{x \rightarrow 0} f(x) = f(0)$$

Therefore, f is continuous at $x = 0$

Case III:

If $0 < c < 1$, then $f(x) = 0$

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (0) = 0$$

$$\therefore \lim_{x \rightarrow c} f(x) = f(c)$$

Therefore, f is continuous in the interval $(0, 1)$.

Case IV:

If $c = 1$, then $f(c) = f(1) = 0$

The left hand limit of f at $x = 1$ is,

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (0) = 0$$

The right hand limit of f at $x = 1$ is,

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (4x) = 4(1) = 4$$

The left and right hand limits of f at $x = 1$ do not coincide.

Therefore, f is not continuous at $x = 1$.

Conclusion

Case V:

If $c < 1$, then $f(c) = 4c$

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (4x) = 4c$$

$$\therefore \lim_{x \rightarrow c} f(x) = f(c)$$

Therefore, f is continuous at all points x , such that $x > 1$.

Hence, f is not continuous only at $x = 1$.

Q
16
Continuity:
piecewise
function
()

$$f(x) = \begin{cases} -2, & \text{if } x \leq -1 \\ 2x, & \text{if } -1 < x \leq 1 \\ 2, & \text{if } x > 1 \end{cases}$$

Given

The given function is

$$f(x) = \begin{cases} -2, & \text{if } x \leq -1 \\ 2x, & \text{if } -1 < x \leq 1 \\ 2, & \text{if } x > 1 \end{cases}$$

The given function f is defined at all the points.

Let c be a point on the real line.

Substitute

Case I:

If $c < -1$, then $f(c) = -2$

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (-2) = -2$$

$$\therefore \lim_{x \rightarrow c} f(x) = f(c)$$

Therefore, f is continuous at all points x , such that $x < -1$.

Case II:

If $c = -1$, then $f(c) = f(-1) = -2$

The left hand limit of f at $x = -1$ is,

$$\lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^-} (-2) = -2$$

The right hand limit of f at $x = -1$ is,

$$\lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow -1^+} (2x) = 2(-1) = -2$$

$$\therefore \lim_{x \rightarrow -1} f(x) = f(-1)$$

Therefore, f is continuous at $x = -1$

Case III:

If $-1 < c < 1$, then $f(c) = 2c$

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (2x) = 2c$$

$$\therefore \lim_{x \rightarrow c} f(x) = f(c)$$

Therefore, f is continuous in the interval $(-1, 1)$.

Case IV:

If $c = 1$, then $f(c) = f(1) = 2(1) = 2$

The left hand limit of f at $x = 1$ is,

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (2x) = 2(1) = 2$$

The right hand limit of f at $x = 1$ is,

$$\begin{aligned} \lim_{x \rightarrow 1^+} f(x) &= \lim_{x \rightarrow 1^+} (2) = 2 \\ \therefore \lim_{x \rightarrow 1} f(x) &= f(c) \end{aligned}$$

Therefore, f is continuous at $x = 2$.

Conclusion

Case V:

If $c > 1$, then $f(c) = 2$

$$\begin{aligned} \lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} (2) = 2 \\ \therefore \lim_{x \rightarrow c} f(x) &= f(c) \end{aligned}$$

Therefore, f is continuous at all points x , such that $x > 1$.

So f is continuous at all points of the real line.

Q

17

Continuity:
piecewise
function



Find the relationship between a and b so that the function f defined by

$$f(x) = \begin{cases} ax+1, & \text{if } x \leq 3 \\ bx+3, & \text{if } x > 3 \end{cases}$$

is continuous at $x = 3$.

Conclusion

The given function is $f(x) = \begin{cases} ax+1, & \text{if } x \leq 3 \\ bx+3, & \text{if } x > 3 \end{cases}$

For f to be continuous at $x = 3$, then

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x) = f(3)$$

Also,

$$\begin{aligned} \lim_{x \rightarrow 3^-} f(x) &= \lim_{x \rightarrow 3^-} (ax+1) = 3a+1 \\ \lim_{x \rightarrow 3^+} f(x) &= \lim_{x \rightarrow 3^+} (bx+3) = 3b+3 \\ f(3) &= 3a+1 \end{aligned}$$

Therefore, from (1), we get

$$\begin{aligned} 3a+1 &= 3b+3 = 3a+1 \\ \Rightarrow 3a+1 &= 3b+3 \\ \Rightarrow 3a &= 3b+2 \\ \Rightarrow a &= b + \frac{2}{3} \end{aligned}$$

Therefore, the required relationship is given by, $a = b + \frac{2}{3}$.

Q

18

Continuity:



For what value of λ is the function defined by

$$f(x) = \begin{cases} \lambda(x^2-2x), & \text{if } x \leq 0 \\ 4x+1, & \text{if } x > 0 \end{cases}$$

continuous at $x = 0$? What about continuity at $x = 1$?

Conclusion

The given function is

$$f(x) = \begin{cases} \lambda(x^2-2x), & \text{if } x \leq 0 \\ 4x+1, & \text{if } x > 0 \end{cases}$$

If f is continuous at $x = 0$, then

$$\begin{aligned} \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} f(x) = f(0) \\ \Rightarrow \lim_{x \rightarrow 0^-} \lambda(x^2-2x) &= \lim_{x \rightarrow 0^-} (4x+1) = \lambda(0^2-2 \times 0) \quad [\text{which is not possible}] \\ \Rightarrow \lambda(0^2-2 \times 0) &= 4(0)+1 = 0 \\ \Rightarrow 0 &= 1 = 0 \end{aligned}$$

Therefore, there is no value of λ for which f is continuous at $x = 0$.

At $x = 1$

$$\begin{aligned} f(1) &= 4(1)+1 = 5 \\ \lim_{x \rightarrow 1} (4x+1) &= 4(1)+1 = 5 \\ \therefore \lim_{x \rightarrow 1} f(x) &= f(1) \end{aligned}$$

Therefore, for any values of λ , f is continuous at $x = 1$.

Q

19

Continuity:
 $g(x)=x-[x]$ 

Show that the function defined by $g(x) = x - [x]$ is discontinuous at all integral points. Here $[x]$ denotes the greatest integer less than or equal to x .

Conclusion

The given function is $g(x) = x - [x]$

Clearly, g is defined at all integral points.

Let n be an integer.

Then,

$$g(n) = n - [n] = n - n = 0$$

The left hand limit of g at $x = n$ is,

$$\lim_{x \rightarrow n^-} g(x) = \lim_{x \rightarrow n^-} (x - [x]) = \lim_{x \rightarrow n^-} (x) - \lim_{x \rightarrow n^-} [x] = n - (n-1) = 1$$

The right hand limit of g at $x = n$ is,

$$\lim_{x \rightarrow n^+} g(x) = \lim_{x \rightarrow n^+} (x - [x]) = \lim_{x \rightarrow n^+} (x) - \lim_{x \rightarrow n^+} [x] = n - n = 0$$

The left and right hand limits of g at $x = n$ do not coincide.

Therefore, g is not continuous at $x = n$.

Hence, g is discontinuous at all integral points.

Q

20

Continuity:
 $f(x)=x^2-\sin x+5$

()

Is the function defined by $f(x) = x^2 - \sin x + 5$ continuous at $x = \pi$?

Conclusion

The given function is $f(x) = x^2 - \sin x + 5$

Clearly, f is defined at $x = \pi$.

$$\text{At } x = \pi, f(\pi) = f(\pi) = \pi^2 - \sin \pi + 5 = \pi^2 - 0 + 5 = \pi^2 + 5$$

$$\text{Consider } \lim_{x \rightarrow \pi} f(x) = \lim_{x \rightarrow \pi} (x^2 - \sin x + 5)$$

Put $x = \pi + h$, it is evident that if $x \rightarrow \pi$, then $h \rightarrow 0$

$$\begin{aligned} \therefore \lim_{x \rightarrow \pi} f(x) &= \lim_{x \rightarrow \pi} (x^2 - \sin x) + 5 \\ &= \lim_{h \rightarrow 0} [(\pi+h)^2 - \sin(\pi+h) + 5] \\ &= \lim_{h \rightarrow 0} (\pi+h)^2 - \lim_{h \rightarrow 0} \sin(\pi+h) + \lim_{h \rightarrow 0} 5 \\ &= (\pi+0)^2 - \lim_{h \rightarrow 0} [\sin \pi \cos h + \cos \pi \sin h] + 5 \\ &= \pi^2 - \lim_{h \rightarrow 0} \sin \pi \cos h - \lim_{h \rightarrow 0} \cos \pi \sin h + 5 \\ &= \pi^2 - \sin \pi \cos 0 - \cos \pi \sin 0 + 5 \\ &= \pi^2 - 0(1)(-1) + 5 \\ &= \pi^2 + 5 \\ &= f(\pi) \end{aligned}$$

Therefore, the given function f is continuous at $x = \pi$.

Q

21

Continuity:
 $f(x)=\sin x+\cos x$ 

Discuss the continuity of the following functions:

Part (a)

$$f(x) = \sin x + \cos x$$

Conclusion

It is known that if g and h are two continuous functions, then $g+h$, $g-h$ and $g \cdot h$ are also continuous.

Let $g(x) = \sin x$ and $h(x) = \cos x$ are continuous functions.

Clearly, $g(x) = \sin x$ is defined for every real number.

Let c be a real number.

Put $x = c + h$

If $x \rightarrow c$, then $h \rightarrow 0$

$$\begin{aligned} g(c) &= \sin c \\ \lim_{x \rightarrow c} g(x) &= \lim_{x \rightarrow c} \sin x \\ &= \lim_{h \rightarrow 0} \sin(c+h) \\ &= \lim_{h \rightarrow 0} [\sin c \cos h + \cos c \sin h] \\ &= \lim_{h \rightarrow 0} (\sin c \cos h) + \lim_{h \rightarrow 0} (\cos c \sin h) \\ &= \sin c \cos 0 + \cos c \sin 0 \\ &= \sin c(1) + \cos c(0) \\ &= \sin c \\ \therefore \lim_{x \rightarrow c} g(x) &= g(c) \end{aligned}$$

Therefore, $g(x) = \sin x$ is a continuous function.

Let $h(x) = \cos x$

Clearly, $h(x) = \cos x$ is defined for every real number.

Let c be a real number.

Put $x = c+h$

If $x \rightarrow c$, then $h \rightarrow 0$

$$h(c) = \cos c$$

$$\begin{aligned}\lim_{x \rightarrow c} h(x) &= \lim_{x \rightarrow c} \cos x \\&= \lim_{h \rightarrow 0} \cos(c+h) \\&= \lim_{h \rightarrow 0} [\cos c \cos h - \sin c \sin h] \\&= \lim_{h \rightarrow 0} (\cos c \cos h) - \lim_{h \rightarrow 0} (\sin c \sin h) \\&= \cos c \cos 0 - \sin c \sin 0 \\&= \cos c(1) - \sin c(0) \\&= \cos c \\ \therefore \lim_{x \rightarrow c} h(x) &= h(c)\end{aligned}$$

Therefore, $h(x) = \cos x$ is a continuous function.

$$\mathbf{f(x) = g(x)+h(x) = \sin x+\cos x \text{ is a continuous function.} //$$

Part (b)

$$\mathbf{f(x) = \sin x - \cos x}$$

Conclusion

It is known that if g and h are two continuous functions, then $g+h$, $g-h$ and $g \cdot h$ are also continuous.

Let $g(x) = \sin x$ and $h(x) = \cos x$ are continuous functions.

Clearly, $g(x) = \sin x$ is defined for every real number.

Let c be a real number.

Put $x = c+h$

If $x \rightarrow c$, then $h \rightarrow 0$

$$\begin{aligned}g(c) &= \sin c \\ \lim_{x \rightarrow c} g(x) &= \lim_{x \rightarrow c} \sin x \\&= \lim_{h \rightarrow 0} \sin(c+h) \\&= \lim_{h \rightarrow 0} [\sin c \cos h + \cos c \sin h] \\&= \lim_{h \rightarrow 0} (\sin c \cos h) + \lim_{h \rightarrow 0} (\cos c \sin h) \\&= \sin c \cos 0 + \cos c \sin 0 \\&= \sin c(1) + \cos c(0) \\&= \sin c \\ \therefore \lim_{x \rightarrow c} g(x) &= g(c)\end{aligned}$$

Therefore, $g(x) = \sin x$ is a continuous function.

Let $h(x) = \cos x$

Clearly, $h(x) = \cos x$ is defined for every real number.

Let c be a real number.

Put $x = c+h$

If $x \rightarrow c$, then $h \rightarrow 0$

$$h(c) = \cos c$$

$$\begin{aligned}\lim_{x \rightarrow c} h(x) &= \lim_{x \rightarrow c} \cos x \\&= \lim_{h \rightarrow 0} \cos(c+h) \\&= \lim_{h \rightarrow 0} [\cos c \cos h - \sin c \sin h] \\&= \lim_{h \rightarrow 0} (\cos c \cos h) - \lim_{h \rightarrow 0} (\sin c \sin h) \\&= \cos c \cos 0 - \sin c \sin 0 \\&= \cos c(1) - \sin c(0) \\&= \cos c \\ \therefore \lim_{x \rightarrow c} h(x) &= h(c)\end{aligned}$$

Therefore, $h(x) = \cos x$ is a continuous function.

$$\mathbf{f(x) = g(x)-h(x) = \sin x-\cos x \text{ is a continuous function.} //$$

Part (c)

$$\mathbf{f(x) = \sin x \cdot \cos x}$$

Conclusion

It is known that if g and h are two continuous functions, then $g+h$, $g-h$ and $g \cdot h$ are also continuous.

Let $g(x) = \sin x$ and $h(x) = \cos x$ are continuous functions.

Clearly, $g(x) = \sin x$ is defined for every real number.

Let c be a real number.

Put $x = c+h$

If $x \rightarrow c$, then $h \rightarrow 0$

$$\begin{aligned}
g(c) &= \sin c \\
\lim_{x \rightarrow c} g(x) &= \lim_{x \rightarrow c} \sin x \\
&= \lim_{h \rightarrow 0} \sin(c+h) \\
&= \lim_{h \rightarrow 0} [\sin c \cos h + \cos c \sin h] \\
&= \lim_{h \rightarrow 0} (\sin c \cos h) + \lim_{h \rightarrow 0} (\cos c \sin h) \\
&= \sin c \cos 0 + \cos c \sin 0 \\
&= \sin c(1) + \cos c(0) \\
&= \sin c \\
\therefore \lim_{x \rightarrow c} g(x) &= g(c)
\end{aligned}$$

Therefore, $g(x) = \sin x$ is a continuous function.

Let $h(x) = \cos x$

Clearly, $h(x) = \cos x$ is defined for every real number.

Let c be a real number.

Put $x = c+h$

If $x \rightarrow c$, then $h \rightarrow 0$

$h(c) = \cos c$

$$\begin{aligned}
\lim_{x \rightarrow c} h(x) &= \lim_{x \rightarrow c} \cos x \\
&= \lim_{h \rightarrow 0} \cos(c+h) \\
&= \lim_{h \rightarrow 0} [\cos c \cos h - \sin c \sin h] \\
&= \lim_{h \rightarrow 0} (\cos c \cos h) - \lim_{h \rightarrow 0} (\sin c \sin h) \\
&= \cos c \cos 0 - \sin c \sin 0 \\
&= \cos c(1) - \sin c(0) \\
&= \cos c \\
\therefore \lim_{x \rightarrow c} h(x) &= h(c)
\end{aligned}$$

Therefore, $h(x) = \cos x$ is a continuous function.

$f(x) = g(x) \times h(x) = \sin x \times \cos x$ is a continuous function. //

Q

22

Continuity:
Discuss the
continuity
of...



Discuss the continuity of the cosine, cosecant, secant and cotangent functions.

Algebra of continuous functions

It is known that if g and h are two continuous functions, then

(i) $\frac{h(x)}{g(x)}$, $g(x) \neq 0$ is continuous.

(ii) $\frac{1}{g(x)}$, $g(x) \neq 0$ is continuous.

(iii) $\frac{1}{h(x)}$, $h(x) \neq 0$ is continuous.

Conclusion

Let $g(x) = \sin x$ and $h(x) = \cos x$ are continuous functions.

Clearly, $g(x) = \sin x$ is defined for every real number.

Let c be a real number.

Put $x = c+h$

If $x \rightarrow c$, then $h \rightarrow 0$

$$\begin{aligned}
g(c) &= \sin c \\
\lim_{x \rightarrow c} g(x) &= \lim_{x \rightarrow c} \sin x \\
&= \lim_{h \rightarrow 0} \sin(c+h) \\
&= \lim_{h \rightarrow 0} [\sin c \cos h + \cos c \sin h] \\
&= \lim_{h \rightarrow 0} (\sin c \cos h) + \lim_{h \rightarrow 0} (\cos c \sin h) \\
&= \sin c \cos 0 + \cos c \sin 0 \\
&= \sin c(1) + \cos c(0) \\
&= \sin c \\
\therefore \lim_{x \rightarrow c} g(x) &= g(c)
\end{aligned}$$

Therefore, $g(x) = \sin x$ is a continuous function.

Let $h(x) = \cos x$

Clearly, $h(x) = \cos x$ is defined for every real number.

Let c be a real number.

Put $x = c+h$

If $x \rightarrow c$, then $h \rightarrow 0$

$$\begin{aligned}
 h(c) &= \cos c \\
 \lim_{x \rightarrow c} h(x) &= \lim_{x \rightarrow c} \cos x \\
 &= \lim_{h \rightarrow 0} \cos(c+h) \\
 &= \lim_{h \rightarrow 0} [\cos c \cos h - \sin c \sin h] \\
 &= \lim_{h \rightarrow 0} (\cos c \cos h) - \lim_{h \rightarrow 0} (\sin c \sin h) \\
 &= \cos c \cos 0 - \sin c \sin 0 \\
 &= \cos c(1) - \sin c(0) \\
 &= \cos c \\
 \therefore \lim_{x \rightarrow c} h(x) &= h(c)
 \end{aligned}$$

Therefore, $h(x) = \cos x$ is a continuous function.

So,

$$\begin{aligned}
 \operatorname{cosec} x &= \frac{1}{\sin x}, \sin x \neq 0 \text{ is continuous.} \\
 \Rightarrow \operatorname{cosec} x, x \neq n\pi (n \in \mathbb{Z}) &\text{ is continuous.}
 \end{aligned}$$

Therefore, cosecant is continuous except at $x = n\pi (n \in \mathbb{Z})$

$$\begin{aligned}
 \sec x &= \frac{1}{\cos x}, \cos x \neq 0 \\
 \Rightarrow \sec x, x \neq (2n+1)\frac{\pi}{2} (n \in \mathbb{Z}) &\text{ is continuous.}
 \end{aligned}$$

Therefore, secant is continuous except at $x = (2n+1)\frac{\pi}{2} (n \in \mathbb{Z})$

$$\begin{aligned}
 \cot x &= \frac{\cos x}{\sin x}, \sin x \neq 0 \text{ is continuous.} \\
 \Rightarrow \cot x, x \neq n\pi (n \in \mathbb{Z}) &\text{ is continuous.}
 \end{aligned}$$

Therefore, cotangent is continuous except at $x = n\pi (n \in \mathbb{Z})$.

Q

23

Continuity:
piecewise
function



Find all points of discontinuity of f , where

$$f(x) = \begin{cases} \frac{\sin x}{x}, & \text{if } x < 0 \\ x+1, & \text{if } x \geq 0 \end{cases}$$

Given

The given function is

$$f(x) = \begin{cases} \frac{\sin x}{x}, & \text{if } x < 0 \\ x+1, & \text{if } x \geq 0 \end{cases}$$

The given function f is defined at all the points of the real line.

Let c be a point on the real line.

Substitute

Case I:

$$\text{If } c < 0, \text{ then } f(c) = \frac{\sin c}{c}$$

$$\begin{aligned}
 \lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} \left(\frac{\sin x}{x} \right) = \frac{\sin c}{c} \\
 \therefore \lim_{x \rightarrow c} f(x) &= f(c)
 \end{aligned}$$

Therefore, f is continuous at all points x , such that $x < 0$.

Case II:

If $c > 0$, then $f(c) = c+1$

$$\begin{aligned}
 \lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} (x+1) = c+1 \\
 \therefore \lim_{x \rightarrow c} f(x) &= f(c)
 \end{aligned}$$

Therefore, f is continuous at all points x , such that $x > 0$.

Conclusion

Case III:

If $c = 0$, then $f(c) = f(0) = 0+1 = 1$

The left hand limit of f at $x = 0$ is,

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \left(\frac{\sin x}{x} \right) = 1$$

The right hand limit of f at $x = 0$ is,

$$\begin{aligned}
 \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} (x+1) = 1 \\
 \therefore \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^+} f(x) = f(0)
 \end{aligned}$$

Therefore, f is continuous at $x = 0$

So f is continuous at all points of the real line.

Thus, f has no point of discontinuity.

Q

24

Continuity:
piecewise
function

(())

Determine if f defined by

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

is a continuous function?

Given

The given function is

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

The given function f is defined at all the points of the real line.

Let c be a point on the real line.

Substitute

Case I:

$$\text{If } c \neq 0, \text{ then } f(c) = c^2 \sin \frac{1}{c}$$

$$\begin{aligned} \lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} (x^2 \sin \frac{1}{x}) = (\lim_{x \rightarrow c} x^2) (\lim_{x \rightarrow c} \sin \frac{1}{x}) = c^2 \sin \frac{1}{c} \\ \therefore \lim_{x \rightarrow c} f(x) &= f(c) \end{aligned}$$

Therefore, f is continuous at all points x , such that $x \neq 0$.

Conclusion

Case II:

If $c = 0$, then $f(0) = 0$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x^2 \sin \frac{1}{x}) = \lim_{x \rightarrow 0^-} (x^2 \sin \frac{1}{x})$$

It is known that, $-1 \leq \sin \frac{1}{x} \leq 1, x \neq 0$

$$\Rightarrow -x^2 \leq x^2 \sin \frac{1}{x} \leq x^2$$

$$\Rightarrow \lim_{x \rightarrow 0^-} (-x^2) \leq \lim_{x \rightarrow 0^-} (x^2 \sin \frac{1}{x}) \leq \lim_{x \rightarrow 0^-} x^2$$

$$\Rightarrow 0 \leq \lim_{x \rightarrow 0^-} (x^2 \sin \frac{1}{x}) \leq 0$$

$$\Rightarrow \lim_{x \rightarrow 0^-} (x^2 \sin \frac{1}{x}) = 0$$

$$\therefore \lim_{x \rightarrow 0^-} f(x) = 0$$

Similarly,

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (x^2 \sin \frac{1}{x}) = \lim_{x \rightarrow 0^+} (x^2 \sin \frac{1}{x}) = 0$$

$$\therefore \lim_{x \rightarrow 0^+} f(x) = f(0) = \lim_{x \rightarrow 0^+} f(x)$$

Therefore, f is continuous at $x = 0$.

So f is continuous at every point of the real line.

Thus, f is a continuous function.

Q

25

Continuity:
piecewise
function

⬠

Examine the continuity of f , where f is defined by

$$f(x) = \begin{cases} \sin x - \cos x, & \text{if } x \neq 0 \\ -1, & \text{if } x = 0 \end{cases}$$

Find the values of k so that the function f is continuous at the indicated point in Exercises 26 to 29.

Given

The given function is $f(x) = \begin{cases} \sin x - \cos x, & \text{if } x \neq 0 \\ -1, & \text{if } x = 0 \end{cases}$

The given function f is defined at all the points of the real line.

Let c be a point on the real line.

Substitute

Case I:

$$\text{If } c \neq 0, \text{ then } f(c) = \sin c - \cos c$$

$$\begin{aligned} \lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} (\sin x - \cos x) = \sin c - \cos c \\ \therefore \lim_{x \rightarrow c} f(x) &= f(c) \end{aligned}$$

Therefore, f is continuous at all points x , such that $x \neq 0$.

Conclusion

Case II:

If $c = 0$, then $f(0) = -1$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0} (\sin x - \cos x) = \sin 0 - \cos 0 = 0 - 1 = -1$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0} (\sin x - \cos x) = \sin 0 - \cos 0 = 0 - 1 = -1$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$$

Therefore, f is continuous at $x = 0$.

So f is continuous at every point of the real line.

Thus, f is a continuous function.

Q

26

Continuity:
piecewise
function



$$f(x) = \begin{cases} \frac{k \cos x}{\pi - 2x}, & \text{if } x \neq \frac{\pi}{2} \\ 3, & \text{if } x = \frac{\pi}{2} \end{cases} \quad \text{at } x = \frac{\pi}{2}$$

Conclusion

The given function is

$$f(x) = \begin{cases} \frac{k \cos x}{\pi - 2x}, & \text{if } x \neq \frac{\pi}{2} \\ 3, & \text{if } x = \frac{\pi}{2} \end{cases}$$

The given function f is continuous at $x = \frac{\pi}{2}$, if f is defined at $x = \frac{\pi}{2}$ and if the

value of the f at $x = \frac{\pi}{2}$ equals the limit of f at $x = \frac{\pi}{2}$.

Clearly, f is defined at $x = \frac{\pi}{2}$ and $f(\frac{\pi}{2}) = 3$

$$\lim_{x \rightarrow \frac{\pi}{2}} f(x) = \lim_{x \rightarrow \frac{\pi}{2}} \frac{k \cos x}{\pi - 2x}$$

$$\text{Put } x = \frac{\pi}{2} + h$$

$$\text{Then } x \rightarrow \frac{\pi}{2} \Rightarrow h \rightarrow 0$$

$$\therefore \lim_{x \rightarrow \frac{\pi}{2}} f(x) = \lim_{x \rightarrow \frac{\pi}{2}} \frac{k \cos x}{\pi - 2x} = \lim_{h \rightarrow 0} \frac{k \cos(\frac{\pi}{2} + h)}{\pi - 2(\frac{\pi}{2} + h)}$$

$$= k \lim_{h \rightarrow 0} \frac{-\sin h}{-2h} = \frac{k}{2} \lim_{h \rightarrow 0} \frac{\sin h}{h} = \frac{k}{2} \cdot 1 = \frac{k}{2}$$

$$\therefore \lim_{x \rightarrow \frac{\pi}{2}} f(x) = f(\frac{\pi}{2})$$

$$\Rightarrow \frac{k}{2} = 3$$

$$\Rightarrow k = 6$$

Therefore, the value of $k = 6$.

Q

27

Continuity:
piecewise
function



$$f(x) = \begin{cases} kx^2, & \text{if } x \leq 2 \\ 3, & \text{if } x > 2 \end{cases} \quad \text{at } x = 2$$

Conclusion

The given function is $f(x) = \begin{cases} kx^2, & \text{if } x \leq 2 \\ 3, & \text{if } x > 2 \end{cases}$

The given function f is continuous at $x = 2$, if f is defined at $x = 2$ and if the value of the f at $x = 2$ equals the limit of f at $x = 2$.

Clearly, f is defined at $x = 2$ and $f(2) = k(2)^2 = 4k$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2)$$

$$\Rightarrow \lim_{x \rightarrow 2^-} (kx^2) = \lim_{x \rightarrow 2^+} (3) = 4k$$

$$\Rightarrow k \times 2^2 = 3 = 4k$$

$$\Rightarrow 4k = 3$$

$$\Rightarrow k = \frac{3}{4}$$

Therefore, the value of $k = \frac{3}{4}$.

Q

28

Continuity:
piecewise
function

()

$$f(x) = \begin{cases} kx+1, & \text{if } x \leq \pi \\ \cos x, & \text{if } x > \pi \end{cases} \quad \text{at } x = \pi$$

Conclusion

The given function is $f(x) = \begin{cases} kx+1, & \text{if } x \leq \pi \\ \cos x, & \text{if } x > \pi \end{cases}$

The given function f is continuous at $x = \pi$, if f is defined at $x = \pi$ and if the value of the f at $x = \pi$ equals the limit of f at $x = \pi$.

Clearly, f is defined at $x = \pi$ and $f(\pi) = k\pi + 1$

$$\begin{aligned} \lim_{x \rightarrow \pi^-} f(x) &= \lim_{x \rightarrow \pi^-} (kx+1) = k\pi + 1 \\ \Rightarrow \lim_{x \rightarrow \pi^-} (kx+1) &= \lim_{x \rightarrow \pi^+} (\cos x) = \cos \pi = -1 \\ \Rightarrow k\pi + 1 &= \cos \pi = -1 \\ \Rightarrow k\pi + 1 &= -1 \\ \Rightarrow k &= -\frac{2}{\pi} \end{aligned}$$

Therefore, the value of $k = -\frac{2}{\pi}$.

Q

29

Continuity:
piecewise
function

$$f(x) = \begin{cases} kx+1, & \text{if } x \leq 5 \\ 3x-5, & \text{if } x > 5 \end{cases} \quad \text{at } x = 5$$

Conclusion

The given function is $f(x) = \begin{cases} kx+1, & \text{if } x \leq 5 \\ 3x-5, & \text{if } x > 5 \end{cases}$

The given function f is continuous at $x = 5$, if f is defined at $x = 5$ and if the value of the f at $x = 5$ equals the limit of f at $x = 5$.

Clearly, f is defined at $x = 5$ and $f(5) = k \cdot 5 + 1 = 5k + 1$

$$\begin{aligned} \lim_{x \rightarrow 5^-} f(x) &= \lim_{x \rightarrow 5^-} (kx+1) = 5k + 1 \\ \Rightarrow \lim_{x \rightarrow 5^-} (kx+1) &= \lim_{x \rightarrow 5^+} (3x-5) = 10 \\ \Rightarrow 5k + 1 &= 10 \\ \Rightarrow 5k &= 9 \\ \Rightarrow k &= \frac{9}{5} \end{aligned}$$

Therefore, the value of $k = \frac{9}{5}$.

Q

30

Continuity:
piecewise
function

Find the values of a and b such that the function defined by

$$f(x) = \begin{cases} 5, & \text{if } x \leq 2 \\ ax+b, & \text{if } 2 < x < 10 \\ 21, & \text{if } x \geq 10 \end{cases}$$

is a continuous function.

Conclusion

The given function is

$$f(x) = \begin{cases} 5, & \text{if } x \leq 2 \\ ax+b, & \text{if } 2 < x < 10 \\ 21, & \text{if } x \geq 10 \end{cases}$$

Clearly, f is defined at all points of the real line.

If f is a continuous function, then f is continuous at all real numbers.

In particular, f is continuous at $x = 2$ and $x = 10$

Since f is continuous at $x = 2$, we get

$$\begin{aligned} \lim_{x \rightarrow 2^-} f(x) &= \lim_{x \rightarrow 2^-} 5 = 5 \\ \Rightarrow \lim_{x \rightarrow 2^-} 5 &= \lim_{x \rightarrow 2^+} (ax+b) = 5 \\ \Rightarrow 5 &= 2a+b \\ \Rightarrow 2a+b &= 5 \end{aligned}$$

Since f is continuous at $x = 10$, we get

$$\begin{aligned} \lim_{x \rightarrow 10^-} f(x) &= \lim_{x \rightarrow 10^-} (ax+b) = 10a+b \\ \Rightarrow \lim_{x \rightarrow 10^-} (ax+b) &= \lim_{x \rightarrow 10^+} 21 = 21 \\ \Rightarrow 10a+b &= 21 \end{aligned}$$

On subtracting equation (1) from equation (2), we get

$$\begin{aligned} 8a &= 16 \\ \Rightarrow a &= 2 \end{aligned}$$

By putting $a = 2$ in equation (1), we get

$$\begin{aligned} 2(2)+b &= 5 \\ \Rightarrow 4+b &= 5 \\ \Rightarrow b &= 1 \end{aligned}$$

Therefore, the values of a and b for which f is a continuous function are 2 and 1 respectively.

Q

31

Continuity:
f(x)=cos
(x²)



Show that the function defined by $f(x) = \cos(x^2)$ is a continuous function.

Conclusion

The given function is $f(x) = \cos(x^2)$.

This function f is defined for every real number and f can be written as the composition of two functions as,

$$\begin{aligned} f &= g \circ h, \text{ where } g(x) = \cos x \text{ and } h(x) = x^2 \\ [\therefore (g \circ h)(x) &= g(h(x)) = g(x^2) = \cos(x^2) = f(x)] \end{aligned}$$

First, we show that $g(x) = \cos x$ and $h(x) = x^2$ are continuous functions.

Clearly, g is defined for every real number.

Let c be a real number.

$$\text{Let } g(c) = \cos c.$$

$$\text{Put } x = c+h$$

$$\text{If } x \rightarrow c, \text{ then } h \rightarrow 0$$

$$\begin{aligned} \lim_{x \rightarrow c} g(x) &= \lim_{x \rightarrow c} \cos x \\ &= \lim_{h \rightarrow 0} \cos(c+h) \\ &= \lim_{h \rightarrow 0} [\cos c \cos h - \sin c \sin h] \\ &= \lim_{h \rightarrow 0} (\cos c \cos h) - \lim_{h \rightarrow 0} (\sin c \sin h) \\ &= \cos c \cos 0 - \sin c \sin 0 \\ &= \cos c(1) - \sin c(0) \\ &= \cos c \\ \therefore \lim_{x \rightarrow c} g(x) &= g(c) \end{aligned}$$

Therefore, $g(x) = \cos x$ is a continuous function.

$$\text{Let } h(x) = x^2$$

Clearly, h is defined for every real number.

$$\text{Let } k \text{ be a real number, then } h(k) = k^2$$

$$\begin{aligned} \lim_{x \rightarrow k} h(x) &= \lim_{x \rightarrow k} x^2 = k^2 \\ \therefore \lim_{x \rightarrow k} h(x) &= h(k) \end{aligned}$$

Therefore, h is a continuous function.

It is known that for real valued functions g and h, such that (g o h) is defined at c, if g is continuous at c and if f is continuous at g(c), then (f o g) is continuous at c.

Therefore, $f(x) = (g \circ h)(x) = \cos(x^2)$ is a continuous function.

Q

32

Continuity:
f(x)=|cos x|

(())

Show that the function defined by $f(x) = |\cos x|$ is a continuous function.

Given

The given function is $f(x) = |\cos x|$.

This function f is defined for every real number and f can be written as the composition of two functions as,

$$\begin{aligned} f &= g \circ h, \text{ where } g(x) = |x| \text{ and } h(x) = \cos x \\ [\therefore (g \circ h)(x) &= g(h(x)) = g(\cos x) = |\cos x| = f(x)] \end{aligned}$$

First, we show that $g(x) = |x|$ and $h(x) = \cos x$ are continuous functions.

$$g(x) = |x| \text{ can be written as } g(x) = \begin{cases} -x, & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases}$$

Clearly, g is defined for every real number.

Let c be a real number.

Substitute

Case I:

$$\text{If } c < 0, \text{ then } g(c) = -c$$

$$\begin{aligned} \lim_{x \rightarrow c} g(x) &= \lim_{x \rightarrow c} (-x) = -c \\ \therefore \lim_{x \rightarrow c} g(x) &= g(c) \end{aligned}$$

Therefore, g is continuous at all points x, such that $x < 0$.

Case II:

If $c > 0$, then $g(c) = c$

$$\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} (x) = c$$
$$\therefore \lim_{x \rightarrow c} g(x) = g(c)$$

Therefore, g is continuous at all points x , such that $x > 0$.

Conclusion

Case III:

If $c = 0$, then $g(c) = g(0) = 0$

$$\lim_{x \rightarrow 0^-} g(x) = \lim_{x \rightarrow 0^-} (-x) = 0$$
$$\lim_{x \rightarrow 0^+} g(x) = \lim_{x \rightarrow 0^+} (x) = 0$$
$$\therefore \lim_{x \rightarrow 0^-} g(x) = \lim_{x \rightarrow 0^+} (x) = g(0)$$

Therefore, g is continuous at all $x = 0$.

From the above three observations, it can be concluded that g is continuous at all points.

Let $h(x) = \cos x$

Clearly, $h(x) = \cos x$ is defined for every real number.

Let c be a real number.

Put $x = c + h$

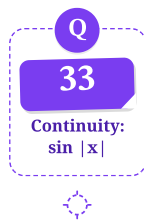
If $x \rightarrow c$, then $h \rightarrow 0$

$$h(c) = \cos c$$
$$\lim_{x \rightarrow c} h(x) = \lim_{x \rightarrow c} \cos x$$
$$= \lim_{h \rightarrow 0} \cos (c+h)$$
$$= \lim_{h \rightarrow 0} [\cos c \cos h - \sin c \sin h]$$
$$= \lim_{h \rightarrow 0} (\cos c \cos h) - \lim_{h \rightarrow 0} (\sin c \sin h)$$
$$= \cos c \cos 0 - \sin c \sin 0$$
$$= \cos c (1) - \sin c (0)$$
$$= \cos c$$
$$\therefore \lim_{x \rightarrow c} h(x) = h(c)$$

Therefore, $h(x) = \cos x$ is a continuous function.

It is known that for real valued functions g and h , such that $(g \circ h)$ is defined at c , if g is continuous at c and if h is continuous at $g(c)$, then $(g \circ h)$ is continuous at c .

Therefore, $f(x) = (g \circ h)(x) = g(h(x)) = g(\cos x) = |\cos x|$ is a continuous function.



Examine that $\sin |x|$ is a continuous function.

Given

The given function is $f(x) = |\sin x|$.

This function f is defined for every real number and f can be written as the composition of two functions as,

$$f = g \circ h, \text{ where } g(x) = |x| \text{ and } h(x) = \sin x$$
$$[\therefore (g \circ h)(x) = g(h(x)) = g(\sin x) = |\sin x| = f(x)]$$

First, we show that $g(x) = |x|$ and $h(x) = \sin x$ are continuous functions.

$$g(x) = |x| \text{ can be written as } g(x) = \begin{cases} -x, & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases}$$

Clearly, g is defined for every real number.

Let c be a real number.

Substitute

Case I:

If $c < 0$, then $g(c) = -c$

$$\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} (-x) = -c$$
$$\therefore \lim_{x \rightarrow c} g(x) = g(c)$$

Therefore, g is continuous at all points x , such that $x < 0$.

Case II:

If $c > 0$, then $g(c) = c$

$$\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} (x) = c$$
$$\therefore \lim_{x \rightarrow c} g(x) = g(c)$$

Therefore, g is continuous at all points x , such that $x > 0$.

Conclusion

Case III:

If $c = 0$, then $g(c) = g(0) = 0$

$$\begin{aligned}\lim_{x \rightarrow 0^-} g(x) &= \lim_{x \rightarrow 0^-} (-x) = 0 \\ \lim_{x \rightarrow 0^+} g(x) &= \lim_{x \rightarrow 0^+} (x) = 0 \\ \therefore \lim_{x \rightarrow 0^-} g(x) &= \lim_{x \rightarrow 0^+} g(x) = g(0)\end{aligned}$$

Therefore, g is continuous at all $x = 0$.

From the above three observations, it can be concluded that g is continuous at all points.

Let $h(x) = \sin x$

Clearly, $h(x) = \sin x$ is defined for every real number.

Let c be a real number.

Put $x = c+k$

If $x \rightarrow c$, then $k \rightarrow 0$

$$\begin{aligned}h(c) &= \sin c \\ \lim_{x \rightarrow c} h(x) &= \lim_{x \rightarrow c} \sin x \\ &= \lim_{k \rightarrow 0} \sin(c+k) \\ &= \lim_{k \rightarrow 0} [\sin c \cos k + \cos c \sin k] \\ &= \lim_{k \rightarrow 0} (\sin c \cos k) + \lim_{k \rightarrow 0} (\cos c \sin k) \\ &= \sin c \cos 0 + \cos c \sin 0 \\ &= \sin c(1) + \cos c(0) \\ &= \sin c \\ \therefore \lim_{x \rightarrow c} h(x) &= h(c)\end{aligned}$$

Therefore, $h(x) = \sin x$ is a continuous function.

It is known that for real valued functions g and h , such that $(g \circ h)$ is defined at c , if g is continuous at c and if h is continuous at $g(c)$, then $(f \circ g)$ is continuous at c .

Therefore, $f(x) = (g \circ h)(x) = g(h(x)) = g(\sin x) = |\sin x|$ is a continuous function.

Q
34
Continuity:
 $f(x) = |x| - |x+1|$

Find all the points of discontinuity of f defined by $f(x) = |x| - |x+1|$.

Given

The given function is $f(x) = |x| - |x+1|$.

The two functions, g and h are defined as $g(x) = |x|$ and $h(x) = |x+1|$.

Then, $f = g - h$

The continuity of g and h are examined first.

$$g(x) = |x| \text{ can be written as } g(x) = \begin{cases} -x, & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases}$$

Clearly, g is defined for every real number.

Let c be a real number.

Substitute

Case I:

If $c < 0$, then $g(c) = -c$

$$\begin{aligned}\lim_{x \rightarrow c} g(x) &= \lim_{x \rightarrow c} (-x) = -c \\ \therefore \lim_{x \rightarrow c} g(x) &= g(c)\end{aligned}$$

Therefore, g is continuous at all points x , such that $x < 0$.

Case II:

If $c > 0$, then $g(c) = c$

$$\begin{aligned}\lim_{x \rightarrow c} g(x) &= \lim_{x \rightarrow c} (x) = c \\ \therefore \lim_{x \rightarrow c} g(x) &= g(c)\end{aligned}$$

Therefore, g is continuous at all points x , such that $x > 0$.

Case III:

If $c = 0$, then $g(c) = g(0) = 0$

$$\begin{aligned}\lim_{x \rightarrow 0^-} g(x) &= \lim_{x \rightarrow 0^-} (-x) = 0 \\ \lim_{x \rightarrow 0^+} g(x) &= \lim_{x \rightarrow 0^+} (x) = 0 \\ \therefore \lim_{x \rightarrow 0^-} g(x) &= \lim_{x \rightarrow 0^+} g(x) = g(0)\end{aligned}$$

Therefore, g is continuous at all $x = 0$.

From the above three observations, it can be concluded that g is continuous at all points.

$$h(x) = |x+1| \text{ can be written as } h(x) = \begin{cases} -(x+1), & \text{if } x < -1 \\ x+1, & \text{if } x \geq -1 \end{cases}$$

Clearly, h is defined for every real number.

Let c be a real number.

Case I:

If $c < -1$, then $h(c) = -(c+1)$

$$\lim_{x \rightarrow c} h(x) = \lim_{x \rightarrow c} [-(x+1)] = -(c+1)$$
$$\therefore \lim_{x \rightarrow c} h(x) = h(c)$$

Therefore, h is continuous at all points x , such that $x < -1$.

Case II:

If $c > -1$, then $h(c) = c+1$

$$\lim_{x \rightarrow c} h(x) = \lim_{x \rightarrow c} (x+1) = c+1$$
$$\therefore \lim_{x \rightarrow c} h(x) = h(c)$$

Therefore, h is continuous at all points x , such that $x > -1$.

Conclusion

Case III:

If $c = -1$, then $h(c) = h(-1) = -1+1 = 0$

$$\lim_{x \rightarrow -1^-} h(x) = \lim_{x \rightarrow -1^-} [-(x+1)] = -(-1+1) = 0$$
$$\lim_{x \rightarrow -1^+} h(x) = \lim_{x \rightarrow -1^+} (x+1) = (-1+1) = 0$$
$$\therefore \lim_{x \rightarrow -1} h(x) = \lim_{x \rightarrow -1^-} h(x) = \lim_{x \rightarrow -1^+} h(x) = h(-1)$$

Therefore, h is continuous at $x = -1$.

From the above three observations, it can be concluded that h is continuous at all points.

It concludes that g and h are continuous functions.

Therefore, $f = g \cdot h$ is also a continuous function.

Therefore, f has no point of discontinuity.

Q

1

Derivative of
composite
function: $\sin$
(x^2+5)



$\sin (x^2+5)$

Conclusion

Let $f(x) = \sin (x^2+5)$, $u(x) = x^2+5$ and $v(t) = \sin t$

Then, $(v \circ u)(x) = v(u(x)) = v(x^2+5) = \sin (x^2+5) = f(x)$

Thus, f is a composite of two functions.

Put $t = u(x) = x^2+5$

Then, we get

$$\begin{aligned}\frac{dv}{dt} &= \frac{d}{dt} (\sin t) = \cos t = \cos (x^2+5) \\ \frac{dt}{dx} &= \frac{d}{dx} (x^2+5) = \frac{d}{dx} (x^2) + \frac{d}{dx} (5) = 2x + 0 = 2x\end{aligned}$$

By chain rule of derivative,

$$\frac{df}{dx} = \frac{dv}{dt} \cdot \frac{dt}{dx} = \cos (x^2+5) \times 2x = 2x \cos (x^2+5)$$

Alternate method:

$$\begin{aligned}\frac{d}{dx} [\sin (x^2+5)] &= \cos (x^2+5) \cdot \frac{d}{dx} (x^2+5) \\ &= \cos (x^2+5) \cdot \left[\frac{d}{dx} (x^2) + \frac{d}{dx} (5) \right] \\ &= \cos (x^2+5) \cdot [2x + 0] \\ &= 2x \cos (x^2+5)\end{aligned}$$

Q

2

Derivative of
composite
function: $\cos$
($\sin x$)



$\cos (\sin x)$

Conclusion

Let $f(x) = \cos (\sin x)$, $u(x) = \sin x$ and $v(t) = \cos t$

Then, $(v \circ u)(x) = v(u(x)) = v(\sin x) = \cos (\sin x) = f(x)$

Here, f is a composite function of two functions.

Put $t = u(x) = \sin x$

$$\begin{aligned}\therefore \frac{dv}{dt} &= \frac{d}{dt} [\cos t] = -\sin t = -\sin (\sin x) \\ \frac{dt}{dx} &= \frac{d}{dx} (\sin x) = \cos x\end{aligned}$$

By chain rule,

$$\frac{df}{dx} = \frac{dv}{dt} \cdot \frac{dt}{dx} = -\sin (\sin x) \cdot \cos x = -\cos x \sin (\sin x)$$

Alternate method:

$$\begin{aligned}\frac{d}{dx} [\cos (\sin x)] &= -\sin (\sin x) \cdot \frac{d}{dx} (\sin x) \\ &= -\sin (\sin x) \times \cos x \\ &= -\cos x \sin (\sin x)\end{aligned}$$

Q

3

Derivative of
composite
function: $\sin$
($a x+b$)



$\sin (a x+b)$

Conclusion

Let $f(x) = \sin (a x+b)$, $u(x) = a x+b$ and $v(t) = \sin t$

Then, $(v \circ u)(x) = v(u(x)) = v(a x+b) = \sin (a x+b) = f(x)$

Here, f is a composite function of two functions u and v .

Put, $t = u(x) = a x+b$

Thus,

$$\begin{aligned}\frac{dv}{dt} &= \frac{d}{dt} (\sin t) = \cos t = \cos (a x+b) \\ \frac{dt}{dx} &= \frac{d}{dx} (a x+b) = \frac{d}{dx} (a x) + \frac{d}{dx} (b) = a + 0 = a\end{aligned}$$

Hence, by chain rule, we get

$$\frac{df}{dx} = \frac{dv}{dt} \cdot \frac{dt}{dx} = \cos (a x+b) \cdot a = a \cos (a x+b)$$

Alternate method:

$$\begin{aligned}\frac{d}{dx} [\sin (a x+b)] &= \cos (a x+b) \cdot \frac{d}{dx} (a x+b) \\ &= \cos (a x+b) \cdot \left[\frac{d}{dx} (a x) + \frac{d}{dx} (b) \right] \\ &= \cos (a x+b) \cdot (a+0) \\ &= a \cos (a x+b)\end{aligned}$$

Q

4

Derivative of
composite
function: sec
(tan (sqrt x))



sec (tan (√(x)))

Conclusion

Let $f(x) = \sec (\tan (\sqrt{x}))$, $u(x) = \sqrt{x}$, $v(t) = \tan t$ and $w(s) = \sec s$

Then, $(w \circ v \circ u)(x) = w[v(u(x))] = w[v(\sqrt{x})] = w(\tan \sqrt{x}) = \sec (\tan \sqrt{x}) = f(x)$

Here, f is a composite function of three functions u , v and w .

Put, $s = v(t) = \tan t$ and $t = u(x) = \sqrt{x}$

Then,

$$\begin{aligned}\frac{dw}{ds} &= \frac{d}{ds} (\sec s) \\ &= \sec s \tan s \\ &= \sec (\tan t) \cdot \tan (\tan t) \quad [s = \tan t] \\ &= \sec (\tan \sqrt{x}) \cdot \tan (\tan \sqrt{x}) \quad [t = \sqrt{x}]\end{aligned}$$

Now,

$$\begin{aligned}\frac{ds}{dt} &= \frac{d}{dt} (\tan t) = \sec^2 t = \sec^2 \sqrt{x} \\ \frac{dt}{dx} &= \frac{d}{dx} (\sqrt{x}) = \frac{d}{dx} \left(x^{\frac{1}{2}}\right) = \frac{1}{2} \cdot x^{\frac{1}{2}-1} = \frac{1}{2 \sqrt{x}}\end{aligned}$$

Hence, by chain rule, we get

$$\begin{aligned}\frac{d}{dx} [\sec (\tan \sqrt{x})] &= \frac{dw}{ds} \cdot \frac{ds}{dt} \cdot \frac{dt}{dx} \\ &= \sec (\tan \sqrt{x}) \cdot \tan (\tan \sqrt{x}) \cdot \sec^2 \sqrt{x} \cdot \frac{1}{2 \sqrt{x}} \\ &= \frac{1}{2 \sqrt{x}} \sec^2 \sqrt{x} \sec (\tan \sqrt{x}) \tan (\tan \sqrt{x}) \\ &= \frac{\sec^2 \sqrt{x} \sec (\tan \sqrt{x}) \tan (\tan \sqrt{x})}{2 \sqrt{x}}\end{aligned}$$

Alternate method:

$$\begin{aligned}\frac{d}{dx} [\sec (\tan \sqrt{x})] &= \sec (\tan \sqrt{x}) \cdot \tan (\tan \sqrt{x}) \cdot \frac{d}{dx} (\tan \sqrt{x}) \\ &= \sec (\tan \sqrt{x}) \cdot \tan (\tan \sqrt{x}) \cdot \sec^2 (\sqrt{x}) \cdot \frac{d}{dx} (\sqrt{x}) \\ &= \sec (\tan \sqrt{x}) \cdot \tan (\tan \sqrt{x}) \cdot \sec^2 (\sqrt{x}) \cdot \frac{1}{2 \sqrt{x}} \\ &= \frac{\sec (\tan \sqrt{x}) \cdot \tan (\tan \sqrt{x}) \cdot \sec^2 (\sqrt{x})}{2 \sqrt{x}}\end{aligned}$$

Q

5

Derivative of
composite
function: sin
(a x+b)/cos
(c x+d)



$$\frac{\sin (a x+b)}{\cos (c x+d)}$$

Conclusion

Given, $f(x) = \frac{\sin (a x+b)}{\cos (c x+d)}$, where $g(x) = \sin (a x+b)$ and $h(x) = \cos (c x+d)$

$$\therefore f' = \frac{g' h - g h'}{h^2}$$

Consider $g(x) = \sin (a x+b)$

Let $u(x) = a x+b$, $v(t) = \sin t$

Then $(v \circ u)(x) = v(u(x)) = v(a x+b) = \sin (a x+b) = g(x)$

$\therefore g$ is a composite function of two functions, u and v .

Put, $t = u(x) = a x+b$

$$\begin{aligned}\frac{dv}{dt} &= \frac{d}{dt} (\sin t) = \cos t = \cos (a x+b) \\ \frac{dt}{dx} &= \frac{d}{dx} (a x+b) = \frac{d}{dx} (a x) + \frac{d}{dx} (b) = a+0 = a\end{aligned}$$

Thus, by chain rule, we get

$$g' = \frac{dg}{dx} = \frac{dv}{dt} \cdot \frac{dt}{dx} = \cos(ax+b) \cdot a = a \cos(ax+b)$$

Consider $h(x) = \cos(cx+d)$

Let $p(x) = cx+d$, $q(y) = \cos y$

Then, $(q \circ p)(x) = q(p(x)) = q(cx+d) = \cos(cx+d) = h(x)$

$\therefore h$ is a composite function of two functions, p and q .

Put, $y = p(x) = cx+d$

$$\begin{aligned} \frac{dq}{dy} &= \frac{d}{dy} (\cos y) = -\sin y = -\sin(cx+d) \\ \frac{dy}{dx} &= \frac{d}{dx} (cx+d) = \frac{d}{dx} (cx) + \frac{d}{dx} (d) = c \end{aligned}$$

Using chain rule, we get

$$\begin{aligned} h' &= \frac{dh}{dx} = \frac{dq}{dy} \cdot \frac{dy}{dx} \\ &= -\sin(cx+d) \times c \\ &= -c \sin(cx+d) \end{aligned}$$

Therefore,

$$\begin{aligned} f' &= \frac{a \cos(ax+b) \cdot \cos(cx+d) - \sin(ax+b) \{-c \sin(cx+d)\}}{[\cos(cx+d)]^2} \\ &= \frac{a \cos(ax+b)}{\cos(cx+d)} + c \sin(ax+b) \cdot \frac{\sin(cx+d)}{\cos(cx+d)} \times \frac{1}{\cos(cx+d)} \\ &= a \cos(ax+b) \sec(cx+d) + c \sin(ax+b) \tan(cx+d) \sec(cx+d) \end{aligned}$$

Q

6

Derivative of composite function: $\cos x^3 \cdot \sin^2(x^5)$

(())

$\cos x^3 \cdot \sin^2(x^5)$

Conclusion

Given, $\cos x^3 \cdot \sin^2(x^5)$

$$\begin{aligned} \frac{d}{dx} [\cos x^3 \cdot \sin^2(x^5)] &= \sin^2(x^5) \times \frac{d}{dx} (\cos x^3) + \cos x^3 \times \frac{d}{dx} [\sin^2(x^5)] \\ &= \sin^2(x^5) \times (-\sin x^3) \times \frac{d}{dx} (x^3) + \cos x^3 \times 2 \sin(x^5) \cdot \frac{d}{dx} [\sin x^5] \\ &= -\sin x^3 \sin^2(x^5) \times 3x^2 + 2 \sin x^5 \cos x^3 \cdot \cos x^5 \times \frac{d}{dx} (x^5) \\ &= -3x^2 \sin x^3 \cdot \sin^2(x^5) + 2 \sin x^5 \cos x^5 \cos x^3 \times 5x^4 \\ &= 10x^4 \sin x^5 \cos x^5 \cos x^3 - 3x^2 \sin x^3 \sin^2(x^5) \end{aligned}$$

Q

7

Derivative of composite function: $2\sqrt{\cot(x^2)}$

☆

$2\sqrt{\cot(x^2)}$

Conclusion

$$\begin{aligned} \frac{d}{dx} [2\sqrt{\cot(x^2)}] &= 2 \cdot \frac{1}{2\sqrt{\cot(x^2)}} \times \frac{d}{dx} [\cot(x^2)] \\ &= \sqrt{\left(\frac{\sin(x^2)}{\cos(x^2)}\right)} \times -\operatorname{cosec}^2(x^2) \times \frac{d}{dx} (x^2) \\ &= \sqrt{\left(\frac{\sin(x^2)}{\cos(x^2)}\right)} \times \frac{-1}{\sin^2(x^2)} \times (2x) \\ &= \frac{-2x}{\sin x^2 \sqrt{\cos x^2 \sin x^2}} \\ &= \frac{-2\sqrt{2}x}{\sin x^2 \sqrt{2 \sin x^2 \cos x^2}} \\ &= \frac{-2\sqrt{2}x}{\sin x^2 \sqrt{\sin 2x^2}} \end{aligned}$$

Q

8

Derivative of composite function: $\cos(\sqrt{x})$

☆

$\cos(\sqrt{x})$

Conclusion

Let $f(x) = \cos(\sqrt{x})$

Also, let $u(x) = \sqrt{x}$ and, $v(t) = \cos t$

Then,

$$\begin{aligned} (v \circ u)(x) &= v(u(x)) \\ &= v(\sqrt{x}) \\ &= \cos \sqrt{x} \\ &= f(x) \end{aligned}$$

Since, f is a composite function of u and v .

$$t = u(x) = \sqrt[2]{x}$$

Then,

$$\begin{aligned} \frac{d t}{d x} &= \frac{d}{d x} (\sqrt[2]{x}) = \frac{d}{d x} \left(x^{\frac{1}{2}} \right) = \frac{1}{2} x^{\frac{-1}{2}} \\ &= \frac{1}{2 \sqrt[2]{x}} \\ \frac{d v}{d t} &= \frac{d}{d t} (\cos t) = -\sin t \end{aligned}$$

$$\text{And, } = -\sin (\sqrt[2]{x})$$

Using chain rule, we get

$$\begin{aligned} \frac{d t}{d x} &= \frac{d v}{d t} \cdot \frac{d t}{d x} \\ &= -\sin (\sqrt[2]{x}) \cdot \frac{1}{2 \sqrt[2]{x}} \\ &= -\frac{1}{2 \sqrt[2]{x}} \sin (\sqrt[2]{x}) \\ &= -\frac{\sin (\sqrt[2]{x})}{2 \sqrt[2]{x}} \end{aligned}$$

Alternate method:

$$\begin{aligned} \frac{d}{d x} [\cos (\sqrt[2]{x})] &= -\sin (\sqrt[2]{x}) \cdot \frac{d}{d x} (\sqrt[2]{x}) \\ &= -\sin (\sqrt[2]{x}) \times \frac{d}{d x} \left(x^{\frac{1}{2}} \right) \\ &= -\sin \sqrt[2]{x} \times \frac{1}{2} x^{\frac{-1}{2}} \\ &= \frac{-\sin \sqrt[2]{x}}{2 \sqrt[2]{x}} \end{aligned}$$

Q

9

Derivative of composite function:
 $f(x) = |x-1|$, $x \in \mathbf{R}$



Prove that the function f given by

$$f(x) = |x-1|, x \in \mathbf{R}$$

is not differentiable at $x = 1$.

Conclusion

Given, $f(x) = |x-1|$, $x \in \mathbf{R}$

It is known that a function f is differentiable at a point $x = c$ in its domain if both $\lim_{h \rightarrow 0^-} \frac{f(c)-f(c-h)}{h}$ and $\lim_{h \rightarrow 0^+} \frac{f(c+h)-f(c)}{h}$ are finite and equal.

$$\lim_{h \rightarrow 0^-} \frac{f(c)-f(c-h)}{h} \quad \text{and} \quad \lim_{h \rightarrow 0^+} \frac{f(c+h)-f(c)}{h} \quad \text{are finite and equal.}$$

To check the differentiability of the given function at $x = 1$,

Consider LHD at $x = 1$

$$\begin{aligned} \lim_{h \rightarrow 0^-} \frac{f(1)-f(1-h)}{h} &= \lim_{h \rightarrow 0^-} \frac{f|1-1|-|1-h-1|}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{0-|h|}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{-h}{h} \quad (h < 0 \Rightarrow |h| = -h) \\ &= -1 \end{aligned}$$

Consider RHD at $x = 1$

$$\begin{aligned} \lim_{h \rightarrow 0^+} \frac{f(1+h)-f(1)}{h} &= \lim_{h \rightarrow 0^+} \frac{f|1+h-1|-|1-1|}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{|h|-0}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{h}{h} \quad (h > 0 \Rightarrow |h| = h) \\ &= 1 \end{aligned}$$

Since LHD and RHD at $x = 1$ are not equal,

Therefore, f is not differentiable at $x = 1$.

ove that the greatest integer function defined by

$$f(x) = [x], 0 < x < 3$$

is not differentiable at $x = 1$ and $x = 2$.

lusion

Given, $f(x) = [x], 0 < x < 3$

It is known that a function f is differentiable at a point

$$x = c \text{ in its domain if both } \lim_{h \rightarrow 0^-} \frac{f(c) - f(c-h)}{h}$$

$$\text{and } \lim_{h \rightarrow 0^+} \frac{f(c+h) - f(c)}{h} \text{ are finite and equal.}$$

At $x = 1$,

Consider the LHD at $x = 1$

$$\lim_{h \rightarrow 0^-} \frac{f(1) - f(1-h)}{h} = \lim_{h \rightarrow 0^-} \frac{[1] - [1-h]}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{1-0}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{1}{h}$$

$$= \infty$$

Consider RHD at $x = 1$

$$\lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0^+} \frac{[1+h] - [1]}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{1-1}{h}$$

$$= \lim_{h \rightarrow 0^+} 0$$

$$= 0$$

Since LHD and RHD at $x = 1$ are not equal,

Hence, f is not differentiable at $x = 1$.

To check the differentiability of the given function at

$$x = 2,$$

Consider LHD at $x = 2$

$$\lim_{h \rightarrow 0^-} \frac{f(2) - f(2-h)}{h} = \lim_{h \rightarrow 0^-} \frac{[2] - [2-h]}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{2-1}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{1}{h}$$

$$= \infty$$

Now, consider RHD at $x = 2$

$$\lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0^+} \frac{[2+h] - [2]}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{2-2}{h}$$

$$= \lim_{h \rightarrow 0^+} 0$$

$$= 0$$

Since, LHD and RHD at $x = 2$ are not equal.

Hence, f is not differentiable at $x = 2$.

Implicit
derivative: 2
 $x+3 y=\sin x$



$2 x+3 y=\sin x$

Conclusion

Given, $2 x+3 y=\sin x$

Differentiating with respect to x , we
get

$$\begin{aligned}\frac{d}{dy} (2x+3y) &= \frac{d}{dx} (\sin x) \\ \Rightarrow \frac{d}{dx} (2x) + \frac{d}{dx} (3y) &= \cos x \\ \Rightarrow 2 + 3 \frac{dy}{dx} &= \cos x \\ \Rightarrow 3 \frac{dy}{dx} &= \cos x - 2 \\ \therefore \frac{dx}{dy} &= \frac{\cos x - 2}{3}\end{aligned}$$

Q

2

Implicit
derivative: $2x+3y=\sin y$



$2x+3y = \sin y$

Conclusion

Given, $2x+3y = \sin y$

Differentiating with respect to x , we get

$$\begin{aligned}\frac{d}{dx} (2x) + \frac{d}{dx} (3y) &= \frac{d}{dx} (\sin y) \\ \Rightarrow 2 + 3 \frac{dy}{dx} &= \cos y \frac{dy}{dx} \quad [\text{By using chain rule}] \\ \Rightarrow 2 &= (\cos y - 3) \frac{dy}{dx} \\ \therefore \frac{dy}{dx} &= \frac{2}{\cos y - 3}\end{aligned}$$

Q

3

Implicit
derivative: $a x + b y^2 = \cos y$



$a x + b y^2 = \cos y$

Conclusion

Given, $a x + b y^2 = \cos y$

Differentiating with respect to x , we get

$$\begin{aligned}\frac{d}{dx} (a x) + \frac{d}{dx} (b y^2) &= \frac{d}{dx} (\cos y) \\ \Rightarrow a + b \frac{d}{dx} (y^2) &= \frac{d}{dx} (\cos y) \\ \frac{d}{dx} (y^2) = 2 y \frac{dy}{dx} \quad \text{and} \quad \frac{d}{dx} (\cos y) &= -\sin y \frac{dy}{dx}\end{aligned}$$

From (1) and (2), we get

$$\begin{aligned}a + b \times 2 y \frac{dy}{dx} &= -\sin y \frac{dy}{dx} \\ \Rightarrow (2 b y + \sin y) \frac{dy}{dx} &= -a \\ \therefore \frac{dy}{dx} &= \frac{-a}{2 b y + \sin y}\end{aligned}$$

Q

4

Implicit
derivative: $x y + y^2 = \tan x + y$



$x y + y^2 = \tan x + y$

Conclusion

Given, $x y + y^2 = \tan x + y$

Differentiating with respect to x , we get

$$\begin{aligned}\frac{d}{dx} (x y + y^2) &= \frac{d}{dx} (\tan x + y) \\ \Rightarrow \frac{d}{dx} (x y) + \frac{d}{dx} (y^2) &= \frac{d}{dx} (\tan x) + \frac{dy}{dx} \\ \Rightarrow [y \cdot \frac{d}{dx} (x) + x \cdot \frac{dy}{dx}] + 2 y \frac{dy}{dx} &= \sec^2 x + \frac{dy}{dx} \quad [\text{using product rule and chain rule}] \\ \Rightarrow y \cdot 1 + x \frac{dy}{dx} + 2 y \frac{dy}{dx} &= \sec^2 x + \frac{dy}{dx} \Rightarrow (x + 2 y - 1) \frac{dy}{dx} = \sec^2 x - y \\ \therefore \frac{dy}{dx} &= \frac{\sec^2 x - y}{(x + 2 y - 1)}\end{aligned}$$

Q

5

Implicit
derivative: $x^2 + x y + y^2 = 100$



$x^2 + x y + y^2 = 100$

Conclusion

Given, $x^2 + x y + y^2 = 100$

Differentiating with respect to x , we get

$$\begin{aligned}
\frac{d}{dx} (x^2 + x y + y^2) &= \frac{d}{dx} (100) \\
\Rightarrow \frac{d}{dx} (x^2) + \frac{d}{dx} (x y) + \frac{d}{dx} (y^2) &= 0 \\
\Rightarrow 2x + [y \cdot \frac{d}{dx} (x) + x \cdot \frac{dy}{dx}] + 2y \frac{dy}{dx} &= 0 \\
\Rightarrow 2x + y \cdot 1 + x \cdot \frac{dy}{dx} + 2y \frac{dy}{dx} &= 0 \\
\Rightarrow 2x + y + (x + 2y) \frac{dy}{dx} &= 0 \\
\therefore \frac{dy}{dx} &= -\frac{2x + y}{x + 2y}
\end{aligned}$$

Q

6

Implicit derivative:
 $x^3 + x^2 y + x y^2 + y^3 = 81$



$$x^3 + x^2 y + x y^2 + y^3 = 81$$

Conclusion

Given, $x^3 + x^2 y + x y^2 + y^3 = 81$

Differentiating with respect to x, we get

$$\begin{aligned}
\frac{d}{dx} (x^3 + x^2 y + x y^2 + y^3) &= \frac{d}{dx} (81) \\
\Rightarrow \frac{d}{dx} (x^3) + \frac{d}{dx} (x^2 y) + \frac{d}{dx} (x y^2) + \frac{d}{dx} (y^3) &= 0 \\
\Rightarrow 3x^2 + [y \frac{d}{dx} (x^2) + x^2 \frac{dy}{dx}] + [y^2 \frac{d}{dx} (x) + x \frac{d}{dx} (y^2)] + 3y^2 \frac{dy}{dx} &= 0 \\
\Rightarrow 3x^2 + [y \cdot 2x + x^2 \frac{dy}{dx}] + [y^2 \cdot 1 + x \cdot 2y \frac{dy}{dx}] + 3y^2 \frac{dy}{dx} &= 0 \\
\Rightarrow (3x^2 + 2xy + 3y^2) \frac{dy}{dx} + (x^2 + 2xy + y^2) &= 0 \\
\therefore \frac{dy}{dx} &= \frac{-(x^2 + 2xy + y^2)}{(3x^2 + 2xy + 3y^2)}
\end{aligned}$$

Q

7

Implicit derivative:
 $\sin^2 y + \cos x y = \pi$



$$\sin^2 y + \cos x y = \pi$$

Conclusion

Given, $\sin^2 y + \cos x y = \pi$

Differentiating with respect to x, we get

$$\begin{aligned}
\frac{d}{dx} (\sin^2 y + \cos x y) &= \frac{d}{dx} (\pi) \\
\Rightarrow \frac{d}{dx} (\sin^2 y) + \frac{d}{dx} (\cos x y) &= 0
\end{aligned}$$

Using chain rule, we get

$$\begin{aligned}
\frac{d}{dx} (\sin^2 y) &= 2 \sin y \frac{dy}{dx} (\sin y) = 2 \sin y \cos y \frac{dy}{dx} \\
\frac{d}{dx} (\cos x y) &= -\sin x y \frac{d}{dx} (x y) = -\sin x y [y \frac{d}{dx} (x) + x \frac{dy}{dx}] \\
&= -\sin x y [y \cdot 1 + x \frac{dy}{dx}] = -y \sin x y - x \sin x y \frac{dy}{dx}
\end{aligned}$$

From (1), (2) and (3), we get

$$\begin{aligned}
2 \sin y \cos y \frac{dy}{dx} + (-y \sin x y - x \sin x y \frac{dy}{dx}) &= 0 \\
\Rightarrow (2 \sin y \cos y - x \sin x y) \frac{dy}{dx} &= y \sin x y \\
\Rightarrow (\sin 2y - x \sin x y) \frac{dy}{dx} &= y \sin x y \\
\therefore \frac{dx}{dy} &= \frac{y \sin x y}{\sin 2y - x \sin x y}
\end{aligned}$$

Q

8

Implicit derivative:
 $\sin^2 x + \cos^2 y = 1$



$$\sin^2 x + \cos^2 y = 1$$

Conclusion

Given, $\sin^2 x + \cos^2 y = 1$

Differentiating with respect to x, we get

$$\begin{aligned}
 \frac{d}{dx} (\sin^2 x + \cos^2 y) &= \frac{d}{dx} (1) \\
 \Rightarrow \frac{d}{dx} (\sin^2 x) + \frac{d}{dx} (\cos^2 y) &= 0 \\
 \Rightarrow 2 \sin x \cdot \frac{d}{dx} (\sin x) + 2 \cos y \cdot \frac{d}{dx} (\cos y) &= 0 \\
 \Rightarrow 2 \sin x \cos x + 2 \cos y (-\sin y) \cdot \frac{dy}{dx} &= 0 \\
 \Rightarrow \sin 2x - \sin 2y \cdot \frac{dy}{dx} &= 0 \\
 \therefore \frac{dy}{dx} &= \frac{\sin 2x}{\sin 2y}
 \end{aligned}$$

Q

9

Implicit derivative:
 $y = \sin^{-1}(2x/(1+x^2))$



$$y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$$

Conclusion

Given,

$$\begin{aligned}
 y &= \sin^{-1}\left(\frac{2x}{1+x^2}\right) \\
 \Rightarrow \sin y &= \frac{2x}{1+x^2}
 \end{aligned}$$

Differentiating with respect to x, we get

$$\begin{aligned}
 \frac{d}{dx} (\sin y) &= \frac{d}{dx} \left(\frac{2x}{1+x^2}\right) \\
 \Rightarrow \cos y \cdot \frac{dy}{dx} &= \frac{d}{dx} \left(\frac{2x}{1+x^2}\right)
 \end{aligned}$$

The function $\frac{2x}{1+x^2}$, is of the form of $\frac{u}{v}$

By quotient rule, we get

$$\begin{aligned}
 \frac{d}{dx} \left(\frac{2x}{1+x^2}\right) &= \frac{(1+x^2) \frac{d}{dx} (2x) - 2x \cdot \frac{d}{dx} (1+x^2)}{(1+x^2)^2} \\
 &= \frac{(1+x^2) \cdot 2 - 2x \cdot [0+2x]}{(1+x^2)^2} \\
 &= \frac{2+2x^2-4x^2}{(1+x^2)^2} \\
 &= \frac{2(1-x^2)}{(1+x^2)^2}
 \end{aligned}$$

$$\text{Also, } \sin y = \frac{2x}{1+x^2}$$

$$\begin{aligned}
 \cos y &= \sqrt{1 - \sin^2 y} = \sqrt{1 - \left(\frac{2x}{1+x^2}\right)^2} \\
 &= \sqrt{\frac{(1+x^2)^2 - 4x^2}{(1+x^2)^2}} \\
 &= \sqrt{\frac{(1-x^2)^2}{(1+x^2)^2}} \\
 &= \frac{1-x^2}{1+x^2}
 \end{aligned}$$

From (1), (2) and (3), we get

$$\begin{aligned}
 \frac{1-x^2}{1+x^2} \times \frac{dy}{dx} &= \frac{2(1-x^2)}{(1+x^2)^2} \\
 \Rightarrow \frac{dy}{dx} &= \frac{2}{1+x^2}
 \end{aligned}$$

Q

10

Implicit derivative:
 $y = \tan^{-1}(3x - x^3)$



$$y = \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right), -\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$$

Conclusion

Given,

$$\begin{aligned}
 y &= \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right) \\
 \Rightarrow \tan y &= \left(\frac{3x-x^3}{1-3x^2}\right)
 \end{aligned}$$

Since, we know that

$$\Rightarrow \tan y = \left(\frac{3 \tan \frac{y}{3} - \tan^3 \frac{y}{3}}{1 - 3 \tan^2 \frac{y}{3}} \right)$$

Comparing (1) and (2) we get,

$$x = \tan \frac{y}{3}$$

Differentiating with respect to x, we get

$$\begin{aligned} \frac{d}{dx} (x) &= \frac{d}{dx} \left(\tan \frac{y}{3} \right) \\ \Rightarrow 1 &= \sec^2 \frac{y}{3} \cdot \frac{d}{dx} \left(\frac{y}{3} \right) \\ \Rightarrow 1 &= \sec^2 \frac{y}{3} \cdot \frac{1}{3} \cdot \frac{dy}{dx} \\ \Rightarrow \frac{dy}{dx} &= \frac{3}{\sec^2 \frac{y}{3}} = \frac{3}{1 + \tan^2 \frac{y}{3}} \\ \therefore \frac{dy}{dx} &= \frac{3}{1 + x^2} \end{aligned}$$

Q

11

Implicit derivative:
 $y = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right), 0 < x < 1$



$$y = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right), 0 < x < 1$$

Conclusion

Given,

$$y = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$$

$$\Rightarrow \cos y = \left(\frac{1-x^2}{1+x^2} \right)$$

$$\Rightarrow \frac{1 - \tan^2 \frac{y}{2}}{1 + \tan^2 \frac{y}{2}} = \frac{1 - x^2}{1 + x^2}$$

Comparing LHS and RHS, we get

$$\tan \frac{y}{2} = x$$

Differentiating with respect to x, we get

$$\begin{aligned} \sec^2 \frac{y}{2} \cdot \frac{d}{dx} \left(\frac{y}{2} \right) &= \frac{d}{dx} (x) \\ \Rightarrow \sec^2 \frac{y}{2} \times \frac{1}{2} \cdot \frac{dy}{dx} &= 1 \\ \Rightarrow \frac{dy}{dx} &= \frac{2}{\sec^2 \frac{y}{2}} \\ \Rightarrow \frac{dy}{dx} &= \frac{2}{1 + \tan^2 \frac{y}{2}} \\ \therefore \frac{dy}{dx} &= \frac{2}{1 + x^2} \end{aligned}$$

$$y = \sin^{-1}\left(\frac{1-x^2}{1+x^2}\right), 0 < x < 1$$

$$\text{Given, } y = \sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)$$

$$y = \sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)$$

$$\Rightarrow \sin y = \frac{1-x^2}{1+x^2}$$

Differentiating with respect to x, we get

$$\frac{d}{dx}(\sin y) = \frac{d}{dx}\left(\frac{1-x^2}{1+x^2}\right)$$

Using chain rule, we get

$$\begin{aligned} \frac{d}{dx}(\sin y) &= \cos y \cdot \frac{dy}{dx} \\ \cos y &= \sqrt{1 - \sin^2 y} = \sqrt{1 - \left(\frac{1-x^2}{1+x^2}\right)^2} \\ &= \sqrt{\frac{(1+x^2)^2 - (1-x^2)^2}{(1+x^2)^2}} = \sqrt{\frac{4x^2}{(1+x^2)^2}} = \frac{2x}{1+x^2} \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{d}{dx}(\sin y) &= \frac{2x}{1+x^2} \cdot \frac{dy}{dx} \\ \frac{d}{dx}\left(\frac{1-x^2}{1+x^2}\right) &= \frac{(1+x^2) \cdot \frac{d}{dx}(1-x^2) - (1-x^2) \cdot \frac{d}{dx}(1+x^2)}{(1+x^2)^2} \\ &= \frac{(1+x^2)(-2x) - (1-x^2)(2x)}{(1+x^2)^2} \\ &= \frac{-2x - 2x^3 - 2x + 2x^3}{(1+x^2)^2} \\ &= \frac{-4x}{(1+x^2)^2} \end{aligned}$$

From equation (1), (2) and (3), we get

$$\begin{aligned} \frac{2x}{1+x^2} \cdot \frac{dy}{dx} &= \frac{-4x}{(1+x^2)^2} \\ \Rightarrow \frac{dy}{dx} &= \frac{-2}{1+x^2} \end{aligned}$$

$$y = \cos^{-1}\left(\frac{2x}{1+x^2}\right), -1 < x < 1$$

$$\text{Given, } y = \cos^{-1}\left(\frac{2x}{1+x^2}\right)$$

$$y = \cos^{-1}\left(\frac{2x}{1+x^2}\right)$$

$$\cos y = \left(\frac{2x}{1+x^2}\right)$$

Differentiating with respect to x, we get

$$\begin{aligned} \frac{d}{dx}(\cos y) &= \frac{d}{dx}\left(\frac{2x}{1+x^2}\right) \\ \Rightarrow -\sin y \cdot \frac{dy}{dx} &= \frac{(1+x^2) \cdot \frac{d}{dx}(2x) - 2x \cdot \frac{d}{dx}(1+x^2)}{(1+x^2)^2} \\ \Rightarrow -\sqrt{1-\cos^2 y} \cdot \frac{dy}{dx} &= \frac{(1+x^2) \times 2 - 2x \times 2x}{(1+x^2)^2} \\ \Rightarrow \left[\sqrt{1-\left(\frac{2x}{1+x^2}\right)^2}\right] \cdot \frac{dy}{dx} &= -\left[\frac{2(1-x^2)}{(1+x^2)^2}\right] \\ \Rightarrow \sqrt{\frac{(1+x^2)^2 - 4x^2}{(1+x^2)^2}} \cdot \frac{dy}{dx} &= \frac{-2(1-x^2)}{(1+x^2)^2} \\ \Rightarrow \sqrt{\frac{(1-x^2)^2}{(1+x^2)^2}} \cdot \frac{dy}{dx} &= \frac{-2(1-x^2)}{(1+x^2)^2} \\ \Rightarrow \frac{1-x^2}{1+x^2} \cdot \frac{dy}{dx} &= \frac{-2(1-x^2)}{(1+x^2)^2} \\ \Rightarrow \frac{dy}{dx} &= \frac{-2}{1+x^2} \end{aligned}$$

Q

14

Implicit derivative:
 $y = \sin^{-1}(2x \sqrt{1-x^2})$, ...



$$y = \sin^{-1}(2x \sqrt{1-x^2}), -\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$$

Conclusion

$$\text{Given, } y = \sin^{-1}(2x \sqrt{1-x^2})$$

$$y = \sin^{-1}(2x \sqrt{1-x^2})$$

$$\Rightarrow \sin y = (2x \sqrt{1-x^2})$$

Differentiating with respect to x, we get

$$\begin{aligned} \cos y \cdot \frac{dy}{dx} &= 2\left[x \cdot \frac{d}{dx}(\sqrt{1-x^2}) + \sqrt{1-x^2} \cdot \frac{dx}{dx}\right] \\ \Rightarrow \sqrt{1-\sin^2 y} \cdot \frac{dy}{dx} &= 2\left[\frac{x}{2} \cdot \frac{-2x}{\sqrt{1-x^2}} + \sqrt{1-x^2}\right] \\ \Rightarrow \sqrt{1-(2x \sqrt{1-x^2})^2} \cdot \frac{dy}{dx} &= 2\left[\frac{-x^2 + 1-x^2}{\sqrt{1-x^2}}\right] \\ \Rightarrow \sqrt{1-4x^2(1-x^2)} \cdot \frac{dy}{dx} &= 2\left[\frac{1-2x^2}{\sqrt{1-x^2}}\right] \\ \Rightarrow \sqrt{(1-2x^2)^2} \cdot \frac{dy}{dx} &= 2\left[\frac{1-2x^2}{\sqrt{1-x^2}}\right] \\ \Rightarrow (1-2x^2) \cdot \frac{dy}{dx} &= 2\left[\frac{1-2x^2}{\sqrt{1-x^2}}\right] \\ \Rightarrow \frac{dy}{dx} &= \frac{2}{\sqrt{1-x^2}} \end{aligned}$$

Q

15

Implicit
derivative:
 $y = \sec^{-1}(2x^2 - 1)$, 0



Part 5 · Exercise 5.4

$$y = \sec^{-1}\left(\frac{1}{2x^2 - 1}\right), 0 < x < \frac{1}{\sqrt{2}}$$

Conclusion

$$\text{Given, } y = \sec^{-1}\left(\frac{1}{2x^2 - 1}\right)$$

$$\Rightarrow y = \sec^{-1}\left(\frac{1}{2x^2 - 1}\right)$$

$$\Rightarrow \sec y = \left(\frac{1}{2x^2 - 1}\right)$$

$$\Rightarrow \cos y = 2x^2 - 1$$

$$\Rightarrow 2x^2 = 1 + \cos y$$

$$\Rightarrow 2x^2 = 2 \cos^2 \frac{y}{2}$$

$$\Rightarrow x = \cos \frac{y}{2}$$

Differentiating with respect to x, we
get

$$\frac{d}{dx}(x) = \frac{d}{dx}\left(\cos \frac{y}{2}\right)$$

$$\Rightarrow 1 = \sin \frac{y}{2} \cdot \frac{d}{dx}\left(\frac{y}{2}\right)$$

$$\Rightarrow \frac{-1}{\sin \frac{y}{2}} = \frac{1}{2} \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-2}{\sin \frac{y}{2}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-2}{\sqrt{1 - \cos^2 \frac{y}{2}}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-2}{\sqrt{1 - x^2}}$$

Q

1

Derivative of
exponential/
log function:
 $e^x \sin x$

()

 e^x $\sin x$

Conclusion

$$\text{Let } y = \frac{e^x}{\sin x}$$

By using the quotient rule, we get

$$\begin{aligned} \frac{dy}{dx} &= \frac{\sin x \cdot \frac{d}{dx}(e^x) - e^x \cdot \frac{d}{dx}(\sin x)}{\sin^2 x} \\ &= \frac{\sin x \cdot (e^x) - e^x \cdot (\cos x)}{\sin^2 x} \\ &= \frac{e^x(\sin x - \cos x)}{\sin^2 x} \end{aligned}$$

Q

2

Derivative of
exponential/
log function:
 $e^{\sin^{-1} x}$

()

 $e^{\sin^{-1} x}$

Conclusion

$$\text{Let } y = e^{\sin^{-1} x}$$

By using the quotient rule,
we get

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx} (e^{\sin^{-1} x}) \\
 &= e^{\sin^{-1} x} \cdot \frac{d}{dx} (\sin^{-1} x) \\
 &= e^{\sin^{-1} x} \cdot \frac{1}{\sqrt{1-x^2}} \\
 &= \frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}} \\
 &= \frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}}, x \in (-1,1)
 \end{aligned}$$

Q

3

Derivative of exponential/log function:
 e^{x^3}



e^{x^3}

Conclusion

Let $y = e^{x^3}$

By using the quotient rule, we get

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx} (e^{x^3}) \\
 &= e^{x^3} \cdot \frac{d}{dx} (x^3) \\
 &= e^{x^3} \cdot 3x^2 \\
 &= 3x^2 e^{x^3}
 \end{aligned}$$

Q

4

Derivative of exponential/log function:
 $\sin(\tan^{-1} e^{-x})$



$\sin(\tan^{-1} e^{-x})$

Conclusion

Let $y = \sin(\tan^{-1} e^{-x})$

By using the chain rule, we get

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx} [\sin(\tan^{-1} e^{-x})] \\
 &= \cos(\tan^{-1} e^{-x}) \cdot \frac{d}{dx} (\tan^{-1} e^{-x}) \\
 &= \cos(\tan^{-1} e^{-x}) \cdot \frac{1}{1+(e^{-x})^2} \cdot \frac{d}{dx} (e^{-x}) \\
 &= \frac{\cos(\tan^{-1} e^{-x})}{1+e^{-2x}} \cdot e^{-x} \cdot \frac{d}{dx} (-x) \\
 &= \frac{e^{-x} \cos(\tan^{-1} e^{-x})}{1+e^{-2x}} \times (-1) \\
 &= \frac{-e^{-x} \cos(\tan^{-1} e^{-x})}{1+e^{-2x}}
 \end{aligned}$$

Q

5

Derivative of exponential/log function:
 $\log(\cos e^x)$



$\log(\cos e^x)$

Conclusion

Let $y = \log(\cos e^x)$

By using the chain rule, we get

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx} [\log(\cos e^x)] \\
 &= \frac{1}{\cos e^x} \cdot \frac{d}{dx} (\cos e^x) \\
 &= \frac{1}{\cos e^x} \cdot (-\sin e^x) \cdot \frac{d}{dx} (e^x) \\
 &= \frac{-\sin e^x}{\cos e^x} \cdot e^x \\
 &= -e^x \tan e^x, e^x \neq (2n+1) \frac{\pi}{2}, n \in \mathbb{N}
 \end{aligned}$$

Q

6

Derivative of exponential/log function:
 $e^x + e^{x^2} + \dots + e^{x^5}$



$e^x + e^{x^2} + \dots + e^{x^5}$

Conclusion

$$\frac{d}{dx} (e^x + e^{x^2} + \dots + e^{x^5})$$

Differentiating wrt x, we get

$$\begin{aligned}
 \frac{d}{dx} (e^x + e^{x^2} + \dots + e^{x^5}) &= \frac{d}{dx} (e^x) + \frac{d}{dx} (e^{x^2}) + \frac{d}{dx} (e^{x^3}) + \frac{d}{dx} (e^{x^4}) + \frac{d}{dx} (e^{x^5}) \\
 &= e^x + [e^{x^2} \times \frac{d}{dx} (x^2)] + [e^{x^3} \times \frac{d}{dx} (x^3)] + [e^{x^4} \times \frac{d}{dx} (x^4)] + [e^{x^5} \times \frac{d}{dx} (x^5)] \\
 &= e^x + (e^{x^2} \times 2x) + (e^{x^3} \times 3x^2) + (e^{x^4} \times 4x^3) + (e^{x^5} \times 5x^4) \\
 &= e^x + 2xe^{x^2} + 3x^2e^{x^3} + 4x^3e^{x^4} + 5x^4e^{x^5}
 \end{aligned}$$

Q

7

Derivative of exponential/log function:
sqrt e^sqrt x,
x>0



$\sqrt{(e^{\sqrt{x}})}, x>0$

Conclusion

Let $y = \sqrt{(e^{\sqrt{x}})}$

Then, $y^2 = e^{\sqrt{x}}$

Differentiating wrt x, we get

$y^2 = e^{\sqrt{x}}$

$$\begin{aligned}
 \frac{d}{dx} (y^2) &= \frac{d}{dx} (e^{\sqrt{x}}) \\
 \Rightarrow 2y \frac{dy}{dx} &= e^{\sqrt{x}} \cdot \frac{d}{dx} (\sqrt{x}) \\
 \Rightarrow 2y \frac{dy}{dx} &= e^{\sqrt{x}} \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{x}} \\
 \Rightarrow \frac{dy}{dx} &= \frac{e^{\sqrt{x}}}{4y\sqrt{x}} \\
 \Rightarrow \frac{dy}{dx} &= \frac{e^{\sqrt{x}}}{4\sqrt{(e^{\sqrt{x}})}\sqrt{x}} \\
 \Rightarrow \frac{dy}{dx} &= \frac{e^{\sqrt{x}}}{4\sqrt{x}e^{\sqrt{x}}}, x>0
 \end{aligned}$$

Q

8

Derivative of exponential/log function:
log (log x),
x>1



$\log (\log x), x>1$

Conclusion

Let $y = \log (\log x)$

By using the chain rule, we get

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx} [\log (\log x)] \\
 &= \frac{1}{\log x} \cdot \frac{d}{dx} (\log x) \\
 &= \frac{1}{\log x} \cdot \frac{1}{x} \\
 &= \frac{1}{x \log x}, x>1
 \end{aligned}$$

Q

9

Derivative of exponential/log function:
cos x/log x,
x>0



$\frac{\cos x}{\log x}, x>0$

Conclusion

Let $y = \frac{\cos x}{\log x}$

By using the quotient rule, we get

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{\frac{d}{dx} (\cos x) \cdot \log x - \cos x \cdot \frac{d}{dx} (\log x)}{(\log x)^2} \\
 &= \frac{-\sin x \log x - \cos x \cdot \frac{1}{x}}{(\log x)^2} \\
 &= -\left[\frac{x \log x \cdot \sin x + \cos x}{x(\log x)^2} \right], x>0
 \end{aligned}$$

Q

10

Derivative of
exponential/
log function:
 $\cos(\log x + e^x)$, $x > 0$



$\cos(\log x + e^x)$, $x > 0$

Part 6 ·
Conclusion
Exercise 5.5

Let $y = \cos(\log x + e^x)$

By using the chain rule, we get

$$\begin{aligned}\frac{dy}{dx} &= -\sin[\log x + e^x] \cdot \frac{d}{dx}(\log x + e^x) \\ &= -\sin(\log x + e^x) \cdot \left[\frac{d}{dx}(\log x) + \frac{d}{dx}(e^x) \right] \\ &= -\sin(\log x + e^x) \cdot \left(\frac{1}{x} + e^x \right) \\ &= -\left(\frac{1}{x} + e^x \right) \sin(\log x + e^x), x > 0\end{aligned}$$

Q

1

Logarithmic
differentiation
n: $\cos x \cdot \cos 2x \cdot \cos 3x$



$\cos x \cdot \cos 2x \cdot \cos 3x$

Conclusion

Let $y = \cos x \cdot \cos 2x \cdot \cos 3x$

Taking logarithm on both the sides, we get

$$\begin{aligned}\log y &= \log(\cos x \cdot \cos 2x \cdot \cos 3x) \\ \Rightarrow \log y &= \log(\cos x) + \log(\cos 2x) + \log(\cos 3x)\end{aligned}$$

Differentiating both sides with respect to x , we get

$$\begin{aligned}\frac{1}{y} \cdot \frac{dy}{dx} &= \frac{1}{\cos x} \cdot \frac{d}{dx}(\cos x) + \frac{1}{\cos 2x} \cdot \frac{d}{dx}(\cos 2x) + \frac{1}{\cos 3x} \cdot \frac{d}{dx}(\cos 3x) \\ \Rightarrow \frac{dy}{dx} &= y \left[-\frac{\sin x}{\cos x} - \frac{\sin 2x}{\cos 2x} - \frac{\sin 3x}{\cos 3x} \right] \\ \therefore \frac{dy}{dx} &= -\cos x \cdot \cos 2x \cdot \cos 3x [\tan x + 2 \tan 2x + 3 \tan 3x]\end{aligned}$$

Q

2

Logarithmic
differentiation
n: $\sqrt{\frac{(x-1)(x-2)}{(x-3)(x-4)(x-5)}}$



$\sqrt{\frac{(x-1)(x-2)}{(x-3)(x-4)(x-5)}}$

Conclusion

Let $y = \sqrt{\frac{(x-1)(x-2)}{(x-3)(x-4)(x-5)}}$

Taking logarithm on both the sides, we get

$$\begin{aligned}\log y &= \log \sqrt{\frac{(x-1)(x-2)}{(x-3)(x-4)(x-5)}} \\ \Rightarrow \log y &= \frac{1}{2} \log \left[\frac{(x-1)(x-2)}{(x-3)(x-4)(x-5)} \right] \\ \Rightarrow \log y &= \frac{1}{2} [\log \{(x-1)(x-2)\} - \log \{(x-3)(x-4)(x-5)\}] \\ \Rightarrow \log y &= \frac{1}{2} [\log(x-1) + \log(x-2) - \log(x-3) - \log(x-4) - \log(x-5)]\end{aligned}$$

Differentiating both sides with respect to x , we get

$$\begin{aligned}\frac{1}{y} \cdot \frac{dy}{dx} &= \frac{1}{2} \left[\frac{1}{x-1} \cdot \frac{d}{dx}(x-1) + \frac{1}{x-2} \cdot \frac{d}{dx}(x-2) - \frac{1}{x-3} \cdot \frac{d}{dx}(x-3) - \frac{1}{x-4} \cdot \frac{d}{dx}(x-4) - \frac{1}{x-5} \cdot \frac{d}{dx}(x-5) \right] \\ \Rightarrow \frac{dy}{dx} &= \frac{y}{2} \left[\frac{1}{x-1} + \frac{1}{x-2} - \frac{1}{x-3} - \frac{1}{x-4} - \frac{1}{x-5} \right] \\ \therefore \frac{dy}{dx} &= \frac{1}{2} \sqrt{\frac{(x-1)(x-2)}{(x-3)(x-4)(x-5)}} \left[\frac{1}{x-1} + \frac{1}{x-2} - \frac{1}{x-3} - \frac{1}{x-4} - \frac{1}{x-5} \right]\end{aligned}$$

Q

3

Logarithmic
differentiation
n: $(\log x)^{\cos x}$



$(\log x)^{\cos x}$

Conclusion

Let $y = (\log x)^{\cos x}$

Taking logarithm on both the sides, we get

$$\log y = \cos x \cdot \log(\log x)$$

Differentiating both sides with respect to x , we get

$$\begin{aligned}\frac{1}{y} \cdot \frac{dy}{dx} &= \frac{d}{dx}(\cos x \cdot \log(\log x)) + \cos x \cdot \frac{d}{dx}[\log(\log x)] \\ \Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} &= -\sin x \log(\log x) + \cos x \cdot \frac{1}{\log x} \cdot \frac{d}{dx}(\log x) \\ \Rightarrow \frac{dy}{dx} &= y \left[-\sin x \log(\log x) + \frac{\cos x}{\log x} \cdot \frac{1}{x} \right] \\ \therefore \frac{dy}{dx} &= (\log x)^{\cos x} \left[\frac{\cos x}{x \log x} - \sin x \log(\log x) \right]\end{aligned}$$

Q

4

Logarithmic
differentiation
n: $x^x \cdot 2^{\sin x}$
x



$x^x \cdot 2^{\sin x}$

Conclusion

Let $y = x^x \cdot 2^{\sin x}$

Also, let $x^x = u$ and $2^{\sin x} = v$

$$\begin{aligned} \therefore y &= u \cdot v \\ \Rightarrow \frac{d y}{d x} &= \frac{d u}{d x} \cdot \frac{d v}{d x} \\ u &= x^x \end{aligned}$$

Taking logarithm on both the sides, we get

$$\log u = x \log x$$

Differentiating both sides with respect to x, we get

$$\begin{aligned} \frac{1}{u} \cdot \frac{d u}{d x} &= \left[\frac{d}{d x} (x) \times \log x + x \times \frac{d}{d x} (\log x) \right] \\ \Rightarrow \frac{d u}{d x} &= u \left[1 \times \log x + x \times \frac{1}{x} \right] \\ \Rightarrow \frac{d u}{d x} &= x^x (\log x + 1) \\ \Rightarrow \frac{d u}{d x} &= x^x (1 + \log x) \\ v &= 2^{\sin x} \end{aligned}$$

Taking logarithm on both the sides, we get

$$\log v = \sin x \cdot \log 2$$

Differentiating both sides with respect to x, we get

$$\begin{aligned} \frac{1}{v} \cdot \frac{d v}{d x} &= \log 2 \cdot \frac{d}{d x} (\sin x) \\ \Rightarrow \frac{d v}{d x} &= v \log 2 \cos x \\ \Rightarrow \frac{d v}{d x} &= 2^{\sin x} \cos x \log 2 \\ \therefore \frac{d y}{d x} &= x^x (1 + \log x) \cdot 2^{\sin x} \cos x \log 2 \end{aligned}$$

Q

5

Logarithmic
differentiation
n: $(x+3)^2 \cdot (x+4)^3 \cdot (x+5)^4$
(x+3)^2 *
(x+4)^3 *
(x+5)^4...



$(x+3)^2 \cdot (x+4)^3 \cdot (x+5)^4$

Conclusion

Let $y = (x+3)^2 \cdot (x+4)^3 \cdot (x+5)^4$

Taking logarithm on both the sides, we get

$$\begin{aligned} \log y &= \log (x+3)^2 + \log (x+4)^3 + \log (x+5)^4 \\ \Rightarrow \log y &= 2 \log (x+3) + 3 \log (x+4) + 4 \log (x+5) \end{aligned}$$

Differentiating both sides with respect to x, we get

$$\begin{aligned} \frac{1}{y} \cdot \frac{d y}{d x} &= 2 \cdot \frac{1}{x+3} \cdot \frac{d}{d x} (x+3) + 3 \cdot \frac{1}{x+4} \cdot \frac{d}{d x} (x+4) + 4 \cdot \frac{1}{x+5} \cdot \frac{d}{d x} (x+5) \\ \Rightarrow \frac{d y}{d x} &= y \left[\frac{2}{x+3} + \frac{3}{x+4} + \frac{4}{x+5} \right] \\ \Rightarrow \frac{d y}{d x} &= (x+3)^2 (x+4)^3 (x+5)^4 \cdot \left[\frac{2}{x+3} + \frac{3}{x+4} + \frac{4}{x+5} \right] \\ \Rightarrow \frac{d y}{d x} &= (x+3)^2 (x+4)^3 (x+5)^4 \cdot \left[\frac{2(x+4)(x+5) + 3(x+3)(x+5) + 4(x+3)(x+4)}{(x+3)(x+4)(x+5)} \right] \\ \Rightarrow \frac{d y}{d x} &= (x+3)(x+4)^2 (x+5)^3 \cdot \left[\frac{2(x^2+9x+20) + 3(x^2+8x+15) + 4(x^2+7x+12)}{(x+3)(x+4)(x+5)} \right] \\ \therefore \frac{d y}{d x} &= (x+3)(x+4)^2 (x+5)^3 (9x^2 + 70x + 133) \end{aligned}$$

Q

6

Logarithmic
differentiation
n: $(x+1/x)^{x+x^{1+1/x}}$
(1+1/x)



$(x + \frac{1}{x})^{x+x^{(1+\frac{1}{x})}}$

Conclusion

Let $y = (x + \frac{1}{x})^{x+x^{(1+\frac{1}{x})}}$

Also, let $u = (x + \frac{1}{x})^x$ and $v = x^{(1+\frac{1}{x})}$

$$\begin{aligned} \therefore y &= u \cdot v \\ \Rightarrow \frac{d y}{d x} &= \frac{d u}{d x} \cdot \frac{d v}{d x} \end{aligned}$$

Then, $u = (x + \frac{1}{x})^x$

Taking logarithm on both the sides, we get

$$\Rightarrow \log u = \log (x + \frac{1}{x})^x$$

$$\Rightarrow \log u = x \log (x + \frac{1}{x})$$

Differentiating both sides with respect to x, we get

$$\begin{aligned} \frac{1}{u} \cdot \frac{d u}{d x} &= \frac{d}{d x} (x) \times \log (x + \frac{1}{x}) + x \times \frac{d}{d x} [\log (x + \frac{1}{x})] \\ \Rightarrow \frac{1}{u} \cdot \frac{d u}{d x} &= 1 \times \log (x + \frac{1}{x}) + x \times \frac{1}{(x + \frac{1}{x})} \cdot \frac{d}{d x} (x + \frac{1}{x}) \\ \Rightarrow \frac{d u}{d x} &= u [\log (x + \frac{1}{x}) + \frac{x}{(x + \frac{1}{x})} \times (1 - \frac{1}{x^2})] \\ \Rightarrow \frac{d u}{d x} &= (x + \frac{1}{x})^x [\log (x + \frac{1}{x}) + \frac{(x - \frac{1}{x})}{(x + \frac{1}{x})}] \\ \Rightarrow \frac{d u}{d x} &= (x + \frac{1}{x})^x [\log (x + \frac{1}{x}) + \frac{x^2 - 1}{x^2 + 1}] \\ \Rightarrow \frac{d u}{d x} &= (x + \frac{1}{x})^x [\frac{x^2 - 1}{x^2 + 1} + \log (x + \frac{1}{x})] \end{aligned}$$

Now, $v = x^{(1 + \frac{1}{x})}$

Taking logarithm on both the sides, we get

$$\Rightarrow \log v = \log [x^{(1 + \frac{1}{x})}]$$

$$\Rightarrow \log v = (1 + \frac{1}{x}) \log x$$

Differentiating both sides with respect to x, we get

$$\begin{aligned} \frac{1}{v} \cdot \frac{d v}{d x} &= [\frac{d}{d x} (1 + \frac{1}{x})] \times \log x + (1 + \frac{1}{x}) \cdot \frac{d}{d x} \log x \\ \Rightarrow \frac{1}{v} \cdot \frac{d v}{d x} &= (-\frac{1}{x^2}) \log x + (1 + \frac{1}{x}) \cdot \frac{1}{x} \\ \Rightarrow \frac{1}{v} \cdot \frac{d v}{d x} &= -\frac{\log x}{x^2} + \frac{1}{x} + \frac{1}{x^2} \\ \Rightarrow \frac{d v}{d x} &= v [\frac{-\log x + x + 1}{x^2}] \\ \Rightarrow \frac{d v}{d x} &= x^{(1 + \frac{1}{x})} [\frac{-\log x + x + 1}{x^2}] \end{aligned}$$

Therefore, from (1), (2) and (3);

$$\frac{d y}{d x} = (x + \frac{1}{x})^x [\frac{x^2 - 1}{x^2 + 1} + \log (x + \frac{1}{x})] + x^{(1 + \frac{1}{x})} [\frac{x + 1 - \log x}{x^2}]$$

Q

7

Logarithmic
differentiation:
 $(\log x)^{x+x^{\log x}}$

(()

$(\log x)^{x+x^{\log x}}$

Conclusion

Let $y = (\log x)^{x+x^{\log x}}$

Also, let $u = (\log x)^x$ and $v = x^{\log x}$

$$\therefore y = u \cdot v$$

$$\Rightarrow \frac{d y}{d x} = \frac{d u}{d x} \cdot v + u \cdot \frac{d v}{d x}$$

Then, $u = (\log x)^x$

Taking logarithm on both the sides, we get

$$\Rightarrow \log u = \log [(\log x)^x]$$

$$\Rightarrow \log u = x \log (\log x)$$

Differentiating both sides with respect to x, we get

$$\begin{aligned}
\frac{1}{u} \cdot \frac{du}{dx} &= \frac{d}{dx} (x) \times \log(\log x) + x \cdot \frac{d}{dx} [\log(\log x)] \\
\Rightarrow \frac{du}{dx} &= u \left[1 \times \log(\log x) + x \cdot \frac{1}{(\log x)} \cdot \frac{d}{dx} (\log x) \right] \\
\Rightarrow \frac{du}{dx} &= (\log x)^x \left[\log(\log x) + \frac{x}{(\log x)} \cdot \frac{1}{x} \right] \\
\Rightarrow \frac{du}{dx} &= (\log x)^x \left[\log(\log x) + \frac{1}{(\log x)} \right] \\
\Rightarrow \frac{du}{dx} &= (\log x)^x \left[\frac{\log(\log x) \cdot \log x + 1}{\log x} \right] \\
\Rightarrow \frac{du}{dx} &= (\log x)^{x-1} [1 + \log x \cdot \log(\log x)] \\
v &= x^{\log x}
\end{aligned}$$

Taking logarithm on both the sides, we get

$$\begin{aligned}
\Rightarrow \log v &= \log (x^{\log x}) \\
\Rightarrow \log v &= \log x \log x = (\log x)^2
\end{aligned}$$

Differentiating both sides with respect to x, we get

$$\begin{aligned}
\frac{1}{v} \cdot \frac{dv}{dx} &= \frac{d}{dx} [(\log x)^2] \\
\Rightarrow \frac{1}{v} \cdot \frac{dv}{dx} &= 2(\log x) \cdot \frac{d}{dx} (\log x) \\
\Rightarrow \frac{dv}{dx} &= 2v(\log x) \cdot \frac{1}{x} \\
\Rightarrow \frac{dv}{dx} &= 2x^{\log x} \cdot \frac{\log x}{x} \\
\Rightarrow \frac{dv}{dx} &= 2x^{\log x-1} \cdot \log x
\end{aligned}$$

Therefore, from (1), (2) and (3);

$$\frac{dy}{dx} = (\log x)^{x-1} [1 + \log x \cdot \log(\log x)] + 2x^{\log x-1} \cdot \log x$$

Q

8

Logarithmic
differentiation:
 $(\sin x)^x + \sin^{-1} \sqrt{x}$



$(\sin x)^x + \sin^{-1} \sqrt{x}$

Conclusion

Let $y = (\sin x)^x + \sin^{-1} \sqrt{x}$

Also, let $u = (\sin x)^x$ and $v = \sin^{-1} \sqrt{x}$

$$\begin{aligned}
\therefore y &= u + v \\
\Rightarrow \frac{dy}{dx} &= \frac{du}{dx} + \frac{dv}{dx}
\end{aligned}$$

Then, $u = (\sin x)^x$

Taking logarithm on both the sides, we get

$$\begin{aligned}
\Rightarrow \log u &= \log (\sin x)^x \\
\Rightarrow \log u &= x \log (\sin x)
\end{aligned}$$

Differentiating both sides with respect to x, we get

$$\begin{aligned}
\frac{1}{u} \cdot \frac{du}{dx} &= \frac{d}{dx} (x) \times \log(\sin x) + x \cdot \frac{d}{dx} [\log(\sin x)] \\
\Rightarrow \frac{du}{dx} &= u \left[1 \times \log(\sin x) + x \cdot \frac{1}{(\sin x)} \cdot \frac{d}{dx} (\sin x) \right] \\
\Rightarrow \frac{du}{dx} &= (\sin x)^x \left[\log(\sin x) + \frac{x}{(\sin x)} \cdot \cos x \right] \\
\Rightarrow \frac{du}{dx} &= (\sin x)^x [x \cot x + \log \sin x]
\end{aligned}$$

$v = \sin^{-1} \sqrt{x}$

Differentiating both sides with respect to x, we get

$$\begin{aligned}
\frac{dv}{dx} &= \frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot \frac{d}{dx} (\sqrt{x}) \\
\Rightarrow \frac{dv}{dx} &= \frac{1}{\sqrt{1-x}} \cdot \frac{1}{2\sqrt{x}} \\
\Rightarrow \frac{dv}{dx} &= \frac{1}{2\sqrt{x-x^2}}
\end{aligned}$$

Therefore, from (1), (2) and (3);

$$\frac{dy}{dx} = (\sin x)^x [x \cot x + \log \sin x] + \frac{1}{2\sqrt{x-x^2}}$$

Q

9

Logarithmic
differentiation
n: $x^{\sin x} + (\sin x)^{\cos x}$



$$x^{\sin x} + (\sin x)^{\cos x}$$

Conclusion

$$\text{Let } y = x^{\sin x} + (\sin x)^{\cos x}$$

$$\text{Also, let } u = x^{\sin x} \text{ and } v = (\sin x)^{\cos x}$$

$$\therefore y = u + v$$

$$\Rightarrow \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

$$\text{Then, } u = x^{\sin x}$$

Taking logarithm on both the sides, we get

$$\Rightarrow \log u = \log (x^{\sin x})$$

$$\Rightarrow \log u = \sin x \log x$$

Differentiating both sides with respect to x, we get

$$\frac{1}{u} \cdot \frac{du}{dx} = \frac{d}{dx} (\sin x) \cdot \log x + \sin x \cdot \frac{d}{dx} (\log x)$$

$$\Rightarrow \frac{du}{dx} = u [\cos x \log x + \sin x \cdot \frac{1}{x}]$$

$$\Rightarrow \frac{du}{dx} = x^{\sin x} [\cos x \log x + \frac{\sin x}{x}]$$

$$v = (\sin x)^{\cos x}$$

Taking logarithm on both the sides, we get

$$\Rightarrow \log v = \log (\sin x)^{\cos x}$$

$$\Rightarrow \log v = \cos x \log (\sin x)$$

Differentiating both sides with respect to x, we get

$$\frac{1}{v} \cdot \frac{dv}{dx} = \frac{d}{dx} (\cos x) \times \log (\sin x) + \cos x \times \frac{d}{dx} [\log (\sin x)]$$

$$\Rightarrow \frac{dv}{dx} = v [-\sin x \cdot \log (\sin x) + \cos x \cdot \frac{1}{\sin x} \cdot \frac{d}{dx} (\sin x)]$$

$$\Rightarrow \frac{dv}{dx} = (\sin x)^{\cos x} [-\sin x \log (\sin x) + \frac{\cos x}{\sin x} \cos x]$$

$$\Rightarrow \frac{dv}{dx} = (\sin x)^{\cos x} [-\sin x \log (\sin x) + \cot x \cos x]$$

$$\Rightarrow \frac{dv}{dx} = (\sin x)^{\cos x} [\cot x \cos x - \sin x \log (\sin x)]$$

Therefore, from (1), (2) and (3);

$$\frac{dy}{dx} = x^{\sin x} [\cos x \log x + \frac{\sin x}{x}] + (\sin x)^{\cos x} [\cot x \cos x - \sin x \log (\sin x)]$$

Q

10

Logarithmic
differentiation
n: $x^x \cos x + \frac{x^2+1}{x^2-1}$



$$x^x \cos x + \frac{x^2+1}{x^2-1}$$

Conclusion

$$\text{Let } y = x^x \cos x + \frac{x^2+1}{x^2-1}$$

$$\text{Also, let } u = x^x \cos x \text{ and } v = \frac{x^2+1}{x^2-1}$$

$$\therefore y = u + v$$

$$\Rightarrow \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

$$\text{Then, } u = x^x \cos x$$

Taking logarithm on both the sides, we get

$$\Rightarrow \log u = \log (x^x \cos x)$$

$$\Rightarrow \log u = x \cos x \log x$$

Differentiating both sides with respect to x, we get

$$\frac{1}{u} \cdot \frac{du}{dx} = \frac{d}{dx} (x) \cdot \cos x \cdot \log x + x \cdot \frac{d}{dx} (\cos x) \cdot \log x + x \cos x \cdot \frac{d}{dx} (\log x)$$

$$\Rightarrow \frac{du}{dx} = u [1 \cdot \cos x \cdot \log x + x \cdot (-\sin x) \log x + x \cos x \cdot \frac{1}{x}]$$

$$\Rightarrow \frac{du}{dx} = x^x \cos x [\cos x \log x - x \cdot \sin x \log x + \cos x]$$

$$\Rightarrow \frac{du}{dx} = x^x \cos x [\cos x (1 + \log x) - x \cdot \sin x \log x]$$

$$v = \frac{x^2+1}{x^2-1}$$

Taking logarithm on both the sides, we get

$$\Rightarrow \log v = \log (x^2+1) - \log (x^2-1)$$

Differentiating both sides with respect to x, we get

$$\begin{aligned} \frac{1}{v} \cdot \frac{dv}{dx} &= \frac{2x}{x^2+1} - \frac{2x}{x^2-1} \\ \Rightarrow \frac{dv}{dx} &= v \left[\frac{2x(x^2-1) - 2x(x^2+1)}{(x^2+1)(x^2-1)} \right] \\ \Rightarrow \frac{dv}{dx} &= \frac{x^2+1}{x^2-1} \times \left[\frac{-4x}{(x^2+1)(x^2-1)} \right] \\ \Rightarrow \frac{dv}{dx} &= \frac{-4x}{(x^2-1)^2} \end{aligned}$$

Therefore, from (1), (2) and (3);

$$\frac{dy}{dx} = x^x \cos x [\cos x (1 + \log x) - x \cdot \sin x \log x] - \frac{4x}{(x^2-1)^2}$$

Q

11

Logarithmic differentiation

n: $(x \cos x)^x + (x \sin x)^{1/x}$

(())

$$(x \cos x)^x + (x \sin x)^{\frac{1}{x}}$$

Conclusion

$$\text{Let } y = (x \cos x)^x + (x \sin x)^{\frac{1}{x}}$$

$$\text{Also, let } u = (x \cos x)^x \text{ and } v = (x \sin x)^{\frac{1}{x}}$$

$$\begin{aligned} \therefore y &= u + v \\ \Rightarrow \frac{dy}{dx} &= \frac{du}{dx} + \frac{dv}{dx} \end{aligned}$$

Then, $u = (x \cos x)^x$

Taking logarithm on both the sides, we get

$$\begin{aligned} \Rightarrow \log u &= (x \cos x)^x \\ \Rightarrow \log u &= x \log (x \cos x) \\ \Rightarrow \log u &= x [\log x + \log \cos x] \\ \Rightarrow \log u &= x \log x + x \log \cos x \end{aligned}$$

Differentiating both sides with respect to x, we get

$$\begin{aligned} \frac{1}{u} \cdot \frac{du}{dx} &= \frac{d}{dx} (x \log x) + \frac{d}{dx} (x \log \cos x) \\ \Rightarrow \frac{du}{dx} &= u \left\{ \log x \cdot \frac{d}{dx} (x) + x \cdot \frac{d}{dx} (\log x) + \log \cos x \cdot \frac{d}{dx} (x) + x \cdot \frac{d}{dx} (\log \cos x) \right\} \\ \Rightarrow \frac{du}{dx} &= (x \cos x)^x \left[(\log x \cdot 1 + x \cdot \frac{1}{x}) + \{ \log \cos x \cdot 1 + x \cdot \frac{1}{\cos x} \cdot \frac{d}{dx} (\cos x) \} \right] \\ \Rightarrow \frac{du}{dx} &= (x \cos x)^x \left[(\log x + 1) + \{ \log \cos x + \frac{x}{\cos x} \cdot (-\sin x) \} \right] \\ \Rightarrow \frac{du}{dx} &= (x \cos x)^x [(1 + \log x) + (\log \cos x - x \tan x)] \\ \Rightarrow \frac{du}{dx} &= (x \cos x)^x [(1 - x \tan x) + (\log x + \log \cos x)] \\ \Rightarrow \frac{du}{dx} &= (x \cos x)^x [1 - x \tan x + \log (x \cos x)] \end{aligned}$$

$$v = (x \sin x)^{\frac{1}{x}}$$

Taking logarithm on both the sides, we get

$$\begin{aligned} \Rightarrow \log v &= \log (x \sin x)^{\frac{1}{x}} \\ \Rightarrow \log v &= \frac{1}{x} \log (x \sin x) \\ \Rightarrow \log v &= \frac{1}{x} (\log x + \log \sin x) \\ \Rightarrow \log v &= \frac{1}{x} \log x + \frac{1}{x} \log \sin x \end{aligned}$$

Differentiating both sides with respect to x, we get

$$\begin{aligned}
\frac{1}{v} \cdot \frac{dv}{dx} &= \frac{d}{dx} \left(\frac{1}{x} \log x \right) + \frac{d}{dx} \left[\frac{1}{x} \log (\sin x) \right] \\
\Rightarrow \frac{1}{v} \cdot \frac{dv}{dx} &= \left[\log x \cdot \frac{d}{dx} \left(\frac{1}{x} \right) + \frac{1}{x} \cdot \frac{d}{dx} (\log x) \right] + \left[\log (\sin x) \cdot \frac{d}{dx} \left(\frac{1}{x} \right) + \frac{1}{x} \cdot \frac{d}{dx} (\log (\sin x)) \right] \\
\Rightarrow \frac{1}{v} \cdot \frac{dv}{dx} &= \left[\log x \cdot \left(-\frac{1}{x^2} \right) + \frac{1}{x} \cdot \frac{1}{x} \right] + \left[\log (\sin x) \cdot \left(-\frac{1}{x^2} \right) + \frac{1}{x} \cdot \frac{1}{\sin x} \cdot \frac{d}{dx} (\sin x) \right] \\
\Rightarrow \frac{1}{v} \cdot \frac{dv}{dx} &= \frac{1}{x^2} (1 - \log x) + \left[-\frac{\log (\sin x)}{x^2} + \frac{1}{x \sin x} \cdot \cos x \right] \\
\Rightarrow \frac{dv}{dx} &= (x \sin x) \cdot \frac{1}{x} \left[\frac{1 - \log x}{x^2} + \frac{-\log (\sin x) + x \cot x}{x^2} \right] \\
\Rightarrow \frac{dv}{dx} &= (x \sin x) \cdot \frac{1}{x} \left[\frac{1 - \log x - \log (\sin x) + x \cot x}{x^2} \right] \\
\Rightarrow \frac{dv}{dx} &= (x \sin x) \cdot \frac{1}{x} \left[\frac{1 - \log (x \sin x) + x \cot x}{x^2} \right]
\end{aligned}$$

Therefore, from (1), (2) and (3);

$$\frac{dy}{dx} = (x \cos x)^x [1 - x \tan x + \log (x \cos x)] + (x \sin x) \cdot \frac{1}{x} \left[\frac{x \cot x + 1 - \log (x \sin x)}{x^2} \right]$$

Q

12

Logarithmic
differentiation
n: $x^y + y^x = 1$



$$x^y + y^x = 1$$

Conclusion

The given function is $x^y + y^x = 1$

Let, $x^y = u$ and $y^x = v$

$$\therefore u + v = 1$$

$$\Rightarrow \frac{du}{dx} + \frac{dv}{dx} = 0$$

Then, $u = x^y$

Taking logarithm on both the sides, we get

$$\Rightarrow \log u = \log (x^y)$$

$$\Rightarrow \log u = y \log x$$

Differentiating both sides with respect to x , we get

$$\begin{aligned}
\frac{1}{u} \cdot \frac{du}{dx} &= \log x \cdot \frac{dy}{dx} + y \cdot \frac{d}{dx} (\log x) \\
\Rightarrow \frac{du}{dx} &= u \left[\log x \cdot \frac{dy}{dx} + y \cdot \frac{1}{x} \right] \\
\Rightarrow \frac{du}{dx} &= x^y \left[\log x \cdot \frac{dy}{dx} + \frac{y}{x} \right]
\end{aligned}$$

Now, $v = y^x$

Taking logarithm on both the sides, we get

$$\Rightarrow \log v = \log (y^x)$$

$$\Rightarrow \log v = x \log y$$

Differentiating both sides with respect to x , we get

$$\begin{aligned}
\frac{1}{v} \cdot \frac{dv}{dx} &= \log y \cdot \frac{d}{dx} (x) + x \cdot \frac{d}{dx} (\log y) \\
\Rightarrow \frac{dv}{dx} &= v \left[\log y \cdot 1 + x \cdot \frac{1}{y} \cdot \frac{dy}{dx} \right] \\
\Rightarrow \frac{dv}{dx} &= y^x \left[\log y + \frac{x}{y} \cdot \frac{dy}{dx} \right]
\end{aligned}$$

Therefore, from (1), (2) and (3);

$$\begin{aligned}
x^y \left[\log x \cdot \frac{dy}{dx} + \frac{y}{x} \right] + y^x \left[\log y + \frac{x}{y} \cdot \frac{dy}{dx} \right] &= 0 \\
\Rightarrow (x^y \log x + x y^{x-1}) \cdot \frac{dy}{dx} &= -(y x^{y-1} + y^x \log y) \\
\therefore \frac{dy}{dx} &= \frac{-(y x^{y-1} + y^x \log y)}{(x^y \log x + x y^{x-1})}
\end{aligned}$$

Q

13

Logarithmic
differentiation
n: $y^x = x^y$



$$y^x = x^y$$

Conclusion

The given function is $y^x = x^y$

Taking logarithm on both the sides, we get

$$x \log y = y \log x$$

Differentiating both sides with respect to x , we get

$$\begin{aligned}
\log y \cdot \frac{d}{dx}(x) + x \cdot \frac{d}{dx}(\log y) &= \log x \cdot \frac{d}{dx}(y) + y \cdot \frac{d}{dx}(\log x) \\
\Rightarrow \log y \cdot 1 + x \cdot \frac{1}{y} \cdot \frac{dy}{dx} &= \log x \cdot \frac{dy}{dx} + y \cdot \frac{1}{x} \\
\Rightarrow \log y + \frac{x}{y} \cdot \frac{dy}{dx} &= \log x \cdot \frac{dy}{dx} + \frac{y}{x} \\
\Rightarrow \left(\frac{x}{y} - \log x \right) \frac{dy}{dx} &= \frac{y}{x} - \log y \\
\Rightarrow \left(\frac{x-y \log x}{y} \right) \frac{dy}{dx} &= \frac{y-x \log y}{x} \\
\therefore \frac{dy}{dx} &= \frac{y}{x} \left(\frac{y-x \log y}{x-y \log x} \right)
\end{aligned}$$

Q

14

Logarithmic differentiation
n: $(\cos x)^y = (\cos y)^x$



$(\cos x)^y = (\cos y)^x$

Conclusion

The given function is $(\cos x)^y = (\cos y)^x$

Taking logarithm on both the sides, we get

$$y \log \cos x = x \log \cos y$$

Differentiating both sides with respect to x, we get

$$\begin{aligned}
\log \cos x \cdot \frac{dy}{dx} + y \cdot \frac{d}{dx}(\log \cos x) &= \log \cos y \cdot \frac{d}{dx}(x) + x \cdot \frac{d}{dx}(\log \cos y) \\
\Rightarrow \log \cos x \cdot \frac{dy}{dx} + y \cdot \frac{1}{\cos x} \cdot \frac{d}{dx}(\cos x) &= \log \cos y \cdot 1 + x \cdot \frac{1}{\cos y} \cdot \frac{d}{dx}(\cos y) \\
\Rightarrow \log \cos x \cdot \frac{dy}{dx} + \frac{y}{\cos x} \cdot (-\sin x) &= \log \cos y + \frac{x}{\cos y} \cdot (-\sin y) \cdot \frac{dy}{dx} \\
\Rightarrow \log \cos x \cdot \frac{dy}{dx} - y \tan x &= \log \cos y - x \tan y \cdot \frac{dy}{dx} \\
\Rightarrow (\log \cos x + x \tan y) \frac{dy}{dx} &= y \tan x + \log \cos y \\
\therefore \frac{dy}{dx} &= \frac{y \tan x + \log \cos y}{x \tan y + \log \cos x}
\end{aligned}$$

Q

15

Logarithmic differentiation
n: $x y = e^{(x-y)}$

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$x y = e^{(x-y)}$

Conclusion

The given function is $x y = e^{(x-y)}$

Taking logarithm on both the sides, we get

$$\begin{aligned}
\log(x y) &= \log(e^{x-y}) \\
\Rightarrow \log x + \log y &= (x-y) \log e \\
\Rightarrow \log x + \log y &= (x-y) \times 1 \\
\Rightarrow \log x + \log y &= (x-y)
\end{aligned}$$

Differentiating both sides with respect to x, we get

$$\begin{aligned}
\frac{d}{dx}(\log x) + \frac{d}{dx}(\log y) &= \frac{d}{dx}(x) - \frac{d}{dx}(y) \\
\Rightarrow \frac{1}{x} + \frac{1}{y} \cdot \frac{dy}{dx} &= 1 - \frac{dy}{dx} \\
\Rightarrow \left(1 + \frac{1}{y}\right) \frac{dy}{dx} &= 1 - \frac{dy}{dx} \\
\Rightarrow \left(\frac{y+1}{y}\right) \frac{dy}{dx} &= \frac{x-1}{x} \\
\therefore \frac{dy}{dx} &= \frac{y(x-1)}{x(y+1)}
\end{aligned}$$

Q

16

Logarithmic differentiation
n: $f(x) = (1+x)(1+x^2)(1+x^4)(1+x^8)$



Find the derivative of the function given by $f(x) = (1+x)(1+x^2)(1+x^4)(1+x^8)$ and hence find $f'(1)$.

Conclusion

The given function is $f(x) = (1+x)(1+x^2)(1+x^4)(1+x^8)$

Taking logarithm on both the sides, we get

$$\log f(x) = \log(1+x) + \log(1+x^2) + \log(1+x^4) + \log(1+x^8)$$

Differentiating both sides with respect to x, we get

$$\begin{aligned}
\frac{1}{f(x)} \cdot \frac{d}{dx}[f(x)] &= \frac{d}{dx} \log(1+x) + \frac{d}{dx} \log(1+x^2) \\
&+ \frac{d}{dx} \log(1+x^4) + \frac{d}{dx} \log(1+x^8) \quad \&\Rightarrow \\
\frac{1}{f(x)} \cdot f'(x) &= \frac{1}{1+x} \cdot \frac{d}{dx}(1+x) + \frac{1}{1+x^2} \cdot \frac{d}{dx}(1+x^2) \&+
\end{aligned}$$

$$\frac{1}{1+x^4} \cdot \frac{d}{dx} (1+x^4) + \frac{1}{1+x^8} \cdot \frac{d}{dx} (1+x^8) \Rightarrow f'(x) = f(x) \left[\frac{1}{1+x} + \frac{1}{1+x^2} \cdot 2x + \frac{1}{1+x^4} \cdot 4x^3 + \frac{1}{1+x^8} \cdot 8x^7 \right] \Rightarrow f'(x) = (1+x) \left[\frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + \frac{8x^7}{1+x^8} \right]$$

Hence,

$$\begin{aligned} f'(1) &= (1+1)(1+1^2)(1+1^4)(1+1^8) \left[\frac{1}{1+1} + \frac{2(1)}{1+1^2} + \frac{4(1)^3}{1+1^4} + \frac{8(1)^7}{1+1^8} \right] \\ &= 2 \times 2 \times 2 \times 2 \left[\frac{1}{2} + \frac{2}{2} + \frac{4}{2} + \frac{8}{2} \right] \\ &= 16 \left(\frac{15}{2} \right) \\ &= 120 \end{aligned}$$

Q

17

Logarithmic
differentiation
n: $(x^2-5x+8)(x^3+7x+9)$



Differentiate $(x^2-5x+8)(x^3+7x+9)$ in three ways mentioned below: (i) by using product rule (ii) by expanding the product to obtain a single polynomial. (iii) by logarithmic differentiation. Do they all give the same answer?

Part (i)

by using product rule

Conclusion

$$\text{Let } y = (x^2-5x+8)(x^3+7x+9)$$

$$\text{Let } u = (x^2-5x+8) \text{ and } v = x^3+7x+9$$

$$\therefore y = u \cdot v$$

$$\Rightarrow \frac{dy}{dx} = \frac{du}{dx} \cdot v + u \cdot \frac{dv}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx} (x^2-5x+8) \cdot (x^3+7x+9) + (x^2-5x+8) \cdot \frac{d}{dx} (x^3+7x+9)$$

$$\Rightarrow \frac{dy}{dx} = (2x-5)(x^3+7x+9) + (x^2-5x+8)(3x^2+7)$$

$$\Rightarrow \frac{dy}{dx} = 2x(x^3+7x+9) - 5(x^3+7x+9) + x^2(3x^2+7) - 5x(3x^2+7) + 8(3x^2+7)$$

$$\Rightarrow \frac{dy}{dx} = (2x^4+14x^2+18x) - 5x^3-35x-45 + (3x^4+7x^2) - 15x^3-35x+24x^2+56$$

$$\Rightarrow \frac{dy}{dx} = -15x^3-35x+24x^2+56$$

$$\therefore \frac{dy}{dx} = 5x^4-20x^3+45x^2-52x+11$$

Part (ii)

by expanding the product to obtain a single polynomial.

Conclusion

$$y = (x^2-5x+8)(x^3+7x+9)$$

$$= x^2(x^3+7x+9) - 5x(x^3+7x+9) + 8(x^3+7x+9)$$

$$= x^5+7x^3+9x^2-5x^4-35x^2-45x+8x^3+56x+72$$

$$= x^5-5x^4+15x^3-26x^2+11x+72$$

Therefore,

$$\frac{dy}{dx} = \frac{d}{dx} (x^5-5x^4+15x^3-26x^2+11x+72)$$

$$= \frac{d}{dx} (x^5) - 5 \frac{d}{dx} (x^4) + 15 \frac{d}{dx} (x^3) - 26 \frac{d}{dx} (x^2) + 11 \frac{d}{dx} (x) + \frac{d}{dx} (72)$$

$$= 5x^4 - 5(4x^3) + 15(3x^2) - 26(2x) + 11(1) + 0$$

$$= 5x^4 - 20x^3 + 45x^2 - 52x + 11$$

Part (iii)

by logarithmic differentiation.

Conclusion

$$y = (x^2-5x+8)(x^3+7x+9)$$

Taking logarithm on both the sides, we get

$$\log y = \log (x^2-5x+8) + \log (x^3+7x+9)$$

Differentiating both sides with respect to x, we get

$$\begin{aligned}
\frac{1}{y} \cdot \frac{dy}{dx} &= \frac{d}{dx} \log(x^2-5x+8) + \frac{d}{dx} \log(x^3+7x+9) \\
\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} &= \frac{1}{x^2-5x+8} \cdot \frac{d}{dx} (x^2-5x+8) + \frac{1}{x^3+7x+9} \cdot \frac{d}{dx} (x^3+7x+9) \\
\Rightarrow \frac{dy}{dx} &= y \left[\frac{1}{x^2-5x+8} \cdot (2x-5) + \frac{1}{x^3+7x+9} \cdot (3x^2+7) \right] \\
\Rightarrow \frac{dy}{dx} &= (x^2-5x+8)(x^3+7x+9) \left[\frac{2x-5}{x^2-5x+8} + \frac{3x^2+7}{x^3+7x+9} \right] \\
\Rightarrow \frac{dy}{dx} &= (x^2-5x+8)(x^3+7x+9) \left[\frac{(2x-5)(x^3+7x+9) + (3x^2+7)(x^2-5x+8)}{(x^2-5x+8)(x^3+7x+9)} \right] \\
\Rightarrow \frac{dy}{dx} &= 2x(x^3+7x+9) - 5(x^3+7x+9) + 3x^2(x^2-5x+8) + 7(x^2-5x+8) \\
\Rightarrow \frac{dy}{dx} &= 2x^4 + 14x^2 + 18x - 5x^3 - 35x - 45 + 3x^5 - 15x^3 + 24x^2 + 7x^2 - 35x + 56 \\
\Rightarrow \frac{dy}{dx} &= 5x^4 - 20x^3 + 45x^2 - 52x + 11
\end{aligned}$$

Conclusion

From the above three observations, it can be concluded that all the results of $\frac{dy}{dx}$ are same.

Q

18

Logarithmic
differentiation

n: u, v



If u, v and w are functions of x, then show that

Part 7 ·
Exercise 5.6

$$\frac{d}{dx} (u \cdot v \cdot w) = \frac{du}{dx} \cdot v \cdot w + u \cdot \frac{dv}{dx} \cdot w + u \cdot v \cdot \frac{dw}{dx}$$

in two ways - first by repeated application of product rule, second by logarithmic differentiation.

ConclusionLet $y = u \cdot v \cdot w = u \cdot (v \cdot w)$

By applying product rule, we get

$$\begin{aligned} \frac{dy}{dx} &= \frac{du}{dx} \cdot (v \cdot w) + u \cdot \frac{d}{dx} (v \cdot w) \\ \Rightarrow \frac{dy}{dx} &= \frac{du}{dx} \cdot (v \cdot w) + u \left[\frac{dv}{dx} \cdot w + v \cdot \frac{dw}{dx} \right] \text{ (Again applying product rule)} \\ \Rightarrow \frac{dy}{dx} &= \frac{du}{dx} \cdot v \cdot w + u \cdot \frac{dv}{dx} \cdot w + u \cdot v \cdot \frac{dw}{dx} \end{aligned}$$

Taking logarithm on both the sides of the equation $y = u \cdot v \cdot w$, we get $\log y = \log u + \log v + \log w$

Differentiating both sides with respect to x, we get

$$\begin{aligned} \frac{1}{y} \cdot \frac{dy}{dx} &= \frac{d}{dx} (\log u) + \frac{d}{dx} (\log v) + \frac{d}{dx} (\log w) \\ \Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} &= \frac{1}{u} \cdot \frac{du}{dx} + \frac{1}{v} \cdot \frac{dv}{dx} + \frac{1}{w} \cdot \frac{dw}{dx} \\ \Rightarrow \frac{dy}{dx} &= y \left(\frac{1}{u} \cdot \frac{du}{dx} + \frac{1}{v} \cdot \frac{dv}{dx} + \frac{1}{w} \cdot \frac{dw}{dx} \right) \\ \Rightarrow \frac{dy}{dx} &= u \cdot v \cdot w \left(\frac{1}{u} \cdot \frac{du}{dx} + \frac{1}{v} \cdot \frac{dv}{dx} + \frac{1}{w} \cdot \frac{dw}{dx} \right) \\ \therefore \frac{d}{dx} (u \cdot v \cdot w) &= \frac{du}{dx} \cdot v \cdot w + u \cdot \frac{dv}{dx} \cdot w + u \cdot v \cdot \frac{dw}{dx} \end{aligned}$$

Q

1

Parametric
derivative:
 $x = 2a t^2, y = a t^4$

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$$x = 2a t^2, y = a t^4$$

ConclusionGiven, $x = 2a t^2, y = a t^4$

Then,

$$\begin{aligned} \frac{dx}{dt} &= \frac{d}{dt} (2a t^2) = 2a \cdot \frac{d}{dt} (t^2) = 2a \cdot 2t = 4at \\ \frac{dy}{dt} &= \frac{d}{dt} (a t^4) = a \cdot \frac{d}{dt} (t^4) = a \cdot 4 \cdot t^3 = 4at^3 \\ \therefore \frac{dy}{dx} &= \frac{\left(\frac{dy}{dt} \right)}{\left(\frac{dx}{dt} \right)} = \frac{4at^3}{4at} = t^2 \end{aligned}$$

Q

2

Parametric
derivative:
 $x = a \cos \theta, y = b \cos \theta$ 

$$x = a \cos \theta, y = b \cos \theta$$

ConclusionGiven, $x = a \cos \theta, y = b \cos \theta$

Then,

$$\begin{aligned} \frac{dx}{d\theta} &= \frac{d}{d\theta} (a \cos \theta) = a(-\sin \theta) = -a \sin \theta \\ \frac{dy}{d\theta} &= \frac{d}{d\theta} (b \cos \theta) = b(-\sin \theta) = -b \sin \theta \\ \therefore \frac{dy}{dx} &= \frac{\left(\frac{dy}{d\theta} \right)}{\left(\frac{dx}{d\theta} \right)} = \frac{-b \sin \theta}{-a \sin \theta} = \frac{b}{a} \end{aligned}$$

Q

3

Parametric
derivative:
 $x = \sin t, y = \cos 2t$ 

$$x = \sin t, y = \cos 2t$$

ConclusionGiven, $x = \sin t, y = \cos 2t$

$$\text{Then, } \frac{dx}{dt} = \frac{d}{dt} (\sin t) = \cos t$$

$$\frac{dy}{dt} = \frac{d}{dt} (\cos 2t) = -\sin 2t \cdot \frac{d}{dt} (2t) = -2 \sin 2t$$

$$\therefore \frac{dy}{dx} = \frac{\left(\frac{dy}{dt} \right)}{\left(\frac{dx}{dt} \right)} = \frac{-2 \sin 2t}{\cos t} = \frac{-2 \cdot 2 \sin t \cos t}{\cos t} = -4 \sin t$$

Q

4

Parametric
derivative:
 $x=4t, y=4/t$



$$x = 4t, y = \frac{4}{t}$$

Conclusion

Given, $x = 4t, y = \frac{4}{t}$

$$\frac{dx}{dt} = \frac{d}{dt}(4t) = 4$$

$$\frac{dy}{dt} = \frac{d}{dt}\left(\frac{4}{t}\right) = 4 \cdot \frac{d}{dt}\left(\frac{1}{t}\right) = 4 \cdot \left(\frac{-1}{t^2}\right) = \frac{-4}{t^2}$$

$$\therefore \frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)} = \frac{\left(\frac{-4}{t^2}\right)}{4} = \frac{-1}{t^2}$$

Q

5

Parametric
derivative:
 $x = \cos \theta - \cos 2\theta$,
 $y = \sin \theta - \sin 2\theta$

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$$x = \cos \theta - \cos 2\theta, y = \sin \theta - \sin 2\theta$$

Conclusion

Given, $x = \cos \theta - \cos 2\theta, y = \sin \theta - \sin 2\theta$

Then,

$$\frac{dx}{d\theta} = \frac{d}{d\theta}(\cos \theta - \cos 2\theta) = \frac{d}{d\theta}(\cos \theta) - \frac{d}{d\theta}(\cos 2\theta)$$

$$= -\sin \theta - (-2 \sin 2\theta) = 2 \sin 2\theta - \sin \theta$$

$$\frac{dy}{d\theta} = \frac{d}{d\theta}(\sin \theta - \sin 2\theta) = \frac{d}{d\theta}(\sin \theta) - \frac{d}{d\theta}(\sin 2\theta)$$

$$= \cos \theta - 2 \cos 2\theta$$

$$\therefore \frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{\cos \theta - 2 \cos 2\theta}{2 \sin 2\theta - \sin \theta}$$

Q

6

Parametric
derivative:
 $x = a(-\sin \theta)$,
 $y = a(1 + \cos \theta)$



$$x = a(-\sin \theta), y = a(1 + \cos \theta)$$

Conclusion

Given, $x = a(-\sin \theta), y = a(1 + \cos \theta)$

Then, $\frac{dx}{d\theta} = a\left[\frac{d}{d\theta}(-\sin \theta) - \frac{d}{d\theta}(\sin \theta)\right] = a(-\cos \theta)$

$$\frac{dy}{d\theta} = a\left[\frac{d}{d\theta}(1) + \frac{d}{d\theta}(\cos \theta)\right] = a[0 + (-\sin \theta)] = -a \sin \theta$$

$$\therefore \frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{-a \sin \theta}{a(-\cos \theta)} = \frac{-2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \sin^2 \frac{\theta}{2}} = \frac{-\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} = -\cot \frac{\theta}{2}$$

Q

7

Parametric
derivative:
 $x = \sin^3 t$,
 $y = \sqrt{\cos 2t}$



$$x = \sin^3 t, y = \sqrt{\cos 2t}$$

Conclusion

Given, $x = \sin^3 t, y = \sqrt{\cos 2t}$

Then,

$$\begin{aligned}
\frac{dx}{dt} &= \frac{d}{dt} \left[\frac{\sin^3 t}{\sqrt{\cos 2t}} \right] \\
&= \frac{\sqrt{\cos 2t} \cdot \frac{d}{dt} (\sin^3 t) - \sin^3 t \cdot \frac{d}{dt} \sqrt{\cos 2t}}{\cos 2t} \\
&= \frac{\sqrt{\cos 2t} \cdot 3 \sin^2 t \cdot \frac{d}{dt} (\sin t) - \sin^3 t \times \frac{1}{2 \sqrt{\cos 2t}} \cdot \frac{d}{dt} (\cos 2t)}{\cos 2t} \\
&= \frac{3 \sqrt{\cos 2t} \cdot \sin^2 t \cdot \cos t - \frac{\sin^3 t}{2 \sqrt{\cos 2t}} \cdot (-2 \sin 2t)}{\cos 2t} \\
&= \frac{3 \cos 2t \cdot \sin^2 t \cos t + \sin^3 t \cdot \sin 2t}{\cos 2t \sqrt{\cos 2t}} \\
\frac{dy}{dt} &= \frac{d}{dt} \left[\frac{\cos^3 t}{\sqrt{\cos 2t}} \right] \\
&= \frac{\sqrt{\cos 2t} \cdot \frac{d}{dt} (\cos^3 t) - \cos^3 t \cdot \frac{d}{dt} (\sqrt{\cos 2t})}{\cos 2t} \\
&= \frac{\sqrt{\cos 2t} \cdot 3 \cos^2 t \cdot \frac{d}{dt} (\cos t) - \cos^3 t \cdot \frac{1}{2 \sqrt{\cos 2t}} \cdot \frac{d}{dt} (\cos 2t)}{\cos 2t} \\
&= \frac{3 \sqrt{\cos 2t} \cdot \cos^2 t (-\sin t) - \cos^3 t \cdot \frac{1}{\sqrt{\cos 2t}} \cdot (-2 \sin 2t)}{\cos 2t} \\
&= \frac{-3 \cos 2t \cdot \cos^2 t \cdot \sin t + \cos^3 t \cdot \sin 2t}{\cos 2t \sqrt{\cos 2t}} \\
\therefore \frac{dy}{dx} &= \frac{\left(\frac{dy}{dt} \right)}{\left(\frac{dx}{dt} \right)} = \frac{\frac{-3 \cos 2t \cdot \cos^2 t \cdot \sin t + \cos^3 t \sin 2t}{\cos 2t \cdot \sqrt{\cos 2t}}}{\frac{3 \cos 2t \cdot \sin^2 t \cdot \cos t + \sin^3 t \sin 2t}{\cos 2t \cdot \sqrt{\cos 2t}}} \\
&= \frac{-3 \cos 2t \cdot \cos^2 t \cdot \sin t + \cos^3 t \sin 2t}{3 \cos 2t \cdot \sin^2 t \cdot \cos t + \sin^3 t \sin 2t} \\
&= \frac{-3 \cos 2t \cdot \cos^2 t \cdot \sin t + \cos^3 t (2 \sin t \cos t)}{3 \cos 2t \cdot \sin^2 t \cdot \cos t + \sin^3 t (2 \sin t \cos t)} \\
&= \frac{\sin t \cos t [3 \cos 2t \sin t + 2 \sin^3 t]}{\sin t \cos t [3 \cos 2t \sin t + 2 \sin^3 t]} \\
&= \frac{[-3(2 \cos^2 t - 1) \cos t + 2 \cos^3 t]}{[3(1 - 2 \sin^2 t) \sin t + 2 \sin^3 t]} \left[\begin{array}{l} \cos 2t = (2 \cos^2 t - 1) \\ \cos 2t = (1 - 2 \sin^2 t) \end{array} \right] \\
&= \frac{-4 \cos^3 t + 3 \cos t}{3 \sin t - 4 \sin^3 t} \left[\begin{array}{l} \cos 3t = 4 \cos^3 t - 3 \cos t \\ \sin 3t = 3 \sin t - 4 \sin^3 t \end{array} \right] \\
&= \frac{-\cos 3t}{\sin 3t} = -\cot 3t
\end{aligned}$$

Q

8

Parametric
derivative:
 $x = a(\cos t + \log \tan t/2)$
 $y = a \sin t$



$x = a(\cos t + \log \tan \frac{t}{2})$ $y = a \sin t$

Conclusion

Given, $x = a(\cos t + \log \tan \frac{t}{2})$, $y = a \sin t$

Then,

$$\begin{aligned}
 \frac{dx}{dt} &= a \cdot \left[\frac{d}{dt} (\cos t) + \frac{d}{dt} \left(\log \tan \frac{t}{2} \right) \right] \\
 &= a \left[-\sin t + \frac{1}{\tan \frac{t}{2}} \cdot \frac{d}{dt} \left(\tan \frac{t}{2} \right) \right] \\
 &= a \left[-\sin t + \cot \frac{t}{2} \cdot \sec^2 \frac{t}{2} \cdot \frac{d}{dt} \left(\frac{t}{2} \right) \right] \\
 &= a \left[-\sin t + \frac{\cos \frac{t}{2}}{\sin \frac{t}{2}} \times \frac{1}{\cos^2 \frac{t}{2}} \times \frac{1}{2} \right] \\
 &= a \left[-\sin t + \frac{1}{2 \sin \frac{t}{2} \cos \frac{t}{2}} \right] \\
 &= a \left(-\sin t + \frac{1}{\sin t} \right) \\
 &= a \left(\frac{-\sin^2 t + 1}{\sin t} \right) \\
 &= a \left(\frac{\cos^2 t}{\sin t} \right) \\
 \frac{dy}{dt} &= a \frac{d}{dt} (\sin t) = a \cos t
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{\left(\frac{dy}{dt} \right)}{\left(\frac{dx}{dt} \right)} = \frac{a \cos t}{\left(a \frac{\cos^2 t}{\sin t} \right)} = \frac{\sin t}{\cos t} = \tan t
 \end{aligned}$$

Q

9

Parametric derivative:
 $x = a \sec \theta$, $y = b \tan \theta$

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$x = a \sec \theta$, $y = b \tan \theta$

Conclusion

Given, $x = a \sec \theta$, $y = b \tan \theta$

Then,

$$\begin{aligned}
 \frac{dx}{d\theta} &= a \cdot \frac{d}{d\theta} (\sec \theta) = a \sec \theta \tan \theta \\
 \frac{dy}{d\theta} &= b \cdot \frac{d}{d\theta} (\tan \theta) = b \sec^2 \theta
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{\left(\frac{dy}{d\theta} \right)}{\left(\frac{dx}{d\theta} \right)} \\
 &= \frac{b \sec^2 \theta}{a \sec \theta \tan \theta} \\
 &= \frac{b}{a} \sec \theta \cot \theta \\
 &= \frac{b \cos \theta}{a \cos \theta \sin \theta} \\
 &= \frac{b}{a} \times \frac{1}{\sin \theta} \\
 &= \frac{b}{a} \operatorname{cosec} \theta
 \end{aligned}$$

Q

10

Parametric derivative:
 $x = a(\cos \theta + \sin \theta)$, $y = a(\sin \theta - \cos \theta)$

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$x = a(\cos \theta + \sin \theta)$, $y = a(\sin \theta - \cos \theta)$

Conclusion

Given, $x = a(\cos \theta + \sin \theta)$, $y = a(\sin \theta - \cos \theta)$

Then,

$$\begin{aligned}
 \frac{dx}{d\theta} &= a \left[\frac{d}{d\theta} \cos \theta + \frac{d}{d\theta} (\sin \theta) \right] \\
 &= a \left[-\sin \theta + \frac{d}{d\theta} (\sin \theta) + \sin \theta \cdot \frac{d}{d\theta} (\theta) \right] \\
 &= a [-\sin \theta + \cos \theta + \sin \theta] \\
 &= a \cos \theta
 \end{aligned}$$

$$\begin{aligned}\frac{dy}{d\theta} &= a \left[\frac{d}{d\theta} (\sin \theta) - \frac{d}{d\theta} (\theta \cos \theta) \right] = a \left[\cos \theta - \left\{ \theta \frac{d}{d\theta} (\cos \theta) + \cos \theta \cdot \frac{d}{d\theta} (\theta) \right\} \right] \\ &= a [\cos \theta - \theta \sin \theta - \cos \theta] \\ &= -a \theta \sin \theta\end{aligned}$$

Therefore,

$$\begin{aligned}\frac{dy}{dx} &= \frac{\left(\frac{dy}{d\theta} \right)}{\left(\frac{dx}{d\theta} \right)} \\ &= \frac{-a \theta \sin \theta}{a \theta \cos \theta} \\ &= -\tan \theta\end{aligned}$$

Q

11

Parametric
derivative:
 $x = \sqrt{a} \sin t$, $y = \sqrt{a} \cos t$



If $x = \sqrt{a} \sin t$, $y = \sqrt{a} \cos t$, show that $\frac{dy}{dx} = -\frac{y}{x}$

Part 8 •

Exercise 5.7

Conclusion

Given, $x = \sqrt{a} \sin t$ and $y = \sqrt{a} \cos t$

Hence,

$$x = \sqrt{a} \sin t = (\sqrt{a})^{\frac{1}{2}} \sin t = \frac{1}{2} \sqrt{a} \sin^2 t \text{ and}$$

$$y = \sqrt{a} \cos t = (\sqrt{a})^{\frac{1}{2}} \cos t = \frac{1}{2} \sqrt{a} \cos^2 t$$

$$\text{Consider } x = \frac{1}{2} \sqrt{a} \sin^2 t$$

Taking log on both sides, we get

$$\log x = \frac{1}{2} \sin^2 t \log a$$

Therefore,

$$\Rightarrow \frac{1}{x} \cdot \frac{dx}{dt} = \frac{1}{2} \log a \cdot \frac{d}{dt} (\sin^2 t)$$

$$\Rightarrow \frac{dx}{dt} = \frac{x}{2} \log a \cdot \frac{1}{\sqrt{1-t^2}}$$

$$\Rightarrow \frac{dx}{dt} = \frac{x \log a}{2 \sqrt{1-t^2}}$$

$$\text{Now, } y = \frac{1}{2} \sqrt{a} \cos^2 t$$

Taking log on both sides, we get

$$\log y = \frac{1}{2} \cos^2 t \log a$$

Therefore,

$$\Rightarrow \frac{1}{y} \cdot \frac{dy}{dt} = \frac{1}{2} \log a \cdot \frac{d}{dt} (\cos^2 t)$$

$$\Rightarrow \frac{dy}{dt} = \frac{y}{2} \log a \cdot \frac{-1}{\sqrt{1-t^2}}$$

$$\Rightarrow \frac{dy}{dt} = \frac{-y \log a}{2 \sqrt{1-t^2}}$$

Hence,

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{dt} \right)}{\left(\frac{dx}{dt} \right)} = \frac{\left(\frac{-y \log a}{2 \sqrt{1-t^2}} \right)}{\left(\frac{x \log a}{2 \sqrt{1-t^2}} \right)} = -\frac{y}{x}$$

Q

1

Second order
derivative:
 $x^2 + 3x + 2$



$x^2 + 3x + 2$

Conclusion

Consider, $y = x^2 + 3x + 2$

Then,

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} (x^2) + \frac{d}{dx} (3x) + \frac{d}{dx} (2) \\ &= 2x + 3 + 0 \\ &= 2x + 3 \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{d}{dx} (2x + 3) \\ &= \frac{d}{dx} (2x) + \frac{d}{dx} (3) \\ &= 2 + 0 \\ &= 2 \end{aligned}$$

Q

2

Second order
derivative:
 x^{20}

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 x^{20}

Conclusion

Consider, $y = x^{20}$

Then,

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} (x^{20}) \\ &= 20 x^{19}\end{aligned}$$

Therefore,

$$\begin{aligned}\frac{d^2 y}{dx^2} &= \frac{d}{dx} (20 x^{19}) \\ &= 20 \frac{d}{dx} (x^{19}) \\ &= 20 \cdot 19 \cdot x^{18} \\ &= 380 x^{18}\end{aligned}$$

Q

3

Second order
derivative: $x \cdot \cos x$

 $x \cdot \cos x$

Conclusion

Consider, $y = x \cos x$

Then,

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} (x \cdot \cos x) \\ &= \cos x \cdot \frac{d}{dx} (x) + x \cdot \frac{d}{dx} (\cos x) \\ &= \cos x \cdot 1 + x(-\sin x) \\ &= \cos x - x \sin x\end{aligned}$$

Therefore,

$$\begin{aligned}\frac{d^2 y}{dx^2} &= \frac{d}{dx} [\cos x - x \sin x] \\ &= \frac{d}{dx} (\cos x) - \frac{d}{dx} (x \sin x) \\ &= -\sin x - \left[\sin x \cdot \frac{d}{dx} (x) + x \cdot \frac{d}{dx} (\sin x) \right] \\ &= -\sin x - (\sin x + x \cos x) \\ &= -(x \cos x + 2 \sin x)\end{aligned}$$

Q

4

Second order
derivative:
 $\log x$

 $\log x$

Conclusion

Let $y = \log x$

Then,

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} (\log x) \\ &= \frac{1}{x}\end{aligned}$$

Therefore,

$$\begin{aligned}\frac{d^2 y}{dx^2} &= \frac{d}{dx} \left(\frac{1}{x} \right) \\ &= \frac{-1}{x^2}\end{aligned}$$

Q

5

Second order
derivative:
 $x^3 \log x$

 $x^3 \log x$

Conclusion

Let $y = x^3 \log x$

Then,

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} [x^3 \log x] \\ &= \log x \cdot \frac{d}{dx} (x^3) + x^3 \cdot \frac{d}{dx} (\log x) \\ &= \log x \cdot 3 x^2 + x^3 \cdot \frac{1}{x} = \log x \cdot 3 x^2 + x^2 \\ &= x^2 (3 \log x + 1)\end{aligned}$$

Therefore,

$$\begin{aligned}
 \frac{d^2 y}{d x^2} &= \frac{d}{d x} [x^2(1+3 \log x)] \\
 &= (1+3 \log x) \cdot \frac{d}{d x} (x^2) + x^2 \cdot \frac{d}{d x} (1+3 \log x) \\
 &= (1+3 \log x) \cdot 2 x + x^2 \cdot \frac{3}{x} \\
 &= 2 x + 6 \log x + 3 x \\
 &= 5 x + 6 \log x \\
 &= x(5 + 6 \log x)
 \end{aligned}$$

Q

6

Second order
derivative:
 $e^x \sin 5 x$

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$e^x \sin 5 x$

Conclusion

Let $y = e^x \sin 5 x$

Then,

$$\begin{aligned}
 \frac{d y}{d x} &= \frac{d}{d x} (e^x \sin 5 x) \\
 &= \sin 5 x \times \frac{d}{d x} (e^x) + e^x \cdot \frac{d}{d x} (\sin 5 x) \\
 &= \sin 5 x \cdot e^x + e^x \cdot \cos 5 x \cdot \frac{d}{d x} (5 x) \\
 &= e^x \sin 5 x + e^x \cos 5 x \cdot 5 \\
 &= e^x (\sin 5 x + 5 \cos 5 x)
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \frac{d^2 y}{d x^2} &= \frac{d}{d x} [e^x (\sin 5 x + 5 \cos 5 x)] \\
 &= (\sin 5 x + 5 \cos 5 x) \cdot \frac{d}{d x} (e^x) + e^x \cdot \frac{d}{d x} (\sin 5 x + 5 \cos 5 x) \\
 &= (\sin 5 x + 5 \cos 5 x) e^x + e^x [\cos 5 x \cdot \frac{d}{d x} (5 x) + 5(-\sin 5 x) \cdot \frac{d}{d x} (5 x)] \\
 &= e^x (\sin 5 x + 5 \cos 5 x) + e^x (5 \cos 5 x - 25 \sin 5 x) \\
 &= e^x (10 \cos 5 x - 24 \sin 5 x) \\
 &= 2 e^x (5 \cos 5 x - 12 \sin 5 x)
 \end{aligned}$$

Q

7

Second order
derivative:
 $e^{6 x} \cos 3 x$



$e^{6 x} \cos 3 x$

Conclusion

Let $y = e^{6 x} \cos 3 x$

Then,

$$\begin{aligned}
 \frac{d y}{d x} &= \frac{d}{d x} (e^{6 x} \cos 3 x) = \cos 3 x \cdot \frac{d}{d x} (e^{6 x}) + e^{6 x} \cdot \frac{d}{d x} (\cos 3 x) \\
 &= \cos 3 x \cdot e^{6 x} \cdot \frac{d}{d x} (6 x) + e^{6 x} \cdot (-\sin 3 x) \cdot \frac{d}{d x} (3 x) \\
 &= 6 e^{6 x} \cos 3 x - 3 e^{6 x} \sin 3 x
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \frac{d^2 y}{d x^2} &= \frac{d}{d x} (6 e^{6 x} \cos 3 x - 3 e^{6 x} \sin 3 x) = 6 \cdot \frac{d}{d x} (e^{6 x} \cos 3 x) - 3 \cdot \frac{d}{d x} (e^{6 x} \sin 3 x) \\
 &= 6 \cdot [6 e^{6 x} \cos 3 x - 3 e^{6 x} \sin 3 x] - 3 \cdot [\sin 3 x \cdot \frac{d}{d x} (e^{6 x}) + e^{6 x} \cdot \frac{d}{d x} (\sin 3 x)] \quad [\text{using (1)}] \\
 &= 36 e^{6 x} \cos 3 x - 18 e^{6 x} \sin 3 x - 3 [\sin 3 x \cdot e^{6 x} \cdot 6 + e^{6 x} \cdot \cos 3 x \cdot 3] \\
 &= 36 e^{6 x} \cos 3 x - 18 e^{6 x} \sin 3 x - 18 e^{6 x} \sin 3 x - 9 e^{6 x} \cos 3 x \\
 &= 27 e^{6 x} \cos 3 x - 36 e^{6 x} \sin 3 x \\
 &= 9 e^{6 x} (3 \cos 3 x - 4 \sin 3 x)
 \end{aligned}$$

Q

8

Second order
derivative:
 $\tan^{-1} x$



$\tan^{-1} x$

Conclusion

Let $y = \tan^{-1} x$

Then,

$$\begin{aligned}
 \frac{d y}{d x} &= \frac{d}{d x} (\tan^{-1} x) \\
 &= \frac{1}{1+x^2}
 \end{aligned}$$

Therefore,

$$\begin{aligned}\frac{d^2 y}{d x^2} &= \frac{d}{d x} \left(\frac{1}{1+x^2} \right) = \frac{d}{d x} (1+x^2)^{-1} \\ &= (-1) \cdot (1+x^2)^{-2} \cdot \frac{d}{d x} (1+x^2) = \frac{-1}{(1+x^2)^2} \times 2x \\ &= \frac{-2x}{(1+x^2)^2}\end{aligned}$$

Q

9

Second order
derivative:
 $\log(\log x)$



$\log(\log x)$

Conclusion

Consider, $y = \log(\log x)$

Then,

$$\begin{aligned}\frac{d y}{d x} &= \frac{d}{d x} [\log(\log x)] \\ &= \frac{1}{\log x} \cdot \frac{d}{d x} (\log x) \\ &= \frac{1}{\log x} \cdot \frac{1}{x} = (x \log x)^{-1}\end{aligned}$$

Therefore,

$$\begin{aligned}\frac{d^2 y}{d x^2} &= \frac{d}{d x} [(x \log x)^{-1}] \\ &= (-1) \cdot (x \log x)^{-2} \cdot \frac{d}{d x} (x \log x) \\ &= \frac{-1}{(x \log x)^2} \cdot \left[\log x \cdot \frac{d}{d x} (x) + x \cdot \frac{d}{d x} (\log x) \right] \\ &= \frac{-1}{(x \log x)^2} \cdot \left[\log x \cdot 1 + x \cdot \frac{1}{x} \right] \\ &= \frac{-(1+\log x)}{(x \log x)^2}\end{aligned}$$

Q

10

Second order
derivative:
 $\sin(\log x)$

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$\sin(\log x)$

Conclusion

Let $y = \sin(\log x)$

Then,

$$\begin{aligned}\frac{d y}{d x} &= \frac{d}{d x} [\sin(\log x)] \\ &= \cos(\log x) \cdot \frac{d}{d x} (\log x) \\ &= \frac{\cos(\log x)}{x}\end{aligned}$$

Therefore,

$$\begin{aligned}\frac{d^2 y}{d x^2} &= \frac{d}{d x} \left[\frac{\cos(\log x)}{x} \right] \\ &= \frac{x \cdot \frac{d}{d x} [\cos(\log x)] - \cos(\log x) \cdot \frac{d}{d x} (x)}{x^2} \\ \frac{d^2 y}{d x^2} &= \frac{x[-\sin(\log x) \cdot \frac{d}{d x} (\log x)] - \cos(\log x) \cdot 1}{x^2} \\ &= \frac{-x \sin(\log x) \cdot \frac{1}{x} - \cos(\log x)}{x^2} \\ &= \frac{-[\sin(\log x) + \cos(\log x)]}{x^2}\end{aligned}$$

Q

11

Second order
derivative:
 $y = 5 \cos x - 3 \sin x$



If $y = 5 \cos x - 3 \sin x$, prove that $\frac{d^2 y}{d x^2} + y = 0$

Conclusion

Given, $y = 5 \cos x - 3 \sin x$

Then,

$$\begin{aligned}\frac{d y}{d x} &= \frac{d}{d x} (5 \cos x) - \frac{d}{d x} (3 \sin x) \\ &= 5 \frac{d}{d x} (\cos x) - 3 \frac{d}{d x} (\sin x) \\ &= 5(-\sin x) - 3 \cos x \\ &= -(5 \sin x + 3 \cos x)\end{aligned}$$

Therefore,

$$\begin{aligned}\frac{d^2 y}{d x^2} &= \frac{d}{d x} [-(5 \sin x + 3 \cos x)] \\ &= -[5 \cdot \frac{d}{d x} (\sin x) + 3 \cdot \frac{d}{d x} (\cos x)] \\ &= -[5 \cos x + 3(-\sin x)] \\ &= -[5 \cos x - 3 \sin x] \\ &= -y\end{aligned}$$

$$\text{Thus, } \frac{d^2 y}{d x^2} + y = 0$$

Hence proved.

Q

12

Second order
derivative:
 $y = \cos^{-1} x$



If $y = \cos^{-1} x$, Find $\frac{d^2 y}{d x^2}$ in terms of y alone.

Conclusion

Given, $y = \cos^{-1} x$

Then,

$$\begin{aligned}\frac{d y}{d x} &= \frac{d}{d x} (\cos^{-1} x) \\ &= \frac{-1}{\sqrt{1-x^2}} \\ &= -(1-x^2)^{-\frac{1}{2}}\end{aligned}$$

Therefore,

$$\begin{aligned}\frac{d^2 y}{d x^2} &= \frac{d}{d x} [-(1-x^2)^{-\frac{1}{2}}] \\ &= -(-\frac{1}{2})(1-x^2)^{-\frac{3}{2}} \cdot \frac{d}{d x} (1-x^2) \\ &= \frac{1}{2 \sqrt{(1-x^2)^3}} \times (-2x) \\ \frac{d^2 y}{d x^2} &= \frac{-x}{\sqrt{(1-x^2)^3}}\end{aligned}$$

But we need to calculate $\frac{d^2 y}{d x^2}$ in terms of y

$$\begin{aligned}\Rightarrow y &= \cos^{-1} x \\ \Rightarrow x &= \cos y\end{aligned}$$

Putting $x = \cos y$ in equation (1), we get

$$\begin{aligned}\frac{d^2 y}{d x^2} &= \frac{-\cos y}{\sqrt{(1-\cos^2 y)^3}} \\ &= \frac{-\cos y}{\sqrt{(\sin^2 y)^3}} \\ &= \frac{-\cos y}{\sin^3 y} \\ &= \frac{-\cos y}{\sin y} \times \frac{1}{\sin^2 y} \\ &= -\cot y \cdot \operatorname{cosec}^2 y\end{aligned}$$

Q

13

Second order
derivative:
 $y = 3 \cos (\log x) + 4 \sin (\log x)$



If $y = 3 \cos (\log x) + 4 \sin (\log x)$, show that $x^2 y_2 + x y_1 + y = 0$

Conclusion

Given, $y = 3 \cos (\log x) + 4 \sin (\log x)$

Then,

$$\begin{aligned}y_1 &= 3 \cdot \frac{d}{d x} [\cos (\log x)] + 4 \cdot \frac{d}{d x} [\sin (\log x)] \\ &= 3 \cdot [-\sin (\log x) \cdot \frac{d}{d x} (\log x)] + 4 \cdot [\cos (\log x) \cdot \frac{d}{d x} (\log x)] \\ &= \frac{-3 \sin (\log x)}{x} + \frac{4 \cos (\log x)}{x} \\ &= \frac{4 \cos (\log x) - 3 \sin (\log x)}{x}\end{aligned}$$

Therefore,

$$\begin{aligned}
y_2 &= \frac{d}{dx} \left(\frac{4 \cos(\log x) - 3 \sin(\log x)}{x} \right) \\
&= \frac{x \cdot \{4 \cos(\log x) - 3 \sin(\log x)\}' - \{4 \cos(\log x) - 3 \sin(\log x)\} \cdot x}{x^2} \\
&= \frac{x \cdot [4\{\cos(\log x)\}' - \{3 \sin(\log x)\}'] - \{4 \cos(\log x) - 3 \sin(\log x)\} \cdot 1}{x^2} \\
&= \frac{x \cdot [-4 \sin(\log x) \cdot (\log x)' - 3 \cos(\log x) \cdot (\log x)'] - 4 \cos(\log x) + 3 \sin(\log x)}{x^2} \\
&= \frac{x \cdot [-4 \sin(\log x) \cdot \frac{1}{x} - 3 \cos(\log x) \cdot \frac{1}{x}] - 4 \cos(\log x) + 3 \sin(\log x)}{x^2} \\
&= \frac{-4 \sin(\log x) - 3 \cos(\log x) - 4 \cos(\log x) + 3 \sin(\log x)}{x^2} \\
&= \frac{-\sin(\log x) - 7 \cos(\log x)}{x^2}
\end{aligned}$$

Thus,

$$\begin{aligned}
x^2 y_2 + x y_1 + y &= \left[x^2 \left(\frac{-\sin(\log x) - 7 \cos(\log x)}{x^2} \right) + x \left(\frac{4 \cos(\log x) - 3 \sin(\log x)}{x} + 3 \cos(\log x) + 4 \sin(\log x) \right) \right] \\
&= \left[-\sin(\log x) - 7 \cos(\log x) + 4 \cos(\log x) - 3 \sin(\log x) + 3 \cos(\log x) + 4 \sin(\log x) \right] \\
&= 0
\end{aligned}$$

Hence proved.

Q

14

Second order
derivative:
 $y = A e^{mx} + B e^{nx}$

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If $y = A e^{mx} + B e^{nx}$, show that $\frac{d^2 y}{dx^2} - (m+n) \frac{dy}{dx} + m n y = 0$

Conclusion

Given, $y = A e^{mx} + B e^{nx}$

Then,

$$\begin{aligned}
\frac{dy}{dx} &= A \cdot \frac{d}{dx} (e^{mx}) + B \cdot \frac{d}{dx} (e^{nx}) \\
&= A \cdot e^{mx} \cdot \frac{d}{dx} (mx) + B \cdot e^{nx} \cdot \frac{d}{dx} (nx) \\
&= A m e^{mx} + B n e^{nx}
\end{aligned}$$

Therefore,

$$\begin{aligned}
\frac{d^2 y}{dx^2} &= \frac{d}{dx} (A m e^{mx} + B n e^{nx}) \\
&= A m \cdot \frac{d}{dx} (e^{mx}) + B n \cdot \frac{d}{dx} (e^{nx}) \\
&= A m \cdot e^{mx} \cdot \frac{d}{dx} (mx) + B n \cdot e^{nx} \cdot \frac{d}{dx} (nx) \\
&= A m^2 e^{mx} + B n^2 e^{nx}
\end{aligned}$$

Thus,

$$\begin{aligned}
\frac{d^2 y}{dx^2} - (m+n) \frac{dy}{dx} + m n y &= A m^2 e^{mx} + B n^2 e^{nx} - (m+n) \cdot (A m e^{mx} + B n e^{nx}) + m n (A e^{mx} + B e^{nx}) \\
&= A m^2 e^{mx} + B n^2 e^{nx} - A m^2 e^{mx} - B m n e^{nx} - A m n e^{mx} - B n^2 e^{nx} + A m n e^{mx} + B m n e^{nx} \\
&= 0
\end{aligned}$$

Hence proved.

Q

15

Second order
derivative:
 $y = 500 e^{7x} + 600 e^{-7x}$

(())

If $y = 500 e^{7x} + 600 e^{-7x}$, show that $\frac{d^2 y}{dx^2} = 49 y$

Conclusion

Given, $y = 500 e^{7x} + 600 e^{-7x}$

Then,

$$\begin{aligned}
\frac{dy}{dx} &= 500 \cdot \frac{d}{dx} (e^{7x}) + 600 \cdot \frac{d}{dx} (e^{-7x}) \\
&= 500 \cdot e^{7x} \cdot \frac{d}{dx} (7x) + 600 \cdot e^{-7x} \cdot \frac{d}{dx} (-7x) \\
&= 3500 e^{7x} - 4200 e^{-7x}
\end{aligned}$$

Therefore,

$$\begin{aligned}
 \frac{d^2 y}{d x^2} &= 3500 e^{7 x} \cdot \frac{d}{d x} (e^{7 x}) - 4200 \cdot \frac{d}{d x} (e^{-7 x}) \\
 &= 3500 \cdot e^{7 x} \cdot \frac{d}{d x} (7 x) - 4200 \cdot e^{-7 x} \cdot \frac{d}{d x} (-7 x) \\
 &= 7 \times 3500 \cdot e^{7 x} + 7 \times 4200 \cdot e^{-7 x} \\
 &= 49 \times 500 \cdot e^{7 x} + 49 \times 600 e^{-7 x} \\
 &= 49(500 e^{7 x} + 600 e^{-7 x}) \\
 &= 49 y
 \end{aligned}$$

Hence proved.

Q

16

Second order
derivative:
 $e^y(x+1)=1$



If $e^y(x+1)=1$, show that $\frac{d^2 y}{d x^2} = \left(\frac{d y}{d x} \right)^2$

Conclusion

Given, $e^y(x+1) = 1$

$$\Rightarrow e^y(x+1) = 1$$

$$\Rightarrow e^y = \frac{1}{x+1}$$

Taking log on both sides, we get

$$y = \log \frac{1}{(x+1)}$$

Differentiating with respect to x, we get

$$\begin{aligned}
 \frac{d y}{d x} &= (x+1) \frac{d}{d x} \left(\frac{1}{x+1} \right) \\
 &= (x+1) \cdot \frac{-1}{(x+1)^2} \\
 &= \frac{-1}{x+1}
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \frac{d^2 y}{d x^2} &= \frac{d}{d x} \left(\frac{-1}{x+1} \right) = - \left(\frac{-1}{(x+1)^2} \right) \\
 &= \frac{1}{(x+1)^2} = \left(\frac{-1}{x+1} \right)^2 \\
 &= \left(\frac{d y}{d x} \right)^2
 \end{aligned}$$

Hence proved.

Q

17

Second order
derivative:
 $y = (\tan^{-1} x)^2$



If $y = (\tan^{-1} x)^2$, show that $(x^2+1)^2 y_2 + 2$

$x(x^2+1) y_1 = 2$

Miscellaneous

Exercise

Conclusion

Given, $y = (\tan^{-1} x)^2$

Then,

$$\Rightarrow y_1 = 2 \tan^{-1} x \cdot \frac{d}{dx} (\tan^{-1} x)$$

$$\Rightarrow y_1 = 2 \tan^{-1} x \cdot \left(\frac{1}{1+x^2} \right)$$

$$\Rightarrow (1+x^2) y_1 = 2 \tan^{-1} x$$

Again, differentiating with respect to x , we get

$$\Rightarrow (1+x^2) y_2 + 2 x y_1 = 2 \left(\frac{1}{1+x^2} \right)$$

$$\Rightarrow (1+x^2)^2 y_2 + 2 x(1+x^2) y_1 = 2$$

Hence proved.

Q

1

Misc: $(3x^2-9x+5)^9$

(())

$(3x^2-9x+5)^9$

Conclusion

Let $y = (3x^2-9x+5)^9$

Using chain rule, we get

$$\frac{dy}{dx} = \frac{d}{dx} (3x^2-9x+5)^9$$

$$= 9(3x^2-9x+5)^8 \cdot \frac{d}{dx} (3x^2-9x+5)$$

$$= 9(3x^2-9x+5)^8 \cdot (6x-9)$$

$$= 9(3x^2-9x+5)^8 \cdot 3(2x-3)$$

$$= 27(3x^2-9x+5)^8(2x-3)$$

Q

2

Misc: $\sin^3 x + \cos^6 x$



$\sin^3 x + \cos^6 x$

Conclusion

Let $y = \sin^3 x + \cos^6 x$

Using chain rule, we get

$$\frac{dy}{dx} = \frac{d}{dx} (\sin^3 x) + \frac{d}{dx} (\cos^6 x)$$

$$= 3 \sin^2 x \cdot \frac{d}{dx} (\sin x) + 6 \cos^5 x \cdot \frac{d}{dx} (\cos x)$$

$$= 3 \sin^2 x \cdot \cos x + 6 \cos^5 x \cdot (-\sin x)$$

$$= 3 \sin x \cos x (\sin x - 2 \cos^4 x)$$

Q

3

Misc: $(5x)^3 \cos 2x$



$(5x)^3 \cos 2x$

Conclusion

Let $y = (5x)^3 \cos 2x$

Taking logarithm on both the sides, we get

$$\log y = 3 \cos 2x \log 5x$$

Differentiating both sides with respect to x , we get

$$\frac{1}{y} \cdot \frac{dy}{dx} = 3[\log 5x \cdot \frac{d}{dx} (\cos 2x) + \cos 2x \cdot \frac{d}{dx} (\log 5x)]$$

$$\frac{dy}{dx} = 3y[\log 5x \cdot (-\sin 2x) \cdot \frac{d}{dx} (2x) + \cos 2x \cdot \frac{1}{5x} \cdot \frac{d}{dx} (5x)]$$

$$= 3y[-2 \sin 2x \cdot \log 5x + \frac{\cos 2x}{x}]$$

$$= y[\frac{3 \cos 2x}{x} - 6 \sin 2x \log 5x]$$

$$= (5x)^3 \cos 2x [\frac{3 \cos 2x}{x} - 6 \sin 2x \log 5x]$$

Q

4

Misc: $\sin^{-1}(x \sqrt{x})$, $0 \leq x \leq 1$



$\sin^{-1}(x \sqrt{x})$, $0 \leq x \leq 1$

Conclusion

Let $y = \sin^{-1}(x \sqrt{x})$

Using chain rule, we get

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \sin^{-1}(x \sqrt{x}) \\ &= \frac{1}{\sqrt{1-(x \sqrt{x})^2}} \times \frac{d}{dx} (x \sqrt{x}) \\ &= \frac{1}{\sqrt{1-x^3}} \cdot \frac{d}{dx} \left(x \frac{3}{2} \right) \\ &= \frac{1}{\sqrt{1-x^3}} \cdot \frac{3}{2} \cdot x^{\frac{1}{2}} \\ &= \frac{3 \sqrt{x}}{2 \sqrt{1-x^3}} \\ &= \frac{3}{2} \sqrt{\frac{x}{1-x^3}} \end{aligned}$$

$$\frac{\cos^{-1} \frac{x}{2}}{\sqrt{2x+7}}, -2 < x < 2$$

$$\text{Let } y = \frac{\cos^{-1} \frac{x}{2}}{\sqrt{2x+7}}$$

Using quotient rule, we get

$$\begin{aligned} & \frac{\sqrt{2x+7} \cdot \frac{d}{dx} \left(\cos^{-1} \frac{x}{2} \right) - \left(\cos^{-1} \frac{x}{2} \right) \cdot \frac{d}{dx} (\sqrt{2x+7})}{(\sqrt{2x+7})^2} \\ &= \frac{\left[\frac{-1}{\sqrt{1-\left(\frac{x}{2}\right)^2}} \cdot \frac{d}{dx} \left(\frac{x}{2} \right) - \left(\cos^{-1} \frac{x}{2} \right) \cdot \frac{1}{2 \sqrt{2x+7}} \cdot \frac{d}{dx} (2x+7) \right]}{\frac{2x+7}{2}} \\ &= \frac{\left[\frac{-1}{\sqrt{4-x^2}} - \left(\cos^{-1} \frac{x}{2} \right) \cdot \frac{2}{2 \sqrt{2x+7}} \right]}{\frac{2x+7}{2}} \\ &= \frac{1}{\sqrt{2x+7}} + \frac{\cos^{-1} \frac{x}{2}}{(2x+7) \frac{3}{2}} \end{aligned}$$

Q

6

Misc: $\cot^{-1}[\sqrt{1+\sin x} + \sqrt{1-\sin x}]$, $0 < x < \frac{\pi}{2}$



$\cot^{-1} \left[\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right]$, $0 < x < \frac{\pi}{2}$

Conclusion

$$y = \cot^{-1} \left[\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right]$$

Then,

$$\begin{aligned}
& \frac{\sqrt{(1+\sin x)} + \sqrt{(1-\sin x)}}{\sqrt{(1+\sin x)} - \sqrt{(1-\sin x)}} = \frac{(\sqrt{(1+\sin x)} + \sqrt{(1-\sin x)})^2}{(\sqrt{(1+\sin x)} - \sqrt{(1-\sin x))}(\sqrt{(1+\sin x)} + \sqrt{(1-\sin x)})} \\
& = \frac{(1+\sin x) + (1-\sin x) + 2\sqrt{(1+\sin x)(1-\sin x)}}{(1+\sin x) - (1-\sin x)} \\
& = \frac{2 + 2\sqrt{(1-\sin^2 x)}}{2\sin x} = \frac{1 + \cos x}{\sin x} \\
& = \frac{1 + 2\cos^2 \frac{x}{2} - 1}{2\sin \frac{x}{2} \cos \frac{x}{2}} = \frac{2\cos^2 \frac{x}{2}}{2\sin \frac{x}{2} \cos \frac{x}{2}} \\
& = \cot \frac{x}{2}
\end{aligned}$$

Therefore, equation (1) becomes,

$$\begin{aligned}
y &= \cot^{-1} \left(\cot \frac{x}{2} \right) \\
\Rightarrow y &= \frac{x}{2}
\end{aligned}$$

Thus,

$$\begin{aligned}
\Rightarrow \frac{dy}{dx} &= \frac{1}{2} \cdot \frac{d}{dx} (x) \\
&= \frac{1}{2}
\end{aligned}$$

Q

7

Misc: $(\log x)^{\log x}$, $x > 1$



$(\log x)^{\log x}$, $x > 1$

Conclusion

Let $y = (\log x)^{\log x}$

Taking logarithm on both the sides, we get

$$\log y = \log x \cdot \log (\log x)$$

Differentiating both sides with respect to x , we get

$$\begin{aligned}
\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} &= \frac{d}{dx} [\log x \cdot \log (\log x)] \\
\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} &= \log (\log x) \cdot \frac{d}{dx} (\log x) + \log x \cdot \frac{d}{dx} [\log (\log x)] \\
\Rightarrow \frac{dy}{dx} &= y \left[\log (\log x) \cdot \frac{1}{x} + \log x \cdot \frac{1}{\log x} \cdot \frac{d}{dx} (\log x) \right] \\
\Rightarrow \frac{dy}{dx} &= y \left[\frac{1}{x} \cdot \log (\log x) + \frac{1}{x} \right] \\
\Rightarrow \frac{dy}{dx} &= (\log x)^{\log x} \left[\frac{1}{x} + \frac{\log (\log x)}{x} \right]
\end{aligned}$$

Q

8

Misc: $\cos (a \cos x + b \sin x)$



$\cos (a \cos x + b \sin x)$, for some constant a and b .

Conclusion

Let $y = \cos (a \cos x + b \sin x)$

Using chain rule, we get

$$\begin{aligned}
\frac{dy}{dx} &= \frac{d}{dx} \cos (a \cos x + b \sin x) \\
&= -\sin (a \cos x + b \sin x) \cdot \frac{d}{dx} (a \cos x + b \sin x) \\
&= -\sin (a \cos x + b \sin x) \cdot [a(-\sin x) + b \cos x] \\
&= (a \sin x - b \cos x) \cdot \sin (a \cos x + b \sin x)
\end{aligned}$$

Q

9

Misc: $(\sin x - \cos x)^{(\sin x - \cos x)}$

(()

$(\sin x - \cos x)^{(\sin x - \cos x)}$, $\frac{\pi}{4} < x < \frac{3\pi}{4}$

Conclusion

Let $y = (\sin x - \cos x)^{(\sin x - \cos x)}$

Taking log on both the sides, we get

$$\begin{aligned}
\log y &= \log [(\sin x - \cos x)^{(\sin x - \cos x)}] \\
&= (\sin x - \cos x) \log (\sin x - \cos x)
\end{aligned}$$

Differentiating both sides with respect to x , we get

$$\begin{aligned}
\frac{1}{y} \cdot \frac{dy}{dx} &= \frac{d}{dx} [(\sin x - \cos x) \log(\sin x - \cos x)] \\
\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} &= \log(\sin x - \cos x) \cdot \frac{d}{dx} (\sin x - \cos x) + (\sin x - \cos x) \cdot \frac{d}{dx} \log(\sin x - \cos x) \\
\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} &= \log(\sin x - \cos x) \cdot (\cos x + \sin x) + (\sin x - \cos x) \cdot \frac{1}{(\sin x - \cos x)} \cdot \frac{d}{dx} (\sin x - \cos x) \\
\Rightarrow \frac{dy}{dx} &= (\sin x - \cos x)^{(\sin x - \cos x)} [(\cos x + \sin x) \cdot \log(\sin x - \cos x) + (\cos x + \sin x)] \\
\Rightarrow \frac{dy}{dx} &= (\sin x - \cos x)^{(\sin x - \cos x)} (\cos x + \sin x) [1 + \log(\sin x - \cos x)]
\end{aligned}$$

Q

10

Misc:
 $x^x + x^a + a^x + a^a$



$x^x + x^a + a^x + a^a$, for some fixed $a > 0$ and $x > 0$

Conclusion

Let $y = x^x + x^a + a^x + a^a$

Also, let $x^x = u$, $x^a = v$, $a^x = w$ and $a^a = s$

Therefore,

$$\begin{aligned}
\Rightarrow y &= u + v + w + s \\
\Rightarrow \frac{dy}{dx} &= \frac{du}{dx} + \frac{dv}{dx} + \frac{dw}{dx} + \frac{ds}{dx}
\end{aligned}$$

Now, $u = x^x$

Taking logarithm on both the sides, we get

$$\begin{aligned}
\Rightarrow \log u &= \log x^x \\
\Rightarrow \log u &= x \log x
\end{aligned}$$

Differentiating both sides with respect to x , we get

$$\begin{aligned}
\frac{1}{u} \cdot \frac{du}{dx} &= \log x \cdot \frac{d}{dx} (x) + x \cdot \frac{d}{dx} (\log x) \\
\frac{du}{dx} &= u \left[\log x \cdot 1 + x \cdot \frac{1}{x} \right] \\
&= x^x [\log x + 1] = x^x (1 + \log x)
\end{aligned}$$

Now, $v = x^a$

Hence,

$$\begin{aligned}
\frac{dv}{dx} &= \frac{d}{dx} (x^a) \\
&= a x^{a-1}
\end{aligned}$$

Now, $w = a^x$

Taking logarithm on both the sides, we get

$$\begin{aligned}
\Rightarrow \log w &= \log a^x \\
\Rightarrow \log w &= x \log a
\end{aligned}$$

Differentiating both sides with respect to x , we get

$$\begin{aligned}
\frac{1}{w} \cdot \frac{dw}{dx} &= \log a \cdot \frac{d}{dx} (x) \\
\frac{dw}{dx} &= w \log a \\
&= a^x \log a
\end{aligned}$$

Now, $s = a^a$

Since a is constant, a^a is also a constant.

Hence,

$$\frac{ds}{dx} = 0$$

From (1), (2), (3), (4) and (5), we get

$$\begin{aligned}
\frac{dy}{dx} &= x^x (1 + \log x) + a x^{a-1} + a^x \log a + 0 \\
&= x^x (1 + \log x) + a x^{a-1} + a^x \log a
\end{aligned}$$

Q

11

Misc: $x^{x^2-3} + (x-3)^{x^2}$



$x^{x^2-3} + (x-3)^{x^2}$, for $x > 3$

Conclusion

Let $y = x^{x^2-3} + (x-3)^{x^2}$

Also, let $u = x^{x^2-3}$ and $v = (x-3)^{x^2}$

Therefore,

$$y = u + v$$

$$\Rightarrow \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

$$\text{Now, } u = x^{x^2-3}$$

Taking logarithm on both the sides, we get

$$\log u = \log (x^{x^2-3})$$

$$= (x^2-3) \log x$$

Differentiating both sides with respect to x, we get

$$\frac{1}{u} \cdot \frac{du}{dx} = \log x \cdot \frac{d}{dx} (x^2-3) + (x^2-3) \cdot \frac{d}{dx} (\log x)$$

$$\Rightarrow \frac{1}{u} \cdot \frac{du}{dx} = \log x \cdot 2x + (x^2-3) \cdot \frac{1}{x}$$

$$\Rightarrow \frac{du}{dx} = x^{x^2-3} \left[\frac{x^2-3}{x} + 2x \log x \right]$$

$$\text{Now, } v = (x-3)^{x^2}$$

Taking logarithm on both the sides, we get

$$\log v = \log (x-3)^{x^2}$$

$$= x^2 \log (x-3)$$

Differentiating both sides with respect to x, we get

$$\frac{1}{v} \cdot \frac{dv}{dx} = \log (x-3) \cdot \frac{d}{dx} (x^2) + (x^2) \cdot \frac{d}{dx} [\log (x-3)]$$

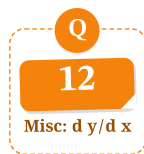
$$\Rightarrow \frac{1}{v} \cdot \frac{dv}{dx} = \log (x-3) \cdot 2x + x^2 \cdot \frac{1}{x-3} \cdot \frac{d}{dx} (x-3)$$

$$\Rightarrow \frac{dv}{dx} = v \left[2x \log (x-3) + \frac{x^2}{x-3} \cdot 1 \right]$$

$$\Rightarrow \frac{dv}{dx} = (x-3)^{x^2} \left[\frac{x^2}{x-3} + 2x \log (x-3) \right]$$

From (1), (2), and (3), we get

$$\frac{dy}{dx} = x^{x^2-3} \left[\frac{x^2-3}{x} + 2x \log x \right] + (x-3)^{x^2} \left[\frac{x^2}{x-3} + 2x \log (x-3) \right]$$



Find $\frac{dy}{dx}$, if $y = 12(1-\cos t)$, $x = 10(t-\sin t)$,

$$\frac{\pi}{2} < t < \frac{\pi}{2}$$

Conclusion

The given function is $y = 12(1-\cos t)$,
 $x = 10(t-\sin t)$

Hence,

$$\frac{dx}{dt} = \frac{d}{dt} [10(t-\sin t)]$$

$$= 10 \cdot \frac{d}{dt} (t-\sin t)$$

$$= 10(1-\cos t)$$

$$\frac{dy}{dt} = \frac{d}{dt} [12(1-\cos t)]$$

$$= 12 \cdot \frac{d}{dt} (1-\cos t)$$

$$= 12 \cdot [0 - (-\sin t)]$$

$$= 12 \sin t$$

Therefore,

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{12 \sin t}{10(1-\cos t)}$$

$$= \frac{12.2 \sin \frac{t}{2} \cdot \cos \frac{t}{2}}{10.2 \sin^2 \frac{t}{2}}$$

$$= \frac{6}{5} \cot \frac{t}{2}$$

Q

13

Misc: $\frac{dy}{dx}$

()

Find $\frac{dy}{dx}$, if $y = \sin^{-1} x + \sin^{-1} \sqrt{1-x^2}$, $0 < x < 1$

Conclusion

The given function is $y = \sin^{-1} x + \sin^{-1} \sqrt{1-x^2}$

Hence,

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx} [\sin^{-1} x + \sin^{-1} \sqrt{1-x^2}] \\
 &= \frac{d}{dx} (\sin^{-1} x) + \frac{d}{dx} (\sin^{-1} \sqrt{1-x^2}) \\
 \frac{dy}{dx} &= \frac{1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-(\sqrt{1-x^2})^2}} \cdot \frac{d}{dx} (\sqrt{1-x^2}) \\
 &= \frac{1}{\sqrt{1-x^2}} + \frac{1}{x} \cdot \frac{1}{2\sqrt{1-x^2}} \cdot \frac{d}{dx} (1-x^2) \\
 &= \frac{1}{\sqrt{1-x^2}} + \frac{1}{2x\sqrt{1-x^2}} (-2x) \\
 &= \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} \\
 &= 0
 \end{aligned}$$

Q

14

Misc: $x \sqrt{1+y} + y \sqrt{1+x} = 0$

✧

If $x \sqrt{1+y} + y \sqrt{1+x} = 0$, for $-1 < x < 1$, prove that

$$\frac{dy}{dx} = -\frac{1}{(1+x)^2}$$

Conclusion

The given function is $x \sqrt{1+y} + y \sqrt{1+x} = 0$

$$\Rightarrow x \sqrt{1+y} = -y \sqrt{1+x}$$

Squaring both sides, we get

$$\begin{aligned}
 x^2(1+y) &= y^2(1+x) \\
 \Rightarrow x^2 + x^2 y &= y^2 + x y^2 \\
 \Rightarrow x^2 - y^2 &= x y^2 - x^2 y \\
 \Rightarrow x^2 - y^2 &= x y(y-x) \\
 \Rightarrow (x+y)(x-y) &= x y(y-x) \\
 \Rightarrow x+y &= -x y \\
 \Rightarrow (1+x) y &= -x \\
 \Rightarrow y &= \frac{-x}{(1+x)}
 \end{aligned}$$

Differentiating both sides with respect to x , we get

$$\begin{aligned}
 \frac{dy}{dx} &= - \left[\frac{(1+x)}{dx} (x) - (x) \cdot \frac{d}{dx} (1+x) \right] \\
 &= - \frac{(1+x) - x}{(1+x)^2} \\
 &= - \frac{1}{(1+x)^2}
 \end{aligned}$$

Hence proved.

Q

15

Misc: $(x-a)^2 + (y-b)^2 = c^2$

✧

If $(x-a)^2 + (y-b)^2 = c^2$, for some $c > 0$, prove that

$$\frac{[1 + (\frac{dy}{dx})^2]^{\frac{3}{2}}}{\frac{d^2 y}{dx^2}}$$

is a constant independent of a and b .

Conclusion

The given function is $(x-a)^2 + (y-b)^2 = c^2$ Differentiating both sides with respect to x , we get

$$\begin{aligned}
 \frac{d}{dx} [(x-a)^2] + \frac{d}{dx} [(y-b)^2] &= \frac{d}{dx} (c^2) \\
 \Rightarrow 2(x-a) \cdot \frac{d}{dx} (x-a) + 2(y-b) \cdot \frac{d}{dx} (y-b) &= 0 \\
 \Rightarrow 2(x-a) \cdot 1 + 2(y-b) \cdot \frac{dy}{dx} &= 0 \\
 \Rightarrow \frac{dy}{dx} &= \frac{-(x-a)}{y-b}
 \end{aligned}$$

Therefore,

$$\begin{aligned}
\frac{d^2 y}{d x^2} &= \frac{d}{d x} \left[\frac{-(x-a)}{y-b} \right] \\
&= - \left[\frac{(y-b) \cdot \frac{d}{d x} (x-a) - (x-a) \cdot \frac{d}{d x} (y-b)}{(y-b)^2} \right] \\
&= - \left[\frac{(y-b) - (x-a) \cdot \frac{d y}{d x}}{(y-b)^2} \right] \\
&= - \left[\frac{(y-b) - (x-a) \cdot \left\{ \frac{-(x-a)}{y-b} \right\}}{(y-b)^2} \right] \quad [\text{Using (1)}] \\
&= - \left[\frac{(y-b)^2 + (x-a)^2}{(y-b)^3} \right]
\end{aligned}$$

Hence,

$$\begin{aligned}
\frac{[1 + (\frac{d y}{d x})^2]^{\frac{3}{2}}}{\frac{d^2 y}{d x^2}} &= \frac{[1 + \frac{(x-a)^2}{(y-b)^2}]^{\frac{3}{2}}}{- \left[\frac{(y-b)^2 + (x-a)^2}{(y-b)^3} \right]} \\
&= \frac{[\frac{(y-b)^2 + (x-a)^2}{(y-b)^2}]^{\frac{3}{2}}}{- \left[\frac{(y-b)^2 + (x-a)^2}{(y-b)^3} \right]} \\
&= \frac{c^2 [\frac{(y-b)^2 + (x-a)^2}{(y-b)^2}]^{\frac{3}{2}}}{- \frac{(y-b)^2 + (x-a)^2}{(y-b)^3}} \\
&= \frac{c^2}{- \frac{(y-b)^3}{(y-b)^3}} \\
&= \frac{c^3}{- \frac{c^2}{(y-b)^3}} \\
&= -c
\end{aligned}$$

-c is a constant and is independent of a and b.

Hence proved.

Q

16

Misc: $\cos y = x \cos (a+y)$



If $\cos y = x \cos (a+y)$, with $\cos a \neq \pm 1$, prove that $\frac{d y}{d x} = \frac{\cos^2(a+y)}{\sin a}$.

Conclusion

The given function is $\cos y = x \cos (a+y)$

Therefore,

$$\begin{aligned}
\Rightarrow \frac{d}{d x} [\cos y] &= \frac{d}{d x} [x \cos (a+y)] \\
\Rightarrow -\sin y \frac{d y}{d x} &= \cos (a+y) \cdot \frac{d}{d x} (x) + x \cdot \frac{d}{d x} [\cos (a+y)] \\
\Rightarrow -\sin y \frac{d y}{d x} &= \cos (a+y) + x \cdot [-\sin (a+y)] \frac{d y}{d x} \\
\Rightarrow [x \sin (a+y) - \sin y] \frac{d y}{d x} &= \cos (a+y)
\end{aligned}$$

$$\text{Since, } \cos y = x \cos (a+y) \Rightarrow x = \frac{\cos y}{\cos (a+y)}$$

Then, equation (1) becomes,

$$\begin{aligned}
[\frac{\cos y}{\cos (a+y)} \cdot \sin (a+y) - \sin y] \frac{d y}{d x} &= \cos (a+y) \\
\Rightarrow [\cos y \cdot \sin (a+y) - \sin y \cdot \cos (a+y)] \cdot \frac{d y}{d x} &= \cos^2(a+y) \\
\Rightarrow \sin (a+y-y) \frac{d y}{d x} &= \cos^2(a+y) \\
\Rightarrow \frac{d y}{d x} &= \frac{\cos^2(a+y)}{\sin a}
\end{aligned}$$

Hence proved.

Q

17

Misc: $x = a(\cos t + t \sin t)$

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If $x = a(\cos t + t \sin t)$ and $y = a(\sin t - t \cos t)$, find $\frac{d^2 y}{d x^2}$.

Conclusion

The given function is $x = a(\cos t + t \sin t)$ and $y = a(\sin t - t \cos t)$

Therefore,

$$\begin{aligned}\frac{dx}{dt} &= a \cdot \frac{d}{dt} (\cos t + t \sin t) \\ &= a[-\sin t + \sin t + t \cdot \frac{d}{dt} (t) + t \cdot \frac{d}{dt} (\sin t)] \\ &= a[-\sin t + \sin t + t \cos t] \\ &= a t \cos t\end{aligned}$$

$$\begin{aligned}\frac{dy}{dt} &= a \cdot \frac{d}{dt} (\sin t - t \cos t) \\ &= a[\cos t - \{\cos t \cdot \frac{d}{dt} (t) + t \cdot \frac{d}{dt} (\cos t)\}] \\ &= a[\cos t - \{\cos t - t \sin t\}] \\ &= a t \sin t\end{aligned}$$

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)} = \frac{a t \sin t}{a t \cos t} = \tan t$$

$$\begin{aligned}\frac{d^2 y}{d x^2} &= \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} (\tan t) = \sec^2 t \cdot \frac{dt}{dx} \\ &= \sec^2 t \cdot \frac{1}{a t \cos t} \left[\frac{dx}{dt} = a t \cos t \Rightarrow \frac{dt}{dx} = \frac{1}{a t \cos t} \right] \\ &= \frac{\sec^3 t}{a t}, 0 < t < \frac{\pi}{2}\end{aligned}$$

Q

18

Misc:
 $f(x) = |x|^3$ 

If $f(x) = |x|^3$, show that $f'(x)$ exists for all real x and find it.

Conclusion

It is known that

$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

Therefore, when

$$x \geq 0,$$

$$f(x) = |x|^3 = x^3$$

In this case, f'

$$(x) = 3x^2 \text{ and}$$

$$\text{hence, } f''(x) = 6x$$

When $x < 0$,

$$f(x) = |x|^3 = (-x)^3 = -x^3$$

In this case, f'

$$(x) = -3x^2 \text{ and}$$

$$\text{hence, } f''(x) = -6x$$

x

Thus, for

$$f(x) = |x|^3, f'(x)$$

exists for all real

x and is given by,

$$f''(x) = \begin{cases} 6x, & \text{if } x \geq 0 \\ -6x, & \text{if } x < 0 \end{cases}$$

Q

19

Misc: $\sin(A+B) = \sin A \cos B + \cos A \sin B$ 

Using the fact that $\sin(A+B) = \sin A \cos B + \cos A \sin B$ and the differentiation, obtain the sum formula for cosines.

Conclusion

Given, $\sin(A+B) = \sin A \cos B + \cos A \sin B$

Differentiating both sides with respect to x , we get

$$\begin{aligned}
\frac{d}{dx} [\sin (A+B)] &= \frac{d}{dx} (\sin A \cos B) + \frac{d}{dx} (\cos A \sin B) \\
\Rightarrow \cos (A+B) \cdot \frac{d}{dx} (A+B) &= \cos B \cdot \frac{d}{dx} (\sin A) + \sin A \cdot \frac{d}{dx} (\cos B) + \sin B \cdot \frac{d}{dx} (\cos A) + \cos A \cdot \frac{d}{dx} (\sin B) \\
\Rightarrow \cos (A+B) \cdot \frac{d}{dx} (A+B) &= \cos B \cdot \cos A \frac{dA}{dx} + \sin A (-\sin B) \frac{dB}{dx} + \sin B (-\sin A) \frac{dA}{dx} + \cos A \cos B \frac{dB}{dx} \\
\Rightarrow \cos (A+B) \cdot \left[\frac{dA}{dx} + \frac{dB}{dx} \right] &= (\cos A \cos B - \sin A \sin B) \cdot \left[\frac{dA}{dx} + \frac{dB}{dx} \right] \\
\Rightarrow \cos (A+B) &= \cos A \cos B - \sin A \sin B
\end{aligned}$$

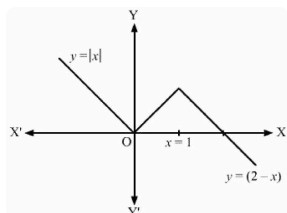
Q

20

Misc: Does there exist a function...



Does there exist a function which is continuous everywhere but not differentiable at exactly two points? Justify your answer.



Conclusion

Consider, $y = \begin{cases} |x| & -\infty < x \leq 1 \\ 2-x & 1 \leq x < \infty \end{cases}$

It can be seen from the above graph that the given function is continuous everywhere but not differentiable at exactly two points which are 0 and 1.

Q

21

Misc: piecewise function

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If $y = \begin{vmatrix} f(x) & g(x) & h(x) \\ l & m & n \\ a & b & c \end{vmatrix}$, prove that $\frac{dy}{dx} = \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ l & m & n \\ a & b & c \end{vmatrix}$

Conclusion

$$y = \begin{vmatrix} f(x) & g(x) & h(x) \\ l & m & n \\ a & b & c \end{vmatrix}$$

$$\Rightarrow y = (m c - n b) f(x) - (l c - n a) g(x) + (l b - m a) h(x)$$

Then,

$$\frac{dy}{dx} = \frac{d}{dx} [(m c - n b) f(x)] - \frac{d}{dx} [(l c - n a) g(x)] + \frac{d}{dx} [(l b - m a) h(x)]$$

$$= (m c - n b) f'(x) - (l c - n a) g'(x) + (l b - m a) h'(x)$$

$$= \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ l & m & n \\ a & b & c \end{vmatrix}$$

$$\frac{dy}{dx} = \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ l & m & n \\ a & b & c \end{vmatrix} \text{ proved.}$$

Q

22

Misc: $y = e^{a \cos^{-1} x}$



If $y = e^{a \cos^{-1} x}$, $-1 \leq x \leq 1$, show that $(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - a^2 y = 0$.

Conclusion

The given function is $y = e^{a \cos^{-1} x}$

Taking logarithm on both the sides, we get

$$\Rightarrow \log y = a \cos^{-1} x \log e$$

$$\Rightarrow \log y = a \cos^{-1} x$$

Differentiating both sides with respect to x, we get

$$\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} = a \cdot \frac{-1}{\sqrt{(1-x^2)}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-a y}{\sqrt{(1-x^2)}}$$

By squaring both the sides, we get

$$\Rightarrow \left(\frac{dy}{dx} \right)^2 = \frac{a^2 y^2}{1-x^2}$$

$$\Rightarrow (1-x^2) \left(\frac{dy}{dx} \right)^2 = a^2 y^2$$

Again, differentiating both sides with respect to x, we get

$$\Rightarrow \left(\frac{dy}{dx} \right)^2 \cdot \frac{d}{dx} (1-x^2) + (1-x^2) \times \frac{d}{dx} \left[\left(\frac{dy}{dx} \right)^2 \right] = a^2 \cdot \frac{d}{dx} (y^2)$$

$$\Rightarrow \left(\frac{dy}{dx} \right)^2 (-2x) + (1-x^2) \times 2 \cdot \frac{dy}{dx} \cdot \frac{d^2 y}{dx^2} = a^2 \cdot 2y \cdot \frac{dy}{dx}$$

$$\Rightarrow -x \cdot \frac{dy}{dx} + (1-x^2) \cdot \frac{d^2 y}{dx^2} = a^2 \cdot y \left[\frac{dy}{dx} \neq 0 \right]$$

$$\Rightarrow (1-x^2) \cdot \frac{d^2 y}{dx^2} - x \cdot \frac{dy}{dx} - a^2 y = 0$$

Hence proved.